

SPATIALAB: CAN VISION–LANGUAGE MODELS PERFORM SPATIAL REASONING IN THE WILD?

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ABSTRACT

Spatial reasoning is a foundational faculty of human cognition, yet it remains among the most persistent obstacles for contemporary vision–language models (VLMs). Progress has been difficult to gauge because prevailing benchmarks are built predominantly on synthetic or LLM-generated environments, organized around narrow task designs and puzzle-like configurations that do not reproduce the visual clutter, perceptual noise, and heterogeneous spatial relations VLMs encounter in realistic scenes. We introduce **SPATIALAB**, a benchmark that evaluates VLM spatial reasoning directly in realistic, unconstrained visual contexts and, by design, resolves this ecological-validity gap rather than merely restating it. SPATIALAB comprises 1,400 visual question–answer pairs organized into six principal categories—*Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*—each decomposed into five subcategories to yield a fine-grained taxonomy of 30 distinct task types. Every subcategory contains at least 25 questions and every principal category at least 200, so that coverage is balanced across the taxonomy and each item can be scored under both multiple-choice and open-ended protocols. Evaluating a broad and representative set of state-of-the-art VLMs—spanning open- and closed-source systems, reasoning-focused models, and models specialized for spatial reasoning—we observe a substantial and consistent gap relative to human competence. On the multiple-choice benchmark, the strongest system, InternVL3.5-72B, attains 54.93% accuracy against 87.57% for humans; under open-ended evaluation every model degrades by a further 10–25%, with GPT-5-mini leading at 40.93% against 64.93% for humans. Because the gap holds across model families, scales, and both evaluation formats, it localizes the shortfall to reasoning over complex spatial relations, depth perception, navigation, and 3D geometry rather than to any single architecture. By coupling a diverse, real-world evaluation framework with fine-grained diagnostic analyses, SPATIALAB renders the specific challenges—and the concrete opportunities—for advancing VLM spatial reasoning explicit, and supplies a reproducible instrument for measuring progress toward robust, human-aligned spatial understanding. We will release the images, questions, annotations, prompts, and evaluation scripts to support faithful replication.

1 INTRODUCTION

Spatial reasoning is a foundational faculty of human cognition, subsuming the interpretation of spatial layouts, the mental imagery and manipulation of objects, and navigation through the environment. It underpins a broad spectrum of applications, from robotics, autonomous driving, and augmented/virtual reality to geospatial analytics, computer graphics, human–computer interaction, and education (Huang et al., 2024b; Qi et al., 2024; 2025). For artificial agents—autonomous systems that must perceive and act within dynamic environments—reliable spatial competence presupposes the joint integration of geometric relations, depth cues, occlusion patterns, and agent-centric perspectives in scenes that are at once visually noisy and structurally complex (Bar-Anan et al., 2006; Cai et al., 2025; Trope & Liberman, 2010; Ramalho et al., 2018). Although recent vision–language models have markedly advanced multimodal representation and language grounding, their spatial judgments remain conspicuously fragile as soon as conditions become realistic (Chen et al., 2025; Kosoy et al., 2025; Pothiraj et al., 2025)—a fragility that current evaluation practice is poorly equipped to expose.

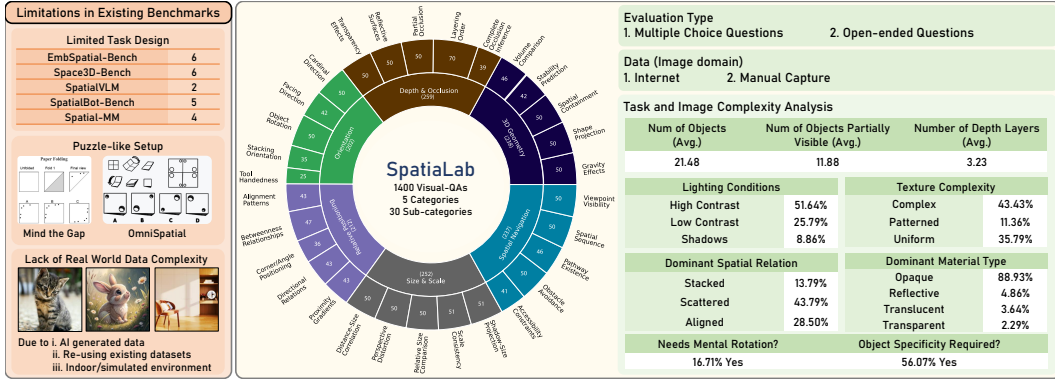


Figure 1: **Overview of SPATIALLAB.** The benchmark targets the limitations of prior datasets (left), contributes 1,400 visual QA pairs spanning six categories and 30 subcategories (center), and scores each item under both multiple-choice and open-ended protocols. Its items exhibit substantial task and image complexity, with wide variation in object counts, depth layers, lighting, textures, spatial relations, and materials (right).

Existing spatial benchmarks, however, are narrow and simplified in ways that systematically obscure this fragility. The majority target elementary tasks—binary spatial relations, low-resolution depth categories, or schematic navigation cues—and situate them in synthetic or simplified settings that suppress visual clutter and homogenize object appearance (Du et al., 2024; Jia et al., 2025; Szymanska et al., 2024; Shiri et al., 2024; Kamath et al., 2023; Fu et al., 2024; Song et al., 2024; Yang et al., 2024; Qi et al., 2025; Cai et al., 2024). Because such controlled setups relax the perceptual and reasoning demands placed on a model, they induce an appearance of saturation that masks failure under distribution shift. Precisely the challenges that matter most for embodied deployment—robust occlusion inference, scale-consistent judgment across viewpoints, and planning under partial observability—remain markedly under-sampled (Kosoy et al., 2025; Pothiraj et al., 2025; Chen et al., 2025; Cai et al., 2025). Consequently, models that appear proficient on existing benchmarks routinely collapse on integrated, multi-step inference, a discrepancy that both distorts measured progress and constrains safe deployment in real-world environments where spatial errors carry tangible consequences.

A meaningful evaluation must therefore span the core axes of spatial reasoning jointly rather than in isolation: relative position, depth and occlusion, orientation, size and scale, navigation, and 3D geometry. Whereas prevailing task-isolated setups probe these axes one at a time, human cognition integrates them seamlessly under noise and ambiguity (Driess et al., 2023; Brohan et al., 2023; Zitkovich et al., 2023; Collaboration et al., 2023; Yuan et al., 2024; Qi et al., 2025). Three design principles follow directly, and together they motivate SPATIALLAB: a benchmark should exercise both discriminative choice and free-form generation, so that selection-based competence is not mistaken for generative grounding; it should calibrate model performance against a human reference, so that the absolute magnitude of the shortfall is legible; and it should draw on real-world scenes that surface the failure modes simulator datasets conceal.

To meet these needs, we introduce **SPATIALLAB**, a benchmark that assesses vision–language spatial reasoning in realistic, unconstrained visual contexts, as outlined in Figure 1. **SPATIALLAB** contributes a balanced suite of visual question–answer items that spans a broad taxonomy of spatial tasks and scores each item under both multiple-choice and open-ended protocols. Its design deliberately foregrounds photographic diversity, compositional spatial-relational queries, and tasks that require the joint integration of depth, perspective, and 3D structure. On this benchmark we conduct a systematic evaluation of a wide collection of state-of-the-art VLMs—open- and closed-source models together with reasoning-focused and spatially specialized systems—calibrate them against a human reference, and pair the results with an extensive diagnostic error analysis. Our contributions are as follows:

1. **A diverse, real-world spatial-reasoning benchmark with a fine-grained taxonomy.** We introduce **SPATIALLAB**, a benchmark of 1,400 visual question–answer pairs spanning six principal categories and thirty task types situated in realistic, unconstrained contexts. With

at least 200 questions per category and 25 per subcategory, it affords balanced taxonomic coverage and scores each item under both multiple-choice (SPATIALAB-MCQ) and open-ended (SPATIALAB-OPEN) protocols, so that discriminative and generative competence are measured on identical content.

2. **A cross-family evaluation against a human reference.** We evaluate more than 25 state-of-the-art vision-language models—open- and closed-source, reasoning-oriented, and spatially specialized systems—against a human baseline, establishing a spatial-reasoning gap that is both substantial and consistent across model families, scales, and evaluation formats, and thereby not attributable to any single design choice.
3. **Fine-grained diagnostic error analyses.** We localize the systematic failure modes underlying this gap—depth perception, navigation, occlusion, and 3D geometry—and translate them into concrete, actionable directions for advancing robust, human-aligned spatial understanding.
4. **A systematic study of enhancement strategies.** We investigate a spectrum of inference- and training-time interventions—**supervised fine-tuning (SFT), chain-of-thought (CoT) prompting, CoT with self-reflection, and a multi-agent architecture (SPATIOXOLVER) built on multi-step reasoning**—and delineate precisely where each helps and where it fails across the multiple-choice and open-ended settings, isolating perceptual grounding as the decisive factor.

In the multiple-choice setting the leading models reach only 50–55% accuracy, whereas human annotators exceed 85%; under open-ended generation the same models shed a further 10–25 percentage points, so that the best systems settle near 40% against roughly 65% for humans. Error analysis localizes these deficits to occlusion inference, scale consistency, and multi-step navigational planning, delineating the limits of current compositional spatial representations. The intervention study sharpens the picture further, distinguishing the strategies that yield robust gains from those that instead amplify existing biases, and identifying perceptual grounding—rather than additional reasoning depth—as the decisive prerequisite for generalizable spatial competence. By jointly furnishing tasks, calibrated baseline results, and diagnostic analyses, SPATIALAB supplies a reproducible foundation for measuring and comparing progress in spatial-reasoning evaluation.

2 PRELIMINARIES: SPATIAL REASONING IN VLMS

Visual-spatial reasoning—the capacity to perceive, represent, and manipulate the spatial relations among objects—is central to human cognition and indispensable for embodied agents that must navigate and act in the world (Palmer, 1999; Chabris et al., 2006). We formalize a scene as a set of objects $O = \{o_1, \dots, o_n\}$ endowed with attributes (e.g., position, orientation, size) within a world coordinate system W . Visual-spatial reasoning is then the problem of estimating a mapping $f : (I, Q) \mapsto R$, in which I denotes an image (or image sequence), Q a linguistic query, and R a structured representation of spatial relations $R = \{r(o_i, o_j) \mid o_i, o_j \in O\}$ that encodes predicates such as *left-of*, *occludes*, or *supports*. For vision-language models (VLMs), estimating f is acutely challenging, since it requires reconciling noisy visual perception with compositional linguistic abstraction while simultaneously respecting embodied constraints such as perspective, gravity, and physical interaction (Dai et al., 2017; Alayrac et al., 2022). SPATIALAB is constructed to probe exactly this mapping under the realistic conditions where it is most likely to break down.

Early approaches to spatial reasoning in VLMs relied predominantly on basic spatial relations or narrow task designs (Cheng et al., 2024; Du et al., 2024; Szymanska et al., 2024), typically situated in synthetic datasets such as CLEVR (Johnson et al., 2016), structured question-answering benchmarks such as GQA (Hudson & Manning, 2019), or puzzle-like simulated setups such as MGT (Stogiannidis et al., 2025) and OmniSpatial (Stogiannidis et al., 2025), as summarized in Table 1¹. While serviceable for controlled evaluation, these settings oversimplify spatial structure, strip away environmental noise, and disregard dynamic factors such as temporal change and viewpoint shift. Overcoming these limitations requires benchmarking spatial reasoning in rich, real-world contexts that preserve scene

¹ Here, E. → Embodied, R.W. → Real World, T.C. → Task Categories, E.D. → Existing Dataset, A. → Annotation, T → Template, M → Manual, P → Puzzles, Md. → Modality (I → Image, V → Video), Bin. → Bin., Nm. → Nm., Df. → Difficulty, Cmpl. → Complexity (E → Easy, Mm → Medium, H → Hard, S → Shallow, L → Low, Hi → Hi, VH/VHi → Very Hard/Very High), A.L. → Analysis Level, B.B.P. → Best Baseline Performance.

Table 1: Comparison of spatial-reasoning benchmarks. Relative to prior datasets, SPATIALAB is the only benchmark that combines the broadest task taxonomy (30 categories) with in-the-wild imagery, paired multiple-choice and open-ended evaluation, and the highest reported difficulty and complexity; even the strongest baselines remain well below 60%.

Dataset	E.	T.C.	Data				QAs			Venue/ Year	Eval			A.L.	Md.	Best Baseline	B.B.P.
			Domain	Source	A.	Size	Count	Df.	Cmpl.	Type	Depth						
ScanQA	Y	7	Indoor	E.D.	M	800	41K	H	E	CVPR'22C	Close-ended	L	L	I	-	-	
Visual Spatial	Y	7	R.W.	E.D.	T	10K	10K	E	E	TACL'23	MCQ, Bin.	L	M	I	LXMERT	70.10	
What's up	N	6	Indoor	E.D.	T	5K	5K	E	E	EMNLP'23C	MCQ	S	Hi	I	XVLM-COCO	60.40	
EmbSpatia-Bench	Y	6	Indoor	E.D.	T	2.2K	3.6K	H	E	ACL'24C	MCQ, %	M	S	I	Qwen-VL-Max	49.11	
Space3D-Bench	N	6	Indoor	E.D.	M	211	1K	Mm	E	ECCV'24W	Mix	L	S	I	RAG3D-Chat	66.80	
SpatialRGPT-Bench	N	12	Mix	E.D.	T	1.5K	1.5K	Mm	E	NeurIPS'24C	MCQ, Bin.	M	L	I	GPT-4o	57.83	
BLINK-Spatial	Y	14	Mix	E.D.	M	7.3K	3.8K	H	E	ECCV'24C	MCQ, Bin.	M	M	I	GPT-4o	59.03	
Spatial-MM	N	4	Internet	Internet	T	2.3K	2.3K	E	Mm	EMNLP'24C	Open	L	Hi	I	GPT-4 Vision	63.82	
RoboSpatial	Y	4	Indoor	E.D.	T	1M	3M	H	E	CVPR'25C	Open, Bin.	L	S	I	LEO	71.90	
SpatialVLM	N	2	Web	E.D.	T	10M	2B	H	E	CVPR'24C	Open	L	S	I	GPT-4V	68.00	
VSI-Bench	Y	8	Indoor	M.E.D.	T	288	5K	Mm	Mm	CVPR'25C	MCQ, Nm.	M	M	V	Gemini-1.5 Pro	45.40	
OmniSpatial	Y	50	Internet	Internet	M.P	1387	1.5K	H	Mm	2025	MCQ, Bin.	M	S	I	o3-2025-04-16	56.33	
Mind the Gap	N	6	Mix	AI-Gen	M.P	1.8K	1.8K	E	Mm	2025	Mix	M	S	I	InternVL2.5-26B	48.83	
VLM4D	N	4	Mix	Mix	M	1K	1.8K	E	H	ICCV'25	MCQ	Hi	M	Mix	Gemini-2.5 Pro	62.00	
SPATIALAB-MCQ	Y	30	Mix	Mix, M	M	1.2K+	1.5K	H	H	2025	MCQ	Hi	H	Mix	InternVL3.5-72B	54.93	
SPATIALAB-OPEN	Y	30	Mix	Mix, M	M	1.2K+	1.5K	H	VH	2025	Open	Hi	H	Mix	GPT-5-mini	40.93	

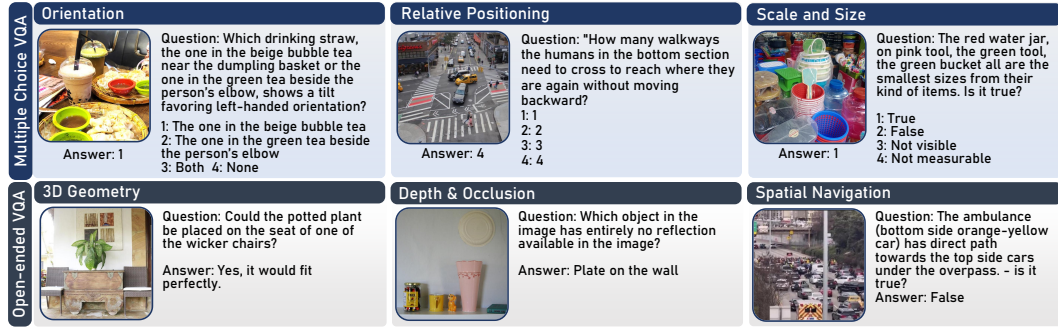


Figure 2: Representative examples drawn from the six categories under both open-ended and MCQ formats, illustrating the visual diversity and reasoning difficulty that characterize SPATIALAB.

complexity, perceptual variation, and multiple perspectives; only such settings confront VLMs with the demands they face in embodied or interactive applications—robotics, AR/VR, and autonomous navigation among them (Mu et al., 2023; Collaboration et al., 2023). SPATIALAB is designed to close precisely this gap.

SPATIALAB organizes spatial reasoning into six principal categories—*Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*—each grounded in a distinct cognitive faculty documented in the psychology and spatial-cognition literature (Baddeley, 1998; Previc, 1998). As shown in Figure 1 and Appendix B, each category is further decomposed into five subcategories, yielding a taxonomy of thirty task types that isolate fine-grained reasoning skills such as directional alignment, occlusion inference, and stability prediction. This classification is designed to span the spectrum of visual-spatial reasoning challenges encountered in realistic scenes; by decomposing complex reasoning into measurable, testable units, it establishes not merely whether a model “solves” spatial tasks but precisely which dimensions remain fragile or underdeveloped (Gardner, 2011; Huang et al., 2022; Gupta & Kembhavi, 2023; Yang et al., 2023).

The benchmark pairs this taxonomy with two complementary evaluation formats, multiple-choice and open-ended question answering, applied to the same items. The multiple-choice format probes recognition and fine-grained discrimination among distractors, whereas the open-ended format measures generative reasoning and compositional description (Driess et al., 2023; Liu et al., 2025; Zheng et al., 2022); jointly they mirror the demands of real-world reasoning and surface complementary strengths and weaknesses in model behavior. By integrating a principled taxonomy, diverse real-world imagery, and this dual-format protocol, SPATIALAB constitutes a structured framework for systematic benchmarking—one that supports fine-grained model comparison, direct calibration against human performance, and sharper insight into how robust spatial cognition might be cultivated in artificial agents.

3 SPATIALAB BENCHMARK

Spatial reasoning remains a persistent challenge for vision–language models (VLMs), most acutely under naturalistic conditions in which occlusion, perspective distortion, and noisy backgrounds co-occur. **SPATIALAB** confronts this challenge directly: it is a benchmark constructed to evaluate, systematically and at fine granularity, the capacity of VLMs for complex spatial cognition under realistic visual conditions. The benchmark comprises 1,400 visual question–answer pairs organized into six major categories—*Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*—each subdivided into five subcategories to yield 30 distinct task types. Every subcategory contains at least 25 QA pairs and every top-level category at least 200, so that coverage is balanced across the taxonomy and each subcategory supports statistically meaningful diagnostic comparison. **SPATIALAB** scores each item under both multiple-choice and open-ended protocols, jointly measuring discriminative accuracy, generative reasoning, and explanatory competence; it thus aligns with established multimodal evaluation practice while foregrounding spatial-specific reasoning. High visual diversity, fine-grained annotation, and a multi-tier quality-control pipeline together render **SPATIALAB** a reliable and extensible testbed. Figure 2 presents representative examples across all six categories in both formats, conveying the richness and difficulty of the real-world spatial reasoning tasks it contains.

3.1 BENCHMARK CONSTRUCTION

Image Collection. The image corpus was curated through three complementary acquisition strategies: (i) automated web crawling at scale, (ii) targeted online retrieval of scene-specific images, and (iii) manual photography across diverse indoor and outdoor environments (Figure 3). This multi-source pipeline secures diversity across object categories, environmental conditions, and spatial configurations that no single source would supply. To render natural variability explicit, every image was profiled along six meta-dimensions: *lighting condition* (high contrast, low contrast, shadows, and reflective settings), *texture complexity* (uniform, patterned, and complex), *edge complexity* (sharp and smooth boundaries), *dominant spatial relation* (stacked, scattered, aligned), *material type* (transparent, translucent, opaque, reflective), and *gravity constraints* (normal, floating, and unconstrained). Because each dimension is systematically represented in the final corpus, the benchmark spans a broad spectrum of real-world, in-the-wild scenarios rather than simplified or synthetic setups. A quantitative complexity analysis substantiates this design: images contain, on average, 21.48 objects, of which 11.88 are partially visible, distributed across 3.23 distinct depth layers. Spatial reference chains (e.g., “object A is left of B, which is behind C”) comprise 2.07 links on average, evidence of genuine multi-step relational structure; moreover, 16.71% of the questions require mental rotation and 56.07% require object specificity (a particular color, pattern, or comparable detail). These statistics confirm that **SPATIALAB** captures the cluttered, layered, and relationally rich scenes characteristic of real-world spatial reasoning—precisely the regime that synthetic datasets leave underrepresented.

Annotation. Annotation proceeded in three sequential phases. First, annotators underwent targeted training (protocol detailed in Appendix C.2) that established a consistent interpretation of the six categories and thirty subcategories through curated examples and counter-examples. Second, each image was paired with one or more spatial QA tasks spanning a range of reasoning types—proximity gradients, alignment patterns, occlusion inference, and stability prediction, among others—with items deliberately balancing perceptual grounding against higher-order inference (e.g., predicting stability under gravity constraints). Third, every QA pair was encoded in parallel in both multiple-choice (four options) and open-ended formats, so that the two evaluation paradigms operate on identical content and remain directly comparable. This process produced approximately 1,500 spatial visual QA pairs distributed across the six categories and thirty subcategories.

Review and Quality Control. To secure reliability, every annotation output was routed through a three-tier review and validation pipeline. The first tier checks each QA item for semantic validity and alignment with its designated subcategory. The second tier has independent annotators verify both the question phrasing and the designated answer, with an explicit emphasis on eliminating ambiguity—including cases in which perspective, camera viewpoint, or occlusion could otherwise admit more than one defensible answer—and on enforcing task clarity. The third tier is a gold-standard adjudication round that consolidates the surviving items into a set of high-confidence QA pairs for evaluation. All image metadata was stripped and the corpus was curated to respect copyright, though minor oversights may remain. This procedure yielded a final benchmark of 1,400 validated

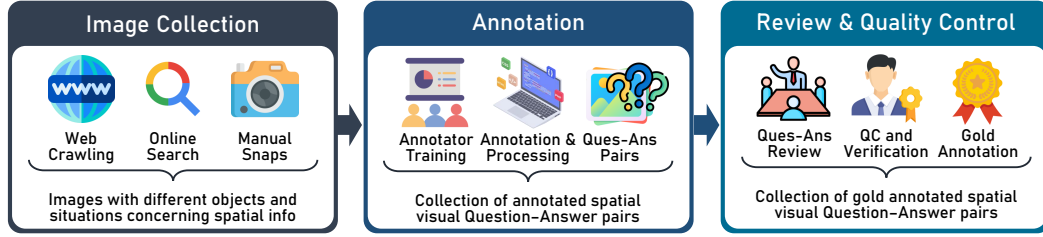


Figure 3: **Data creation pipeline for SPATIALAB.** Images are acquired through web crawling, targeted search, and manual photography, then subjected to structured annotation of spatial question-answer pairs and, finally, to a three-tier quality-control process culminating in gold-standard adjudication.

Table 2: Multiple-choice evaluation accuracy (% , $\uparrow$) on SPATIALAB-MCQ, reported per category and overall for every model group and calibrated against the human baseline. Even the best system, InternVL3.5-72B, reaches only 54.93% overall—more than 32 points below the 87.57% human baseline against a 25% random-choice floor—and every model group leaves comparable headroom; per-group best scores are bold-underlined.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
<i>Random Choice</i>	25.00	25.00	25.00	25.00	25.00	25.00	25.00
<i>Proprietary Models</i>							
GPT-4o-mini	47.06	39.00	47.03	47.17	49.60	49.79	46.50
GPT-5-mini	48.74	54.83	60.40	62.74	44.84	56.54	54.29
Gemini-2.0-Flash	47.06	55.21	53.96	58.02	54.37	46.84	52.50
Gemini-2.5-Flash	44.96	48.26	48.02	56.13	42.46	51.05	48.29
Gemini-2.5-Pro	47.48	50.19	49.50	58.49	43.65	52.32	50.07
Claude 3.5 Haiku	42.44	42.08	46.53	46.23	35.71	45.99	42.93
Mistral Medium 3.1	46.64	49.81	47.52	61.79	41.67	41.77	47.93
<i>Open-Source Models</i>							
InternVL3.5-1B	33.61	32.43	23.27	37.26	31.75	30.80	31.64
InternVL3.5-2B	34.03	31.66	31.68	40.57	32.54	32.49	33.71
Qwen-VL2.5-3B-Instruct	41.18	35.52	46.04	40.09	47.22	39.24	41.43
InternVL3.5-4B	42.86	42.86	42.08	54.72	36.51	42.19	43.29
Gemma-3-4B-it	43.70	34.36	46.53	45.75	37.30	37.97	40.57
Qwen-VL2.5-7B-Instruct	42.86	37.84	42.57	46.23	42.06	35.44	41.00
Llama-3.2-11B-Vision-Instruct	26.47	30.50	20.30	42.92	30.56	32.07	30.50
Gemma-3-27B-it	43.28	40.15	48.02	54.25	48.02	47.26	46.57
Qwen-VL2.5-32B-Instruct	41.18	40.15	46.53	45.28	45.24	41.77	43.21
InternVL3.5-72B	50.00	57.14	53.47	66.04	49.21	54.85	54.93
Qwen-VL2.5-72B-Instruct	47.06	48.65	51.98	54.25	43.65	48.95	48.86
Llama-3.2-90B-Vision-Instruct	46.22	52.12	50.50	58.96	46.83	48.52	50.36
<i>Reasoning Models</i>							
o3-mini	50.00	54.83	51.98	61.32	38.89	50.21	50.93
o4-mini-medium	51.26	58.30	54.95	64.15	40.87	51.48	53.21
Gemini-2-Flash-Thinking	37.82	41.31	41.58	45.75	50.40	43.04	43.36
Gemini-2.5-Flash-Thinking	45.80	53.67	52.97	56.60	55.16	53.59	52.93
Kimi-VL-A3B-Thinking-2506	42.86	41.31	40.59	51.42	39.68	41.35	42.71
<i>Spatial Reasoning Models</i>							
SpaceOm	42.44	38.61	48.02	37.74	42.86	39.24	41.36
SpaceThinker-Qwen2.5VL-3B	40.34	37.84	47.03	38.21	43.25	37.97	40.64
SpaceQwen2.5-VL-3B-Instruct	31.51	35.14	37.62	37.74	50.79	47.26	40.14
<i>Human Baseline</i>	93.70	74.13	91.58	91.51	88.89	87.76	87.57

QA items, whose measurement stability we further quantify through repeated-run and resampling analyses.

4 EXPERIMENTS AND EVALUATION

Models. To secure broad and representative coverage, we organize the evaluated models into four groups. **Proprietary models** comprise GPT-4o-mini (OpenAI, 2024), GPT-5-nano, Gemini-2-Flash and Gemini-2.5-Flash (Team, 2025a), and Claude 3.5 Haiku (Anthropic, 2024). **Open-source models** comprise Intern-VL3 (1B, 2B, 4B) (Zhu et al., 2025), Qwen-VL2.5 (3B, 7B, 32B, 72B Instruct) (Bai et al., 2025), GLM 4.5V-3.06B and GLM-4.1V-9B-Thinking (Team et al., 2025), Gemma-3 (4B, 27B)-it (Gemma Team, Google DeepMind, 2025), Llama-3.2 (11B, 90B) Vision-Instruct (Grattafiori et al., 2024), and Step-3-321B-MoE (StepFun, 2025). **Reasoning models** include o4-mini (OpenAI, 2025), the Gemini-2-Flash and Gemini-2.5-Flash thinking variants (Team,

2025a), and Kimi-VL-A3B-Thinking-2506 (Team, 2025b). **Spatial reasoning specialists** comprise SpaceOm, SpaceThinker-Qwen2.5VL-3B, and SpaceQwen2.5-VL-3B-Instruct (Chen et al., 2024a). This design deliberately crosses three axes—open- versus closed-source provenance, small- versus large-scale capacity, and general-purpose versus reasoning-oriented and spatially specialized training—so that the resulting comparisons isolate the factors that actually govern spatial competence rather than confounding them. For GLM-4.5V-106B-MoE, GLM-4.1V-9B-Thinking, and Step-3-321B-MoE we report open-ended evaluation only: these models are not instruction-tuned and were trained primarily on open-ended tasks, so MCQ prompting induces excessive refusals and would misrepresent their competence. To aid comparison across the results tables, we adopt a consistent color coding: **proprietary**, **open-source up to 7B**, **larger than 7B**, **proprietary reasoning**, **open-source reasoning**, **spatial specialists**, and **human baseline**.

Evaluation Process and Metrics. Accuracy is the primary metric in both the MCQ and open-ended settings. For **MCQ evaluation** we use direct prompting: each model returns the option number of its selected answer, and responses are matched automatically against the ground-truth key. The prompts and full protocol are provided in Appendix D.1. For **open-ended evaluation** each model generates a free-form answer, which is then scored by a large-language-model judge; prompts and protocol details appear in Appendix D.2. Because using an automated judge to score spatial answers could in principle introduce systematic bias, we validate it directly through an LLM-human agreement study (Appendix D.2). Table 5 shows that the judge (Gemini-2.5-Flash) reaches substantial agreement with human annotators overall—a Cohen’s kappa of 0.738 against the majority vote and 0.681–0.795 against individual annotators, a raw majority-vote accuracy of 0.880, and a Fleiss’ kappa of 0.774 among the annotators themselves; agreement is highest in well-defined categories (perfect on 3D Geometry) and only moderate in the more subjective Orientation and Size & Scale categories, a pattern we take into account when interpreting per-category open-ended scores. Additional details on experimental setup, prompts, and evaluation procedures are given in Appendix D.

Improving Visual Reasoning Capabilities. To quantify how much of the observed gap inference- and training-time interventions can close, we study a spectrum of strategies: the models’ inherent reasoning (Appendix H.1), Chain-of-Thought (CoT) prompting (Appendix H.2), CoT with self-reflection (Appendix H.3), supervised fine-tuning (SFT) (Appendix H.4), and a multi-agent system (Appendix H.5). For SFT, we fine-tuned Qwen-VL2.5-3B-Instruct on a stratified 40% split of the benchmark and evaluated strictly on the disjoint held-out 60%, observing improved MCQ generalization across relational and geometric tasks together with reduced errors in navigation and orientation. The agentic analysis, by contrast, yields a sharply uneven picture: orientation benefits substantially while other categories stagnate or degrade—direct evidence that the residual gap demands deeper perceptual and spatial grounding rather than additional layers of multi-step reasoning.

5 FINDINGS AND ANALYSIS

5.1 OVERALL RESULTS

SPATIALAB-MCQ. Table 2 reports multiple-choice performance, and the results reveal pronounced variation across models, tasks, and families: overall accuracy spans roughly 30–55% against a 25% random-choice floor, while the human baseline of 87.57% leaves more than thirty points of headroom above the best system. That best system is the open-source InternVL3.5-72B at 54.93%. Reasoning- and instruction-tuned models (o4-mini at 53.21%, Gemini-2.5-Flash-Thinking at 52.93%, GPT-5-mini at 54.29%) cluster near the top alongside several very large open-source models (InternVL3.5-72B at 54.93%, Llama-3.2-90B at 50.36%), yet scale alone is plainly not decisive: Llama-3.2-11B attains only 30.50% despite its considerable size. The category breakdown is equally diagnostic. Orientation and 3D Geometry are where the strongest models excel—several exceed 60% on one or both—whereas Spatial Navigation, Depth & Occlusion, and Size & Scale are markedly more variable and harder across the board. Tellingly, the spatially specialized models (SpaceOm, SpaceThinker, SpaceQwen) reach only the low-to-mid 40s overall, so domain specialization does not by itself confer higher MCQ accuracy on this diverse question set. Depth & Occlusion and Size & Scale yield mixed outcomes—some architectures handle occlusion competently while others fail—pointing to divergent inductive biases and training distributions rather than a shared capability. Taken together, these patterns indicate that instruction-tuning and reasoning capability preferentially benefit geometric and

Table 3: Open-ended evaluation accuracy (% , $\uparrow$) on SPATIALAB-OPEN, reported per category and overall for every model group and calibrated against the human baseline. Scores fall uniformly below the corresponding MCQ results: the strongest system, GPT-5-mini, attains 40.93% overall against a 64.93% human baseline, and every model degrades by roughly 10–25 points relative to its MCQ score—demonstrating that selection-based accuracy systematically overstates generative spatial competence. Per-group best scores are bold-underlined.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Proprietary Models							
GPT-4o-mini	23.53	16.60	23.27	30.66	17.86	21.94	26.00
GPT-5-mini	45.38	34.75	37.13	49.53	42.46	37.13	40.93
Gemini-2.0-Flash	31.93	24.32	27.23	31.13	26.19	24.47	27.43
Gemini-2.5-Flash	34.03	26.64	31.68	38.68	26.59	29.54	30.93
Gemini-2.5-Pro	37.14	45.45	36.36	37.14	23.91	24.44	33.61
Claude 3.5 Haiku	26.05	18.92	24.75	25.94	20.24	21.10	22.64
Mistral Medium 3.1	25.21	19.31	21.78	29.25	15.08	16.88	21.00
Open-Source Models							
InternVL3.5-1B	05.88	09.65	9.90	13.68	09.13	10.13	09.64
InternVL3.5-2B	12.18	11.20	10.89	23.58	11.90	18.14	14.50
Qwen-VL2.5-3B-Instruct	15.55	8.49	15.35	10.85	18.25	9.28	12.93
InternVL3.5-4B	19.33	17.76	15.84	19.81	16.27	18.99	18.00
Gemma-3-4B-it	20.17	13.13	14.85	23.58	15.08	19.83	17.64
Qwen-VL2.5-7B-Instruct	15.13	15.83	20.30	27.83	15.87	19.83	18.86
Llama-3.2-11B-Vision-Instruct	16.81	16.99	22.28	25.00	13.49	18.57	18.57
Gemma-3-27B-it	22.69	16.22	24.75	34.43	22.62	21.94	23.43
Qwen-VL2.5-32B-Instruct	16.39	09.65	14.85	16.98	09.92	13.50	13.36
InternVL3.5-72B	22.69	20.46	20.30	31.60	19.84	26.16	23.36
Qwen-VL2.5-72B-Instruct	26.89	20.85	25.25	30.66	24.60	20.68	24.64
Llama-3.2-90B-Vision-Instruct	22.69	23.17	21.29	28.30	21.83	27.00	24.00
GLM-4.5V-106B-MoE	31.09	20.46	25.25	26.42	24.21	24.47	25.21
Reasoning Models							
o3-mini	39.08	30.12	29.70	39.15	41.27	30.38	35.00
o4-mini-medium	40.76	32.82	32.18	42.92	44.05	34.18	37.86
Gemini-2-Flash-Thinking	31.09	27.41	31.19	34.43	29.37	29.54	30.36
Gemini-2.5-Flash-Thinking	37.14	45.45	36.36	37.14	21.74	22.22	32.77
Kimi-VL-A3B-Thinking-2506	13.45	14.29	11.88	17.92	12.70	18.57	14.79
GLM-4.1V-9B-Thinking	21.43	19.31	26.73	24.53	19.84	22.78	22.21
Step-3-321B-MoE	32.35	18.53	27.72	27.36	29.76	27.85	27.14
Spatial Reasoning Models							
SpaceOm	12.61	06.95	15.84	11.79	18.65	12.24	12.93
SpaceThinker-Qwen2.5VL-3B	13.45	09.27	17.82	10.38	19.44	10.13	13.36
SpaceQwen2.5-VL-3B-Instruct	12.61	03.86	13.86	09.43	11.90	11.39	10.36
Human Baseline	73.53	50.19	70.30	69.81	65.48	62.87	64.93

orientation reasoning, whereas multi-step grounding tasks such as navigation remain a decisive and, on this benchmark, unresolved weakness.

SPATIALAB-OPEN. The open-ended scores in Table 3 lie uniformly below the MCQ results and exhibit a wide spread: overall accuracy runs from about 9.6% (InternVL3.5-1B) to 40.9% (GPT-5-mini) against a 64.9% human baseline, so the shortfall relative to humans remains large across the entire field. Proprietary models lead this setting (GPT-5-mini at 40.93%, GPT-4o-mini at 26.00%, Gemini-2.5-Flash at 30.93%), but the reasoning-tuned variants are the most instructive: o4-mini (37.86%) and Gemini-2.5-Flash-Thinking (32.77%) decisively outperform many open-source counterparts, evidence that instruction- and CoT-style tuning materially improves generative output. Most small open-source models remain at the bottom, many in the teens (InternVL3.5-1B at 9.64%, Qwen-VL2.5-3B at 12.93%), while the largest open models rise only to the low 20s (InternVL3.5-72B at 23.36%, Qwen-VL2.5-72B at 24.64%), confirming that scale alone buys only modest gains. The spatial specialists (SpaceOm, SpaceThinker, SpaceQwen) reach merely ~ 10 –13% overall, so specialization again fails to translate into robust open-ended generation on this diverse question set. At the subtask level, the strongest models fare comparatively well on orientation and certain relative-position items (GPT-5-mini scores 49.53% on Relative Positioning and 37.13% on Orientation), whereas Depth & Occlusion, Size & Scale, and especially Spatial Navigation stay consistently difficult, with many models below 30%. This pattern reinforces the MCQ finding: multi-step grounding and perceptual reasoning are the dominant bottlenecks, and they bind most tightly when the answer must be generated rather than selected.

5.2 PERFORMANCE DROP IN OPEN-ENDED EVALUATION COMPARED TO MCQ

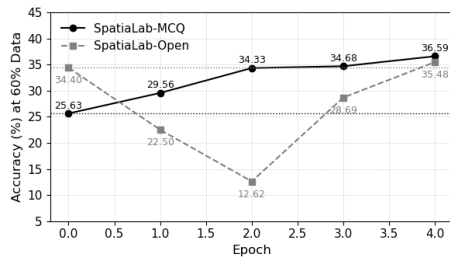
Across 25 models, the average MCQ→open-ended gap on SPATIALAB is 23.0% ($\sigma = 5.5\%$), with subtask means ranging narrowly from 22.89% (spatial navigation) to 24.57% (3D geometry). The gap is systematic rather than uniform: specialist spatial reasoning models incur the largest gaps (about 27%), concentrated in spatial navigation (up to 36.68%) and orientation (34.44%), whereas reasoning-oriented models exhibit the smallest gaps (about 19%) together with the lowest subtask variance—a direct signature of the stabilizing effect of instruction-tuning and CoT decoding. Correlational analysis confirms that spatial navigation dominates the overall disparity (Pearson $r = 0.99$), with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) contributing substantially. The occasional negative or near-zero gaps (−1.98 for Llama-3.2-11B in Depth & Occlusion, −3.18 for o4-mini in Relative Positioning) further reveal that poorly designed MCQ distractors can misrepresent true competence. We attribute this format sensitivity to four interacting factors: the structural advantage of the MCQ format, a specialization bias toward categorical tasks, the sequential-reasoning burden of spatial navigation, and calibration differences in generative output. The implication is unambiguous: MCQ evaluation alone systematically overestimates practical spatial reasoning ability, and reporting it in isolation would misstate the state of the field. We therefore recommend complementing MCQ with open-ended evaluation, auditing distractors, employing stepwise generation or instruction-tuning, and reporting per-subtask diagnostics, so that measured performance reflects genuine competence rather than format-dependent artifacts. Full details appear in Appendix G and Table 18.

5.3 ERROR ANALYSIS

SPATIALAB-MCQ. Our analysis (Appendix E) shows that VLMs attain selective peaks without holistic spatial competence. Closed-source models reach the highest single scores (85.71% in *Stacking Orientation*) yet collapse toward chance in *Relative Size Comparison*. Scaling open-source models raises ceilings (80.56% in *Corner/Angle Positioning*) without averting catastrophic failures (2.0% in *Object Rotation*). Reasoning-augmented and spatially specialized models add localized gains—in occlusion inference or navigation, for instance—but remain capped below 55% on many physical-abstraction tasks such as *Gravity Effects* and *Stability Prediction*. The error distributions expose recurrent weaknesses in embodied reasoning (*Tool Handedness*), recursive relational chaining (*Pathway Existence*), and non-local cues (*Reflective Surfaces*), while the comparatively strong *Obstacle Avoidance* results are best read as shortcut exploitation rather than robust planning. In aggregate, these patterns indicate that the models evaluated here lean heavily on surface-level correlations and lack stable encodings of orientation, physics, and compositional logic. Closing this gap will require both richer training distributions—embodied and physics-driven data in particular—and architectural mechanisms that enforce geometric and reference-frame grounding.

SPATIALAB-OPEN. Error analysis of SPATIALAB-OPEN (Appendix F) shows that even the top closed-source models (GPT-5-mini at 58.14% in *Directional Relations*) collapse on tasks such as *Proximity Gradients*, so their dominance is strictly task-specific. Large open-source models achieve strong *Depth & Occlusion* results (60.0%) yet still fall to 9.3% on *Proximity*, confirming that scale alone does not guarantee robustness. Smaller models exhibit isolated strengths (26.0% in *Relative Size Comparison*) but frequently break down entirely (0.0% in *Betweenness*), a direct reflection of the limits of compact architectures. Reasoning-tuned systems are promising—Gemini-2.5-Flash-Thinking reaches 75.0% in *Tool Handedness*—yet comparable models fall to single digits, exposing fragile integration. The specialized spatial models perform worst overall, rarely exceeding 20% and frequently failing on core relational tasks. Across the board the failure patterns concentrate in occlusion handling, orientation, and multi-step relational chaining, whereas the partial success on size-scale cues betrays reliance on superficial correlations. Three root causes follow: the absence of representations for spatial analogies, architectural limits tied to scale and training diversity, and brittle reasoning pipelines—together motivating unified geometric encodings, physics-aware reasoning, and embodied data as the route from isolated peaks toward consistent spatial competence.

Qualitative Error Analysis. To characterize these weaknesses at the mechanism level, we conducted a qualitative error analysis (Appendix I) and found that failures cluster into a small number of recurring, interpretable classes rather than constituting random noise. The dominant classes are spatial mislocalization, in which models confuse referents in crowded scenes; perspective and scale errors, in which reliance on object-size priors overrides image-based cues; and occlusion or ordering failures, most pronounced for thin or partially hidden structures. We further observe attribute



(a) SFT performance trends over epochs.

Category	MCQ			Open-ended		
	Before	After	Gain	Before	After	Gain
3D Geometry	30.07	37.76	7.69	20.28	23.78	3.50
Depth & Occlusion	23.87	32.90	9.03	43.23	45.81	2.58
Orientation	23.33	36.67	13.33	39.67	42.15	2.48
Relative Positioning	26.19	27.78	1.59	51.18	50.39	-0.79
Size & Scale	27.63	44.74	17.11	18.42	14.47	-3.95
Spatial Navigation	22.38	38.46	16.08	36.62	39.44	2.82
Grand Total	25.63	36.59	10.97	34.40	35.48	1.07

(b) Per-category accuracy before versus after SFT for MCQ and open-ended formats, with the gain highlighted. MCQ improves uniformly, whereas open-ended gains are marginal and category-dependent.

Figure 4: SFT training results for Qwen2.5-VL-3B-Instruct: (a) learning trends over epochs, and (b) final per-category accuracy before and after fine-tuning. SFT improves MCQ accuracy in every category (grand-total gain +10.97%) but yields only a negligible open-ended gain (+1.07%), with declines in Relative Positioning and Size & Scale.

confusion—the conflation of perceptual with functional properties—and open-ended rationalizations that produce fluent yet visually ungrounded narratives. Because these errors recur across architectures and prompting styles, they reflect shared inductive biases in the evaluated VLMs rather than dataset-specific noise. Critically, failure rates spike precisely when multiple cues must be fused—combining depth ordering with relative size, for example—and confidence calibration is especially poor in open-ended settings. Our diagnostic protocols confirm that models routinely ignore minimal but decisive visual features, fall back on brittle heuristics, and fail to update under counterfactual edits. The proximate causes are insufficient object-centric binding, the absence of geometric supervision, and training objectives that reward plausibility over grounding. Collectively, these findings establish that the evaluated VLMs achieve strong coarse perception yet falter on multi-cue integration and grounded reasoning, motivating geometry-aware supervision, multi-scale feature retention, and explicit verification pipelines.

5.4 PERFORMANCE IMPROVEMENT APPROACHES

Inherent Reasoning Mechanisms. Reasoning-enabled models consistently outperform their non-reasoning baselines in both formats, with the largest gains concentrated in relational and orientation tasks (+13.1% in Relative Positioning for MCQ). Reasoning also stabilizes open-ended performance, yet the improvements remain uneven across categories—notably declining in Size & Scale—which establishes that reasoning modules sharpen logical consistency without resolving the underlying grounding and scale-sensitivity deficits. Detailed per-category comparisons appear in Appendix H.1.

Chain-of-Thought (CoT) Prompting. Across models, CoT prompting confers little benefit and often reduces accuracy (Table 19), with orientation the sole category to show consistent gains. Step-by-step reasoning thus aids directional alignment but fails wherever robust perceptual grounding is required—depth, scale, and spatial navigation. Unlike in textual reasoning, CoT here tends to amplify flawed priors rather than correct them, which makes explicit that solving SPATIALAB demands perceptual understanding beyond logical chaining (details in Appendix H.2).

Chain-of-Thought (CoT) with Self-Reflection. Augmenting CoT with a self-reflection stage—in which the model is explicitly prompted to review its prior reasoning step by step and correct mistakes (prompt in Figure 8)—yields modest benefits in the multiple-choice setting, particularly for geometry and depth, yet does not generalize to open-ended tasks (Appendix H.3). This dissociation confirms that reflective reasoning cannot compensate for absent perceptual grounding, and locates the limitation in perception rather than in the reasoning loop itself.

Supervised Fine-Tuning (SFT). As shown in Figure 4, supervised fine-tuning of Qwen2.5-VL-3B-Instruct on a stratified 40% split of SPATIALAB (~560 samples) consistently improves MCQ accuracy across all six spatial reasoning categories, yet delivers little—and occasionally negative—transfer to open-ended tasks. This divergence is diagnostic of overfitting to task-specific answer distributions, and substantiates the concern that SFT tends to *teach to the test* rather than cultivate generalizable reasoning (Wang et al., 2025). The sharp open-ended drop points to internal representations biased toward categorical discrimination and unstable in generative settings, most plausibly through catastrophic forgetting of linguistic priors during fine-tuning (Huang et al., 2024a), compounded by objectives that privilege discrete-choice alignment over spatial reasoning

expressed in language. In short, SFT strengthens performance in structured formats while risking an algorithmic straitjacket that boosts constrained-task accuracy and stalls naturalistic spatial reasoning. Further details appear in Appendix H.4.

AI Agents for Spatial Reasoning (SPATIOXOLVER). To assess agentic reasoning, we employ SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025) for structured spatial reasoning over images. The framework decomposes complex visual reasoning into specialized sub-tasks—object segmentation, attribute extraction, spatial-relation mapping, and transformation tracking—each delegated to a dedicated agent, and consolidates their outputs into a unified structured representation that propagates low-level perceptual cues into higher-order relational inference. We implemented the pipeline with Gemini-2.5-Flash at low temperature to render the multi-stage refinement deterministic. The results are sharply category-dependent: Orientation improves substantially, by +8.00% in MCQ and a striking +36.00% in open-ended evaluation, whereas Depth & Occlusion and Spatial Navigation deteriorate severely under open-ended evaluation, by −24.00% and −12.00% respectively. The dissociation is telling: when the requisite perceptual priors—depth or occlusion cues—are weak, multi-step reasoning reinforces misconceptions instead of correcting them. The broader conclusion is decisive: agentic workflows help on tasks reducible to sequential alignment but cannot substitute for perceptual grounding in complex 3D scenes. Further details appear in Appendix H.5.

6 DISCUSSION

Spatial reasoning is central to embodied intelligence, yet the benchmarks that measure it have largely relied on synthetic scenes, templated questions, or narrowly scoped task designs that abstract away the visual variability of the real world. SPATIALAB was constructed to close precisely this gap: its 1,400 visual question-answer pairs span six principal spatial categories and thirty subcategories—among them 3D geometry, occlusion, and navigation—and each item is scored under paired multiple-choice and open-ended protocols, so that option-based pattern matching can be dissociated from genuine generative grounding on identical content. A multi-stage review pipeline, coupled with quantitative complexity checks over lighting, texture, and spatial-relational diversity, underwrites the reliability of this content. Evaluated against a human reference across more than 25 contemporary systems, the benchmark functions as a precise diagnostic instrument: rather than merely reporting an aggregate score, it localizes where—and along which spatial axis—current vision-language models fail.

Our fine-grained error analysis traces these failures to a common source: across transparency, multi-step navigation, and occlusion, models lean on surface-level texture cues in place of internalized 3D geometric representations. The intervention study sharpens rather than softens this diagnosis. Supervised fine-tuning (SFT) yielded clear gains on the structured MCQ format while simultaneously destabilizing open-ended reasoning—a signature of catastrophic forgetting of linguistic priors when optimization is directed at discrete answers. Chain-of-thought prompting benefited sequential tasks such as Orientation yet left the underlying perceptual errors in Depth and Scale uncorrected, frequently amplifying rather than resolving hallucinations. Agentic decomposition reproduced the same trade-off, improving narrowly alignment-reducible tasks while degrading holistic scene understanding. That three otherwise distinct interventions converge on the same limitation is telling: within the settings we examine, additional scale and reasoning scaffolding do not by themselves supply the missing perceptual grounding. The evidence thus points toward architectural innovations that explicitly integrate geometric grounding and physics-aware representations as the more promising route to human-aligned spatial intelligence.

7 CONCLUDING REMARKS

Spatial reasoning remains a core bottleneck for VLMs, most acutely in the visually complex, real-world settings that prior benchmarks largely omit. SPATIALAB answers this need with 1,400 diverse QA pairs across six spatial categories and thirty subcategories, enabling fine-grained evaluation under paired MCQ and open-ended formats. Even after a spectrum of performance-improving interventions, our experiments expose a substantial and consistent shortfall between state-of-the-art models and human competence, with open-ended generation shedding a further 10–25% relative to multiple-choice accuracy. Error analysis localizes the persistent failures to depth, occlusion, 3D geometry,

and multi-step navigation, and attributes them to a reliance on superficial cues in place of grounded spatial representations. Because the gap holds across model families, scales, and both formats, **SPATIALAB** renders spatial competence directly diagnosable rather than merely scored. The benchmark accordingly supports holistic evaluation and yields actionable guidance for developing VLMs with human-aligned spatial understanding. On the evidence assembled here, continued progress will depend on richer training signals—embodied interaction and action-conditioned feedback among them—coupled with architectures capable of geometric grounding, relational chaining, and consistent multi-scale reasoning.

ETHICS STATEMENT

We took several concrete steps to ensure ethical compliance throughout the development of this work. All metadata associated with the images was removed to protect privacy, and personally identifiable features—human faces or license plates, for instance—were minimized or excluded wherever possible. The dataset was curated with careful attention to sourcing and licensing so as to avoid copyright concerns, though minor oversights may remain despite our best efforts. The construction process was designed for transparency and fairness, ensuring that no demographic group was singled out or negatively represented. We adhered to the ICLR Code of Ethics and confirm that no human subjects or sensitive personal data were involved.

REPRODUCIBILITY STATEMENT

To support faithful replication, we document the dataset construction process, annotation pipeline, and evaluation protocol in detail across the main text and appendix. Every filtering, quality-control, and validation step is recorded explicitly, alongside precise definitions of the task categories and annotation criteria, and the hyperparameters, model configurations, and training procedures are reported in the appendix. To this end we will release the images, questions, annotation metadata, human labels, prompts, and evaluation scripts under a license that permits reproducible academic use.

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A RELATED WORK

Spatial reasoning in multimodal models. Spatial reasoning—the capacity to interpret geometric structure and relational layout in visual scenes—is a constitutive component of embodied intelligence. Early multimodal benchmarks such as CLEVR (Johnson et al., 2016) and GQA (Hudson & Manning, 2019) probed compositional reasoning through synthetic or curated templates, purchasing controlled analysis at the expense of ecological validity. The emergence of large vision–language models (VLMs) (Li et al., 2023; Alayrac et al., 2022; OpenAI, 2023) motivated a return to more naturalistic data, yet a systematic weakness persists beneath their otherwise strong captioning and visual-QA performance: these models routinely falter on tasks that require depth ordering, occlusion inference, or consistent reference-frame alignment. The pattern is diagnostic of a broader limitation—scale alone does not confer structured spatial understanding.

Spatial reasoning benchmarks. A succession of benchmarks has recently been advanced to probe these gaps (Table 1), each addressing part of the problem while leaving a characteristic dimension unresolved. SCANQA (Azuma et al., 2022), among the first embodied QA datasets to demand reasoning over 3D indoor environments, is constrained in both scale (800 scenes) and complexity, which limits its generalization. VISUAL SPATIAL (Liu et al., 2023) and WHAT’S UP (Kamath et al., 2023) extended coverage to real-world and household contexts, but their template-driven questions suppressed linguistic variability and overrepresented simple binary or multiple-choice relations. More recent efforts, including EMBSPATIAL-BENCH (Du et al., 2024) and SPACE3D-BENCH (Szymanska et al., 2024), broadened task variety yet remained bounded by scale (2.2K and 211 samples, respectively) and by template-heavy or narrowly manual annotation. SPATIALRGPT-BENCH and BLINK-SPATIAL widened diversity through mixed domains and up to 14 categories, but continued to rely on either static snapshots or limited multi-view simulation. Across this progression, breadth, naturalism, and evaluative depth have advanced individually but seldom in concert.

Embodiment, difficulty, and diversity. The move toward embodiment is exemplified by ROBOSPATIAL (Song et al., 2024), which scales to 1M scenes with robot-centric questions; its template-based annotation, however, renders those questions unnatural and susceptible to heuristic shortcuts. SPATIALVLM (Chen et al., 2024b) attains unprecedented scale (billions of QA pairs) but confines itself to two categories, precluding fine-grained diagnostic evaluation. At the opposite pole, high-fidelity manual datasets such as VSI-BENCH (Yang et al., 2024) improve annotation quality yet, with only 288 scenes, are too small for training or robust cross-model comparison. These extremes expose a recurring tension between scale and annotation fidelity that no prior benchmark reconciles. A parallel gap opens along the complexity axis: most existing benchmarks cap at “medium” (Mm in Table 1) and rarely reach the “hard” or “extremely hard” levels that require multi-step causal inference, leaving the most demanding regime under-evaluated for the next generation of models.

Evaluation protocols. Evaluation format and analytic depth are no less consequential than scale or taxonomy. Most prior work rests on closed-ended QA or binary judgments (e.g., VISUAL SPATIAL, EMBSPATIAL-BENCH), which streamline scoring at the cost of output expressivity. A handful of datasets—ROBOSPATIAL, SPATIAL-MM, and SPATIALAB-OPEN—introduce open-ended evaluation, but only SPATIALAB-OPEN conjoins it with high task complexity, manual puzzle-style annotation, and comprehensive error analysis. This conjunction is what makes systematic weaknesses legible beyond raw accuracy, surfacing the calibration and cross-model divergence that existing datasets seldom capture. Absent such granularity, reported gains can mislead, since a model may be overfitting to dataset-specific regularities rather than acquiring robust, transferable spatial competence.

Positioning of SPATIALAB. SPATIALAB addresses these limitations jointly rather than piecemeal. Where prior datasets span between 2 and 14 task categories, SPATIALAB covers 30 distinct categories across geometry, occlusion, orientation, relative positioning, size and scale, and navigation—an unprecedented breadth of task diversity. In deliberate contrast to template-based benchmarks, its questions are manually authored and puzzle-inspired, a design that preserves linguistic richness while curtailing the shortcut opportunities on which templated items founder. It is, moreover, fully embodied and multi-modal, spanning both static and dynamic visual contexts, and thereby bridges the robotics-style regime of ROBOSPATIAL and the web-scale regime of SPATIALVLM that earlier benchmarks treat as alternatives. As Table 1 shows, even the strongest baselines—InternVL3.5-72B

and GPT-5-mini—remain well below 60%, and the open-ended variant admits only 40.93% accuracy. This pronounced gap is corroborating evidence on two counts: it establishes the difficulty of the benchmark and, in turn, its headroom for driving future research.

Toward next-generation evaluation. By uniting high difficulty, broad coverage, manual annotation, and open-ended evaluation within a single resource, SPATIALAB moves the benchmark landscape from template-driven, narrow evaluation toward comprehensive, embodied testing. The design supports not only adjudicating whether a model answers correctly, but identifying systematic error patterns, probing causal and counterfactual reasoning, and quantifying calibration reliability. It thereby furnishes a rigorous foundation for the next phase of multimodal research and complements datasets such as MIND THE GAP and VLM4D, which target temporal or generative aspects but remain narrower in coverage. We accordingly expect SPATIALAB to serve as a high-bar diagnostic instrument that pushes models beyond surface-level recognition toward genuinely grounded spatial intelligence.

A.1 DETAILED ANALYSIS OF LIMITATIONS IN PRIOR BENCHMARKS

To situate SPATIALAB within the broader landscape, we conducted a detailed examination of existing spatial reasoning benchmarks. Earlier efforts established indispensable foundations, yet they exhibit structural and methodological constraints that circumscribe their value for evaluating current VLMs. These constraints cluster into three recurring areas—synthetic-domain bias, limited task scope, and restrictive evaluation formats—which Table 4 summarizes.

Synthetic and Template-Based Constraints. Benchmarks such as CLEVR and GQA rest on synthetic environments or template-generated linguistic structures. Although these settings offer controlled conditions for isolating compositional reasoning, their visual domain lacks real-world complexity and their linguistic space is narrow, so that a model can exploit template regularities rather than perform grounded spatial interpretation. Simulated benchmarks such as ROBOSPATIAL inherit the same weakness and incur a substantial sim-to-real gap.

Domain and Scale Limitations. Datasets grounded in 3D reconstruction, including ScanQA and Space3D-Bench, provide depth supervision but are confined to indoor scenes, which forecloses assessment in the outdoor and mixed environments that embodied agents must navigate. High-fidelity manual datasets such as VSI-Bench afford richer annotation yet remain small in scale, which limits their utility for high-resolution subcategory analysis.

Taxonomic and Evaluation Narrowness. Large-scale datasets such as SpatialVLM span broad tasks but lack fine-grained taxonomies, rendering them ill-suited to diagnosing specific weaknesses in orientation, occlusion, or layout reasoning. Many recent benchmarks, including EMBSPATIAL-BENCH and WHAT’S UP, depend heavily on binary or MCQ formats that can inflate measured performance by admitting option elimination in place of genuine generative spatial reasoning. SPATIALAB resolves these issues in a single design by combining MCQ and Open-Ended formats across a detailed taxonomy of 30 spatial subcategories.

Table 4: Limitations of representative prior spatial benchmarks relative to SPATIALAB, organized by domain constraint, evaluation format, and task taxonomy. SPATIALAB is unique in coupling in-the-wild imagery with combined MCQ and open-ended evaluation over a 30-subcategory taxonomy.

Benchmark	Domain	Eval. Format	Taxonomy	Primary Limitations Identified
CLEVR and GQA	Synthetic	Closed-Ended	Relational	Artificial visual domain; template-driven language; prone to shortcut learning.
ScanQA	Indoor 3D	Open-Ended	Navigation	Restricted to indoor point clouds; limited scale (< 1K scenes).
SpatialVLM	Web Data	Open-Ended	General	Covers only a small number of high-level spatial categories.
EmbSpatial-Bench	Indoor	MCQ / Binary	6 Categories	Relies on closed-ended formats; limited complexity in reasoning chains.
RoboSpatial	Simulator	Hybrid	Embodied	Template-generated questions in simulated scenes produce a sim-to-real gap.
Space3D-Bench	Indoor	MCQ	6 Categories	Very small scale (~200 samples) prevents robust statistical evaluation.
OmniSpatial	Mix	MCQ / Binary	Metric	Emphasizes metric outputs; limited support for explanatory reasoning.
SpatialLab (Ours)	In-the-Wild	MCQ + Open	30 Sub-cats	Addresses identified limitations with real-world images, detailed taxonomy, and diverse evaluation formats.

B TASK DESIGN AND DESCRIPTIONS

Spatial reasoning demands far more than the recognition of individual objects; it turns on how those objects relate, align, and interact within physical space. *SpatialLab* is designed to interrogate precisely

these fine-grained aspects of spatial understanding across a deliberately diverse set of categories. Its tasks are framed as humans naturally describe and reason about environments, ranging from relative positioning (e.g., *What’s to the left of the microwave?*) to geometric inference (e.g., *Would this stack fall if another book is added?*). Each category decomposes into subcategories that isolate a subtle but essential dimension of reasoning—occlusion, orientation, navigation, or physical effects such as gravity. By casting every subtask in accessible, everyday terms, SpatiaLab compels models to move beyond object detection toward structured spatial interpretation, and thereby measures how well a system generalizes across the contexts that matter for robotics, AR/VR, autonomous driving, and human–computer interaction. The following sections provide concise, task-level descriptions of every category and subcategory in SpatiaLab.

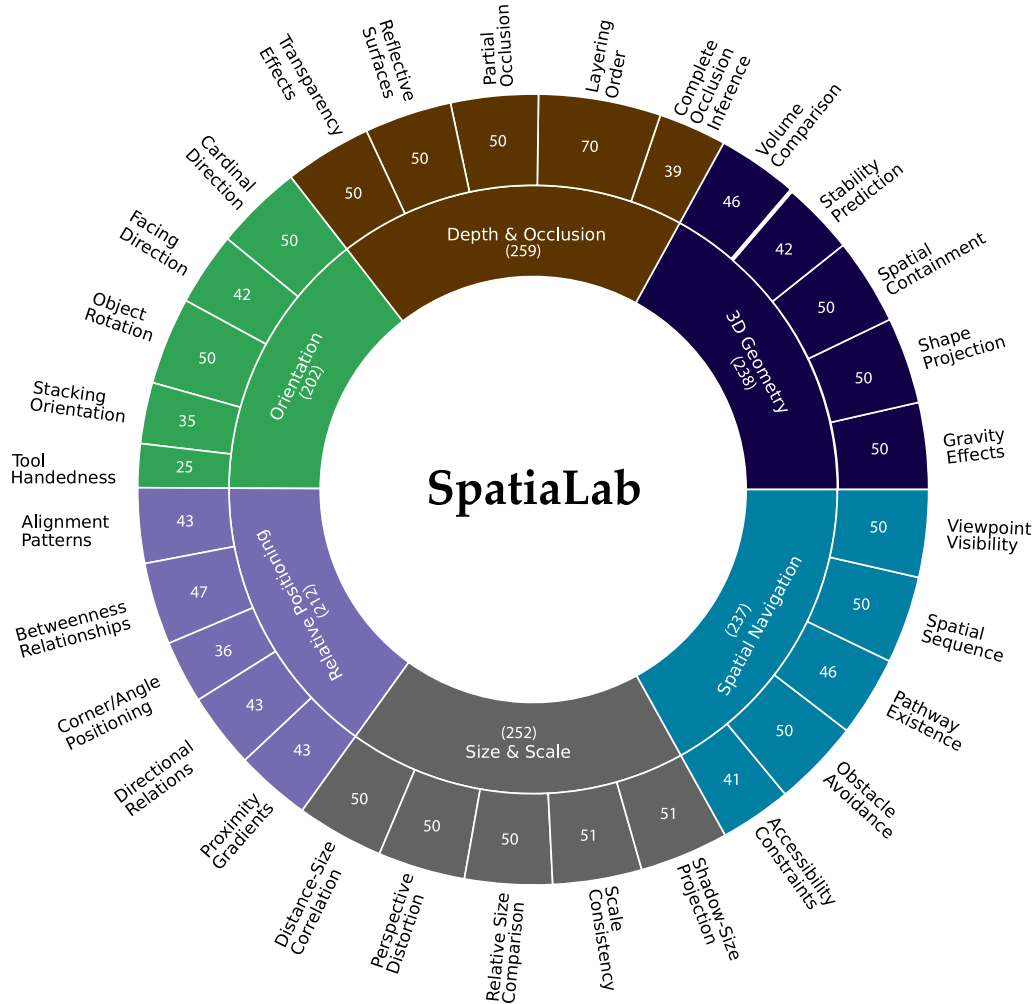


Figure 5: **Categories and subcategories of spatial reasoning in SPATIALAB.** Each of the six principal categories decomposes into five subcategories, yielding a taxonomy of thirty distinct task types.

The descriptions that follow articulate, category by category, the rationale and real-world utility that motivate each task design.

B.1 RELATIVE POSITIONING

Directional Relations. Directional relations specify how objects are positioned relative to one another—left, right, above, or below. Modeling them enables natural interpretation of spatial layouts,

1296 much as humans describe scenes; identifying what lies immediately to the left of an appliance, for
1297 example, supports scene understanding and robot navigation.

1298 **Proximity Gradients.** Proximity reasoning determines which objects are nearest to or farthest from
1299 a reference point, as in “Which object is nearest to the doorway?” Models that estimate graded
1300 closeness can support navigation and assistive technologies.

1301 **Alignment Patterns.** Alignment concerns whether objects follow a consistent orientation or or-
1302 der—stacked vertically or arranged in a straight line. Recognizing such patterns is central to
1303 understanding organization in shelves, tables, and warehouses, and it underpins robotic sorting and
1304 automated inspection.

1305 **Betweenness Relationships.** Betweenness identifies objects occupying the middle ground between
1306 two references; asking what sits between the lamp and the armchair, for instance, requires reasoning
1307 about spatial continuity, which in turn aids navigation, retrieval, and situational awareness.

1308 **Corner/Angle Positioning.** Corners and angled positions serve as reference anchors in spatial
1309 reasoning. Questions such as “What’s positioned in the back-right corner?” demand localization
1310 within constrained regions—a capability vital to robotic mapping and placement.

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B.2 DEPTH & OCCLUSION

Layering Order. Layering requires distinguishing which objects appear in front of others; identifying the frontmost item on a table exemplifies depth reasoning and is particularly valuable in AR/VR applications where visual stacking shapes user experience.

Partial Occlusion. Partial occlusion arises when one object hides part of another. Recognizing the item concealed behind a coffee cup requires inferring missing information, which strengthens the robustness of perception systems in cluttered environments.

Complete Occlusion Inference. When an object is entirely hidden, indirect cues such as shadows or surface bulges may nonetheless reveal its presence. Inferring what lies beneath a blanket demands reasoning from indirect evidence, mirroring human inference under visual uncertainty.

Transparency Effects. Transparent surfaces such as glass or water permit partial visibility of objects behind them. Determining what is visible through a fish tank fuses occlusion reasoning with transparency perception, a capability central to simulation, graphics, and embodied AI.

Reflective Surfaces. Reflective reasoning requires interpreting mirrors, shiny metals, and glass reflections. Identifying what a mirror reflects entails indirect spatial mapping, which is essential to safety in autonomous vehicles and situational awareness in robotics.

B.3 ORIENTATION

Cardinal Direction. Cardinal reasoning concerns compass directions—north, south, east, west. Determining which way a front door faces requires integrating global orientation cues, a prerequisite for navigation and geospatial alignment.

Object Rotation. Rotation reasoning quantifies how far an object has been turned about an axis. Estimating that a chair is rotated by a given angle is crucial for accurate 3D modeling and enables robots to align objects for manipulation.

Facing Direction. Facing direction specifies the orientation of an object’s front. Deciding whether a monitor faces north requires mapping its surface orientation onto global references, which matters in surveillance and robotics.

Stacking Orientation. Stacking reasoning determines whether objects are piled vertically, horizontally, or in mixed orientations. Recognizing such layouts ensures stability in warehouses and homes and is critical for safe robotic stacking.

Tool Handedness. Certain tools, such as scissors or gloves, are designed for left- or right-handed use. Identifying handedness requires fine-grained shape and orientation perception, which enhances assistive robotics in human-centered environments.

1350 B.4 SIZE & SCALE

1351

1352 **Relative Size Comparison.** Relative comparisons assess whether one object is larger, taller, or

1353 smaller than another; judging whether a vase is taller than a statue requires proportional reasoning

1354 that is crucial for packing, fitting, and recognition.

1355 **Perspective Distortion.** Perspective can make distant objects appear smaller than nearby ones

1356 irrespective of true size. Recognizing illusions—a faraway car appearing smaller than a nearby

1357 bicycle—requires compensating for depth cues, which renders models robust to camera perspective.

1358

1359 **Distance-Size Correlation.** Here size judgments depend on distance estimation: a nearby bicycle

1360 appearing larger than a distant car must not confound true size reasoning. This capability aids

1361 navigation, AR/VR, and visual grounding.

1362 **Scale Consistency.** Scale reasoning checks whether object sizes are plausible in context; a mouse

1363 larger than a chair signals an inconsistency that models must detect, strengthening realism in simulated

1364 environments.

1365 **Shadow-Size Projection.** Shadows furnish indirect cues to object size and light-source position.

1366 Estimating a light’s height from shadow length requires geometric reasoning valuable in graphics,

1367 architecture, and physics-based modeling.

1368

1369 B.5 SPATIAL NAVIGATION

1370

1371 **Pathway Existence.** Pathway reasoning tests whether a clear route connects two points—for instance,

1372 whether one can reach the desk without moving chairs. This capability underpins indoor robot

1373 planning.

1374 **Obstacle Avoidance.** Avoidance requires selecting routes that minimize collisions; asking for the

1375 clearest path from door to window maps directly onto robotic and human navigation, and is essential

1376 in autonomous driving and indoor mobility aids.

1377 **Viewpoint Visibility.** From a given viewpoint, particular objects may be occluded or visible.

1378 Determining what is obscured when sitting on the couch requires simulating perspective, which

1379 enhances AR/VR realism and robotic vision.

1380

1381 **Spatial Sequence.** Sequence reasoning recovers the order of encounters along a path; walking from

1382 the kitchen to the balcony, one might pass the dining table first. Such reasoning emulates human

1383 navigational memory.

1384 **Accessibility Constraints.** Accessibility reasoning identifies objects blocked from reach by obsta-

1385 cles—for example, a cabinet rendered unreachable by a table in front—which is crucial for assistive

1386 AI and ergonomic design.

1387

1388 B.6 3D GEOMETRY

1389

1390 **Volume Comparison.** Volume comparison asks which object occupies more three-dimensional

1391 space; judging whether a box exceeds a sphere supports packing and containment, and matters for

1392 logistics and simulation.

1393 **Stability Prediction.** Stability reasoning predicts whether adding or moving objects will topple a

1394 stack—for instance, whether placing a book will collapse a pile—mirroring human physical intuition

1395 and supporting safety in robotics and construction.

1396 **Shape Projection.** Projection tasks ask what shape an object would cast under a given rotation or

1397 lighting: a sphere always projects a circle, whereas a cube may project a square or hexagon. Such

1398 reasoning strengthens geometric understanding.

1399

1400 **Spatial Containment.** Containment reasoning determines whether one object can fit inside another,

1401 as in whether a ball fits in a box; this supports packing and design and aligns with intuitive geometry.

1402 **Gravity Effects.** Gravity reasoning predicts motion outcomes under physical forces—for example,

1403 a rolling ball halting beneath a table through friction and barriers—which strengthens real-world

simulation capabilities.

C MORE DETAILS ON BENCHMARK CONSTRUCTION

C.1 ANNOTATOR RECRUITMENT

We recruited annotators aged between 24 and 30, all engineering graduates, who participated voluntarily and worked independently without external incentives. Before annotation commenced, we conducted structured training sessions to establish a shared interpretation of the task requirements and to enforce consistency across annotators, and we upheld ethical and quality-control standards throughout the process. This protocol is what renders the resulting annotations both reliable and faithful to the dataset design.

C.2 ANNOTATOR TRAINING AND PROTOCOL

To secure consistency and reliability across the full task taxonomy, we designed a structured training program administered prior to annotation. Annotators were first introduced to the six core categories (*Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*) and their thirty subcategories. For each subcategory we supplied curated examples illustrating the corresponding spatial relations (e.g., directional relations, occlusion inference, stability prediction), paired with counter-examples that made common pitfalls explicit. Beyond the taxonomy proper, we trained annotators to reason about six meta-dimensions of image variability—lighting, texture complexity, edge density, dominant relation, material type, and gravity constraints—so that each question deliberately exercised visual diversity rather than incidental cues. We further demonstrated how these factors interact under realistic conditions, as when occlusion must be inferred under low-contrast lighting or scale consistency preserved under perspective distortion.

We then introduced the two evaluation modes. For the multiple-choice mode, annotators learned to construct precise distractors that probe fine-grained discrimination; for the open-ended mode, they practiced formulating prompts that elicit compositional reasoning. Calibration sessions grounded this training: annotators generated both question types for sample images and iteratively refined their designs under feedback until their judgments converged. To situate task design within the broader research landscape, we reviewed prior benchmarks (e.g., SpatialBot-Bench, EmbSpatial, SpatialMM) with the annotators and dissected their limitations. Such resources typically privilege simplified relations, synthetic or staged scenes, and narrow reasoning types, while neglecting the natural complexity introduced by shadows, reflections, or gravity-sensitive stability; this gap analysis directly motivated our demand for richer, more ambiguous, and dynamically layered spatial contexts.

Finally, we provided detailed demonstrations of the annotation protocol itself, specifying how to select images of sufficient spatial complexity, how to map each question unambiguously onto a single subcategory, and how to avoid trivial formulations. Annotators were instructed to favor tasks that demand multi-step reasoning—depth layering, relational chaining, or cross-view scale inference—over binary labels. It is this systematic training and iterative feedback that allow the final dataset to balance taxonomic coverage against real-world relevance.

D MORE DETAILS ON EXPERIMENTS AND EVALUATION

D.1 MULTIPLE CHOICE EVALUATION

The prompts supplied to the answer model are shown in Figure 6. For deterministic, single-token selection we set `temperature` to 0 and `max_tokens` to 10, leaving all other generation parameters at their defaults.

D.2 OPEN-ENDED EVALUATION

The prompts supplied to both the answer model and the judge model are provided in Figure 6. To accommodate free-form generation we set `temperature` to 0.7 and `max_tokens` to 1200, again leaving all other parameters at their defaults.



Figure 6: Prompts used for evaluation.

D.2.1 EVALUATION OF LLM-AS-A-JUDGE FOR OPEN-ENDED EVALUATION

To validate *Gemini-2.5-Flash* as an automatic judge, we assembled a category-balanced set of 240 samples spanning all six categories. Each sample presented the full evaluation context—question, ground truth, and model output—and was scored independently by three human annotators, who assigned binary labels (1 for correct, 0 for incorrect). We then compared the judge’s decisions against the human annotations using three complementary agreement metrics, computed both overall and per category so as to expose any category-specific divergence: **Cohen’s Kappa**, **Accuracy with Majority Vote**, and **Fleiss’ Kappa**, defined below.

Cohen’s Kappa. Cohen’s kappa (κ) quantifies pairwise agreement between two raters while correcting for chance. For the binary task it is defined as $\kappa = \frac{p_o - p_e}{1 - p_e}$, where p_o denotes the observed agreement and p_e the agreement expected by chance: $p_e = \sum_{c \in \{0,1\}} p_c^{(1)} p_c^{(2)}$. Here $p_c^{(1)}$ and $p_c^{(2)}$ are the marginal probabilities that each rater assigns class c ($\kappa = 1$: perfect agreement; $\kappa = 0$: agreement at chance; $\kappa < 0$: systematic disagreement). We computed Cohen’s kappa both between the LLM and each annotator individually (κ_{Human1} , κ_{Human2} , κ_{Human3}) and between the LLM and the annotator majority vote (κ_{majority}).

Accuracy with Majority Vote. We additionally measured the raw agreement between the LLM and the annotator majority. Accuracy is defined as $\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(y_i^{\text{LLM}} = y_i^{\text{maj}})$, where y_i^{LLM} is the LLM’s decision for sample i and y_i^{maj} the majority vote among annotators. Unlike Cohen’s kappa, accuracy does not correct for chance and simply records the proportion of identical labels.

Fleiss’ Kappa. To assess inter-annotator reliability, we computed Fleiss’ kappa (κ_F), which generalizes Cohen’s kappa to more than two raters. Let n_{ij} denote the number of annotators assigning category j to item i , and define $P_i = \frac{1}{n(n-1)} \sum_j n_{ij}(n_{ij} - 1)$, where n is the number of annotators. The observed agreement is $\bar{P} = \frac{1}{N} \sum_{i=1}^N P_i$, and the expected agreement is $\bar{P}_e =$

$\sum_j p_j^2$, with $p_j = \frac{1}{Nn} \sum_i n_{ij}$. Fleiss’ kappa is then $\kappa_F = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$, with higher values denoting stronger inter-annotator reliability beyond chance.

Table 5: Agreement between Gemini-2.5-Flash as an automatic judge and three human annotators across all categories. Reported are Cohen’s kappa against the majority vote and each annotator, raw accuracy against the majority, and Fleiss’ kappa among the annotators. The substantial overall agreement ($\kappa_{\text{majority}} = 0.738$, accuracy = 0.880) confirms the judge’s reliability; bold values mark the highest agreement within each category.

Category	κ_{Majority}	κ_{Human1}	κ_{Human2}	κ_{Human3}	Acc. (majority)	Fleiss κ (all)
Overall	0.738	0.738	0.681	0.795	0.880	0.774
3D Geometry	1.000	1.000	0.875	1.000	1.000	0.916
Depth & Occlusion	0.805	0.805	0.899	0.697	0.909	0.871
Orientation	0.500	0.500	0.426	0.588	0.810	0.571
Relative Positioning	0.833	0.657	0.676	0.833	0.917	0.778
Size & Scale	0.417	0.571	0.417	0.696	0.714	0.704
Spatial Navigation	0.815	0.815	0.667	1.000	0.933	0.773

Interpretation Table 5 establishes that the LLM judge (Gemini-2.5-Flash) attains substantial overall agreement with the human annotators: a Cohen’s kappa of 0.738 against the majority vote and 0.681–0.795 against individual annotators, a raw majority accuracy of 0.880, and a Fleiss’ kappa of 0.774 among the annotators themselves—figures that jointly evidence consistency well beyond chance. Agreement is not uniform across the taxonomy, and the variation is itself informative. Categories with clearly bounded answers reach the ceiling: 3D Geometry attains perfect agreement ($\kappa = 1.000$, accuracy = 1.000) and Depth & Occlusion follows closely ($\kappa_{\text{majority}} = 0.805$, accuracy = 0.909). Orientation ($\kappa = 0.500$, accuracy = 0.810) and Size & Scale ($\kappa = 0.417$, accuracy = 0.714) exhibit only moderate kappa, though their raw accuracies remain high and both stay within accepted reliability ranges.

Read together, these results indicate that Gemini-2.5-Flash scores correctness broadly on a par with human evaluators: its Cohen’s kappa values fall predominantly in the substantial range (0.61–0.80), and several categories approach near-perfect agreement (> 0.81). The Fleiss’ kappa of 0.774 confirms that the annotators are themselves highly consistent, which underwrites the reliability of the reference scores against which the judge is measured. The judge reaches perfect or near-perfect agreement in well-delineated categories such as 3D Geometry ($\kappa = 1.000$, accuracy = 1.000) and Spatial Navigation ($\kappa = 0.815$, accuracy = 0.933); the more modest agreement in Orientation and Size & Scale co-occurs with lower inter-annotator agreement in the same categories, suggesting that residual disagreement tracks the intrinsic ambiguity of those tasks rather than a systematic failure of the judge. Taken as a whole, the LLM-as-judge protocol demonstrates reliability sufficient to serve as a practical automated scoring instrument that closely approximates human evaluation, while the category-level variation delimits where its judgments should be read with corresponding caution.

E ERROR ANALYSIS OF SPATIALAB-MCQ

E.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models. Closed-source models (Table 6) produce some of the strongest results on the benchmark, yet their strength is uneven. The single best case is **GPT-4o-mini**, which reaches 85.71% in *Stacking Orientation*—the highest score overall. **Gemini-2.0-Flash** is comparably strong on occlusion and projection, attaining 64.0% in *Partial Occlusion* and 64.71% in *Shadow-Size Projection*, while **GPT-5-mini** records 68.0% in *Tool Handedness* and 55.71% in *Layering Order*. Pronounced weaknesses nonetheless persist: **Claude 3.5 Haiku** reaches only 32.0% in *Relative Size Comparison*. Averaged across categories, closed-source systems lead their open-source counterparts by 15–25 percentage points, but the within-model spread remains large, frequently exceeding 50 percentage points. Proprietary training pipelines therefore raise the ceiling without resolving the harder problem of uniform spatial competence.

Table 6: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Closed-source Models. We bold and underline the best score within each model category.

Category	Sub-Category	GPT-4o-mini	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	52.0	50.0	46.0	52.0	52.0	46.0
	Shape Projection	54.0	48.0	50.0	50.0	46.0	60.0
	Spatial Containment	36.0	46.0	44.0	38.0	34.0	38.0
	Stability Prediction	52.38	52.38	59.52	50.0	45.24	52.38
	Volume Comparison	41.3	47.83	39.13	34.78	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	61.54	69.23	61.54	61.54	61.54	46.15
	Layering Order	32.86	55.71	50.0	42.86	32.86	45.71
	Partial Occlusion	42.0	54.0	64.0	50.0	44.0	52.0
	Reflective Surfaces	34.0	54.0	46.0	48.0	44.0	44.0
	Transparency Effects	32.0	44.0	60.0	44.0	36.0	62.0
Orientation	Cardinal Direction	26.0	48.0	44.0	30.0	44.0	38.0
	Facing Direction	45.24	57.14	57.14	47.62	35.71	45.24
	Object Rotation	40.0	60.0	50.0	46.0	38.0	44.0
	Stacking Orientation	85.71	77.14	62.86	65.71	68.57	62.86
	Tool Handedness	52.0	68.0	64.0	64.0	56.0	56.0
Relative Positioning	Alignment Patterns	44.19	58.14	55.81	53.49	46.51	58.14
	Betweenness Relationships	51.06	63.83	70.21	63.83	44.68	61.7
	Corner/Angle Positioning	47.22	63.89	50.0	58.33	50.0	77.78
	Directional Relations	48.84	79.07	67.44	69.77	53.49	69.77
	Proximity Gradients	44.19	48.84	44.19	34.88	37.21	44.19
Size and Scale	Distance-Size Correlation	56.0	56.0	56.0	42.0	34.0	42.0
	Perspective Distortion	50.0	40.0	46.0	36.0	38.0	34.0
	Relative Size Comparison	42.0	44.0	58.0	46.0	32.0	38.0
	Scale Consistency	47.06	45.1	47.06	39.22	33.33	41.18
	Shadow-Size Projection	52.94	39.22	64.71	49.02	41.18	52.94
Spatial Navigation	Accessibility Constraints	56.1	60.98	65.85	65.85	48.78	56.1
	Obstacle Avoidance	38.0	48.0	34.0	34.0	46.0	40.0
	Pathway Existence	54.35	41.3	43.48	45.65	39.13	36.96
	Spatial Sequence	66.0	78.0	48.0	62.0	56.0	44.0
	Viewpoint Visibility	36.0	54.0	48.0	50.0	40.0	34.0

Table 7: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Open-source Large Models. We bold and underline the best score within each model category.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-72b-instruct	InternVL3.5-72B	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct
3D Geometry	Gravity Effects	22.0	48.0	54.0	52.0	52.0	56.0
	Shape Projection	32.0	42.0	44.0	50.0	46.0	50.0
	Spatial Containment	32.0	34.0	36.0	44.0	46.0	36.0
	Stability Prediction	33.33	54.76	38.1	52.38	47.62	52.38
	Volume Comparison	13.04	39.13	32.61	52.17	43.48	36.96
Depth and Occlusion	Complete Occlusion Inference	30.77	61.54	58.97	66.67	48.72	66.67
	Layering Order	34.29	34.29	37.14	50.0	50.0	38.57
	Partial Occlusion	32.0	34.0	44.0	60.0	48.0	58.0
	Reflective Surfaces	28.0	38.0	26.0	54.0	50.0	44.0
	Transparency Effects	26.0	40.0	40.0	60.0	46.0	62.0
Orientation	Cardinal Direction	22.0	48.0	34.0	46.0	48.0	40.0
	Facing Direction	21.43	42.86	52.38	54.76	50.0	50.0
	Object Rotation	2.0	40.0	38.0	48.0	48.0	44.0
	Stacking Orientation	28.57	65.71	65.71	68.57	62.86	71.43
	Tool Handedness	40.0	48.0	52.0	56.0	56.0	56.0
Relative Positioning	Alignment Patterns	27.91	55.81	48.84	53.49	55.81	53.49
	Betweenness Relationships	42.55	55.32	55.32	65.96	61.7	61.7
	Corner/Angle Positioning	58.33	55.56	36.11	80.56	50.0	75.0
	Directional Relations	60.47	62.79	51.16	83.72	62.79	65.12
	Proximity Gradients	27.91	41.86	32.56	48.84	39.53	41.86
Size and Scale	Distance-Size Correlation	36.0	50.0	54.0	52.0	44.0	46.0
	Perspective Distortion	22.0	42.0	36.0	46.0	34.0	52.0
	Relative Size Comparison	28.0	48.0	40.0	48.0	52.0	46.0
	Scale Consistency	29.41	47.06	41.18	43.14	37.25	37.25
	Shadow-Size Projection	37.25	52.94	54.9	56.86	50.98	52.94
Spatial Navigation	Accessibility Constraints	29.27	58.54	43.9	75.61	53.66	43.9
	Obstacle Avoidance	24.0	38.0	38.0	46.0	42.0	40.0
	Pathway Existence	28.26	54.35	36.96	39.13	54.35	47.83
	Spatial Sequence	34.0	52.0	56.0	66.0	54.0	62.0
	Viewpoint Visibility	44.0	36.0	34.0	50.0	42.0	48.0

Open-source Large Models. Scaling to large open-source models (Table 7) yields clear improvements that coexist with severe fragilities. **InternVL3.5-72B** achieves 80.56% in *Corner/Angle Positioning*, among the highest open-source results, and both **InternVL3.5-72B** and **llama-3.2-90b-vision-instruct** reach 66.67% in *Complete Occlusion Inference*. The weaknesses nonetheless remain acute—**llama-3.2-11b-vision-instruct** scores only 2.0% in *Object Rotation*—and even the gains are localized: **llama-3.2-90b-vision-instruct** attains 71.43% in *Stacking Orientation* while conceding losses elsewhere. The within-model spread often exceeds 70 percentage points, betraying brittle specialization, and gravity reasoning remains modest at 56.0%. Scale thus elevates the peaks without repairing the structural weaknesses: it amplifies selective strengths while leaving the foundational representational gaps intact.

Open-source Small Models. Open-source small models (Table 8) exhibit the sharpest contrasts across sub-categories. **qwen-2.5-vl-7b-instruct** reaches 74.29% in *Stacking Orientation*, whereas **InternVL3.5-1B** collapses to 12.0% in *Cardinal Direction*. Scores in *Pathway Existence* remain modest, with the best models tied at 34.78%, and gravity-related reasoning is similarly weak, peaking

Table 8: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Open-source Small Models. We bold and underline the best score within each model category.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	46.0	38.0	48.0	50.0	50.0	50.0
	Shape Projection	30.0	42.0	38.0	42.0	40.0	44.0
	Spatial Containment	30.0	28.0	34.0	34.0	36.0	34.0
	Stability Prediction	30.95	35.71	50.0	52.38	45.24	52.38
	Volume Comparison	30.43	26.09	36.96	36.96	47.83	34.78
Depth and Occlusion	Complete Occlusion Inference	30.77	30.77	41.03	64.1	33.33	41.03
	Layering Order	30.0	24.29	30.0	32.86	22.86	27.14
	Partial Occlusion	38.0	30.0	38.0	54.0	46.0	42.0
	Reflective Surfaces	38.0	34.0	36.0	32.0	42.0	40.0
	Transparency Effects	26.0	42.0	36.0	40.0	32.0	44.0
Orientation	Cardinal Direction	12.0	18.0	36.0	24.0	38.0	30.0
	Facing Direction	21.43	28.57	52.38	40.48	42.86	38.1
	Object Rotation	12.0	28.0	42.0	34.0	42.0	34.0
	Stacking Orientation	45.71	57.14	62.86	68.57	68.57	74.29
	Tool Handedness	40.0	36.0	40.0	60.0	48.0	48.0
Relative Positioning	Alignment Patterns	27.91	30.23	39.53	44.19	41.86	41.86
	Betweenness Relationships	38.3	38.3	46.81	65.96	46.81	53.19
	Corner/Angle Positioning	50.0	55.56	47.22	69.44	55.56	50.0
	Directional Relations	51.16	58.14	27.91	60.47	46.51	48.84
	Proximity Gradients	20.93	23.26	39.53	34.88	39.53	37.21
Size and Scale	Distance-Size Correlation	32.0	20.0	40.0	36.0	38.0	36.0
	Perspective Distortion	26.0	40.0	42.0	38.0	44.0	44.0
	Relative Size Comparison	34.0	32.0	46.0	38.0	30.0	42.0
	Scale Consistency	27.45	35.29	41.18	31.37	27.45	41.18
	Shadow-Size Projection	39.22	35.29	66.67	39.22	47.06	47.06
Spatial Navigation	Accessibility Constraints	36.59	29.27	51.22	41.46	39.02	43.9
	Obstacle Avoidance	24.0	26.0	38.0	38.0	38.0	36.0
	Pathway Existence	34.78	30.43	32.61	34.78	34.78	28.26
	Spatial Sequence	24.0	42.0	40.0	56.0	42.0	44.0
	Viewpoint Visibility	36.0	34.0	36.0	40.0	36.0	26.0

at 50.0% for **InternVL3.5-4B** and **Gemma-3-4B-it**. The within-model spread often exceeds 60 percentage points, a signature of brittle generalization. Read together, these patterns indicate that smaller models are strongly cue-dependent—competent where a visual template is salient, failing where deeper spatial abstraction is required—so that representation learning proceeds unevenly across sub-categories. Small models thus offer localized competence at the expense of robustness across the full spectrum of spatial reasoning.

Reasoning Models. Reasoning-enhanced models (Table 9) confer notable benefits on selected tasks yet remain inconsistent in aggregate. **o4-mini** reaches 65.12% in *Alignment Patterns* and 56.0% in *Gravity Effects*, surpassing most open baselines; **Gemini-2.0-Flash-Thinking** attains 62.75% in *Shadow-Size Projection* and 52.17% in *Volume Comparison*, evidence of stronger geometric reasoning. The strongest results belong to **Gemini-2.5-Flash-Thinking**, at 71.79% in *Complete Occlusion Inference* and 70.21% in *Betweenness Relationships*—yet the same family falls to 34.0% in both *Cardinal Direction* and *Shape Projection*. This variance shows that reasoning mechanisms deepen competence within particular sub-categories without broadening coverage, with within-model spreads frequently exceeding 35 percentage points. Reasoning augmentation thus supplies localized capacity while leaving the structural gaps in spatial understanding untouched: these models evidence promise rather than generalizable competence.

Spatial Reasoning Models. Spatially specialized models (Table 10) exhibit targeted improvement without generality. **SpaceQwen2.5-VL-3B-Instruct** achieves 48.0% in *Distance-Size Correlation* and 49.02% in *Scale Consistency*, clearly surpassing many open-source baselines on size–scale reasoning, and leads on navigation with 53.66% in *Accessibility Constraints* and 44.0% in *Obstacle Avoidance*. **SpaceOm** and **SpaceThinker-Qwen2.5VL-3B** reach 50.0% in *Stability Prediction*, consistent with their domain-specific objectives. Performance nonetheless drops sharply elsewhere—only 24.0% in *Spatial Containment*—and these models rarely exceed 55.0% even in their strongest areas, with sub-category spreads frequently above 30 percentage points. Specialization therefore yields narrow peaks in place of holistic competence: domain targeting sharpens selected abilities but does not integrate them into a robust, general framework.

Overall Discussion. Across all model groups (Table 8, Table 7, Table 6, Table 10, Table 9), the picture is one of consistent fragmentation. Small open-source models peak at 74.29% in *Stacking Orientation* but collapse to 12.0% in *Cardinal Direction*; larger models raise the ceiling—80.56% in *Corner/Angle Positioning*—yet still fail at 2.0% in *Object Rotation*. Closed-source systems achieve the strongest single scores, with **GPT-4o-mini** reaching 85.71% in *Stacking Orientation*, but fall to 32.0% in *Relative Size Comparison*. Specialized models show strengths in navigation (53.66%)

Table 9: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	o4-mini	Gemini-2.0-Flash-Thinking	Gemini-2.5-Flash-Thinking	Kimi-VL-A3B-Thinking-2506
3D Geometry	Gravity Effects	56.0	38.0	46.0	50.0
	Shape Projection	50.0	36.0	40.0	34.0
	Spatial Containment	56.0	28.0	48.0	42.0
	Stability Prediction	50.0	35.71	50.0	50.0
	Volume Comparison	43.48	52.17	45.65	39.13
Depth and Occlusion	Complete Occlusion Inference	69.23	51.28	71.79	51.28
	Layering Order	57.14	30.0	41.43	35.71
	Partial Occlusion	50.0	42.0	58.0	40.0
	Reflective Surfaces	54.0	40.0	52.0	42.0
	Transparency Effects	64.0	50.0	54.0	42.0
Orientation	Cardinal Direction	46.0	34.0	50.0	28.0
	Facing Direction	50.0	42.86	54.76	45.24
	Object Rotation	56.0	36.0	48.0	30.0
	Stacking Orientation	71.43	57.14	57.14	65.71
	Tool Handedness	56.0	44.0	60.0	44.0
Relative Positioning	Alignment Patterns	65.12	41.86	44.19	34.88
	Betweenness Relationships	65.96	55.32	70.21	63.83
	Corner/Angle Positioning	75.0	55.56	61.11	47.22
	Directional Relations	65.12	44.19	65.12	60.47
	Proximity Gradients	51.16	32.56	41.86	48.84
Size and Scale	Distance-Size Correlation	44.0	42.0	56.0	44.0
	Perspective Distortion	40.0	44.0	50.0	36.0
	Relative Size Comparison	46.0	46.0	58.0	32.0
	Scale Consistency	39.22	56.86	52.94	43.14
	Shadow-Size Projection	35.29	62.75	58.82	43.14
Spatial Navigation	Accessibility Constraints	56.1	41.46	53.66	31.71
	Obstacle Avoidance	32.0	44.0	44.0	36.0
	Pathway Existence	45.65	41.3	47.83	34.78
	Spatial Sequence	64.0	48.0	68.0	54.0
	Viewpoint Visibility	60.0	40.0	54.0	48.0

Table 10: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Spatial Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	48.0	46.0	32.0
	Shape Projection	38.0	36.0	36.0
	Spatial Containment	40.0	36.0	24.0
	Stability Prediction	50.0	50.0	28.57
	Volume Comparison	36.96	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	38.46	35.9	38.46
	Layering Order	30.0	32.86	24.29
	Partial Occlusion	44.0	46.0	38.0
	Reflective Surfaces	44.0	44.0	38.0
	Transparency Effects	40.0	32.0	42.0
Orientation	Cardinal Direction	40.0	38.0	30.0
	Facing Direction	59.52	54.76	50.0
	Object Rotation	40.0	42.0	28.0
	Stacking Orientation	62.86	62.86	40.0
	Tool Handedness	40.0	40.0	48.0
Relative Positioning	Alignment Patterns	39.53	44.19	39.53
	Betweenness Relationships	44.68	42.55	53.19
	Corner/Angle Positioning	41.67	41.67	38.89
	Directional Relations	25.58	25.58	20.93
	Proximity Gradients	37.21	37.21	34.88
Size and Scale	Distance-Size Correlation	40.0	44.0	48.0
	Perspective Distortion	40.0	38.0	42.0
	Relative Size Comparison	42.0	46.0	48.0
	Scale Consistency	33.33	33.33	49.02
	Shadow-Size Projection	58.82	54.9	66.67
Spatial Navigation	Accessibility Constraints	48.78	46.34	53.66
	Obstacle Avoidance	36.0	32.0	44.0
	Pathway Existence	32.61	34.78	41.3
	Spatial Sequence	48.0	44.0	50.0
	Viewpoint Visibility	32.0	34.0	48.0

and stability (50.0%) while remaining capped below 55% overall, and reasoning models post gains such as 71.79% in *Complete Occlusion Inference* alongside others stalled at 34.0%. Within-model spreads routinely exceed 50 percentage points, and the physical-abstraction sub-categories—*Gravity Effects*, *Stability Prediction*, *Spatial Containment*—remain below 35–40% across every model group. Two proximate causes are consistent with this pattern: the absence of stable encodings of orientation

Table 11: Distribution of question counts by the number of models (0–5) that correctly answered each item across categories and sub-categories in SPATIALAB-MCQ. “0” indicates no model answered correctly, while “5” indicates only five models succeeded to answer these among all models tested.

Category	Sub-Category	Number of Models Successfully Answered						Total
		0	1	2	3	4	5	
3D Geometry	Gravity Effects	9	7	3	2	3	1	25
	Shape Projection	4	6	0	9	5	7	31
	Spatial Containment	6	9	6	3	6	1	31
	Stability Prediction	5	6	4	2	2	3	22
	Volume Comparison	6	5	6	5	4	2	28
3D Geometry Total		30	33	19	21	20	14	137
Depth & Occlusion	Complete Occlusion Inference	0	4	0	6	3	4	17
	Layering Order	9	8	7	8	6	5	43
	Partial Occlusion	2	2	5	7	5	2	23
	Reflective Surfaces	6	5	5	7	4	1	28
	Transparency Effects	4	6	3	2	5	7	27
Depth & Occlusion Total		21	25	20	30	23	19	138
Orientation	Cardinal Direction	1	7	10	7	3	4	32
	Facing Direction	2	3	8	5	3	1	22
	Object Rotation	4	6	7	3	5	4	29
	Stacking Orientation	2	3	0	2	1	1	9
	Tool Handedness	2	1	4	3	1	1	12
Orientation Total		11	20	29	20	13	11	104
Relative Positioning	Alignment Patterns	3	4	4	4	4	1	20
	Betweenness Relationships	3	1	2	2	5	7	20
	Corner/Angle Positioning	0	1	3	2	3	3	12
	Directional Relations	1	4	1	2	3	2	13
	Proximity Gradients	4	6	6	5	2	3	26
Relative Positioning Total		11	16	16	15	17	16	91
Size & Scale	Distance-Size Correlation	6	5	6	1	1	7	26
	Perspective Distortion	8	7	3	3	6	4	31
	Relative Size Comparison	5	5	7	4	4	4	29
	Scale Consistency	7	6	7	4	5	2	31
	Shadow-Size Projection	5	7	2	1	1	4	20
Size & Scale Total		31	30	25	13	17	21	137
Spatial Navigation	Accessibility Constraints	3	2	1	5	4	3	18
	Obstacle Avoidance	2	6	9	6	10	2	35
	Pathway Existence	2	10	4	1	4	3	24
	Spatial Sequence	3	2	3	2	7	8	25
	Viewpoint Visibility	2	2	5	9	8	5	31
Spatial Navigation Total		12	22	22	23	33	21	133
Grand Total		116	146	131	122	123	102	740

and physics, and a reliance on surface-level correlation. The variance itself, from 85.71% highs to 2.0% lows, is more readily explained by pattern matching than by robust spatial reasoning. Scaling, proprietary training, and reasoning augmentation each help, but none delivers general coverage; each instead produces isolated peaks that coexist with pronounced blind spots. On the evidence assembled here, competence is fragmented rather than integrated, and closing the gap will plausibly require architectural changes that embed consistent geometric grounding and physics-aware mechanisms. Our MCQ results thus document both remarkable selective gains and the persistence of substantial gaps in holistic spatial intelligence.

E.2 MODEL FAILURE ANALYSIS

The distribution across categories and sub-categories in Table 11 exposes several notable patterns in model performance. 3D Geometry and Size & Scale show relatively balanced distributions, yet both retain a substantial proportion of failures on which no model answered correctly, an indication that even basic geometric invariances are not reliably internalized. Depth & Occlusion, by contrast, exhibits a heavier tail in the mid-range (2–4 correct), pointing to partial but inconsistent reasoning about layered scenes. Orientation emerges as a weak category overall, with especially low totals in Stacking Orientation and Tool Handedness, which suggests that subtle directional and embodied affordances are poorly captured. Relative Positioning presents a mixed picture: tasks such as Proximity Gradients show moderate success, whereas Corner/Angle Positioning accumulates many near-zero correct counts, reflecting brittleness in compositional reasoning over reference frames. Spatial Navigation performs comparatively better, particularly in Obstacle Avoidance and Viewpoint Visibility, but retains substantial errors in Pathway Existence, where relational chaining is required. In aggregate, no category attains uniformly high accuracy, and the Grand Total spreads evenly across the 0–5 bins—a signature of task-dependent success rather than general spatial competence. Models thus display isolated strengths without systematic reliability across spatial dimensions.

These results trace the models’ difficulties to a small set of representational and reasoning bottlenecks. High error rates in Stacking Orientation and Tool Handedness indicate limited grounding in physics-based and embodied reasoning, consistent with training predominantly on static internet imagery rather than interactive or task-driven data. The difficulty in Corner/Angle Positioning and Pathway Existence reflects inadequate multi-step relational chaining, pointing to a weakness in handling recursive spatial logic. The inconsistency in Transparency Effects and Reflective Surfaces underscores the challenge posed by visual phenomena that hinge on non-local cues, such as secondary reflections or occluded geometry. The comparatively stronger performance in Obstacle Avoidance is, by contrast, plausibly attributable to shortcut learning from dataset regularities—frequent co-occurrence patterns of walkable paths—rather than to genuine planning ability. Taken together, these failure modes indicate training regimes that overweight correlation-driven vision–language alignment while under-representing embodied, counterfactual, and physics-based reasoning. Addressing them will plausibly require both richer training distributions (e.g., embodied simulation data, physics-augmented learning) and architectural innovations that explicitly model reference frames and relational constraints; absent such advances, the brittle competence documented here is likely to persist under compositional or physically grounded tasks.

F ERROR ANALYSIS OF SPATIALAB-OPEN

F.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models (Table 12). Closed-source models achieve the strongest overall performance in the open-ended evaluation, yet their results remain uneven across sub-categories. GPT-5-mini is the clear leader, consistently securing the best scores in **3D Geometry** and **Relative Positioning**, with peaks of 58.14% in Directional Relations and 55.81% in Alignment Patterns. Gemini-2.5-Flash and Claude 3.5 Haiku, by contrast, register very weak results in Proximity Gradients (16.28%) and Perspective Distortion (12.0%), respectively. This wide gap exposes the fragility of even the strongest models when reasoning turns on subtle, continuous visual cues. The disparities are most consistent with differing sophistication in the models’ spatial reasoning pipelines, whereby some architectures capture relational and geometric structure more effectively than others. Tasks such as Proximity Gradients are especially demanding, requiring analogical, graded reasoning rather than discrete classification; that GPT-5-mini succeeds where others fail points to a distinct advantage conferred by scale and proprietary training corpora. Even the leader, however, exhibits clear vulnerabilities, which frames the central observation: within this evaluation, closed-source dominance is relative rather than absolute, and these models still fall short of robust, human-like spatial reasoning.

Open-source Large Models (Table 13). Large open-source models present a more distributed performance landscape, with different systems excelling in isolated sub-categories. llama-3.2-90b-vision-instruct achieves 60.0% in Depth and Occlusion, while GLM-4.5V-106B-MoE leads in Spatial Containment at 38.0%. These highlights notwithstanding, several models collapse on Proximity Gradients, with qwen-2.5-vl-32b-instruct and qwen-2.5-vl-72b-instruct both scoring only

Table 12: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Closed-source Models. We bold and underline the best score within each model category.

Category	Sub-Category	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	48.0	36.0	30.0	24.0	20.0
	Shape Projection	40.0	18.0	22.0	16.0	16.0
	Spatial Containment	50.0	38.0	50.0	36.0	36.0
	Stability Prediction	50.0	26.19	35.71	30.95	26.19
	Volume Comparison	39.13	41.3	32.61	23.91	28.26
Depth and Occlusion	Complete Occlusion Inference	46.15	35.9	38.46	28.21	25.64
	Layering Order	30.0	22.86	21.43	17.14	15.71
	Partial Occlusion	38.0	20.0	28.0	14.0	14.0
	Reflective Surfaces	32.0	28.0	32.0	22.0	22.0
	Transparency Effects	32.0	18.0	18.0	16.0	22.0
Orientation	Cardinal Direction	34.0	40.0	34.0	26.0	30.0
	Facing Direction	28.57	28.57	26.19	19.05	14.29
	Object Rotation	42.0	14.0	28.0	30.0	14.0
	Stacking Orientation	37.14	20.0	31.43	22.86	28.57
	Tool Handedness	48.0	36.0	44.0	24.0	24.0
Relative Positioning	Alignment Patterns	55.81	34.88	37.21	37.21	34.88
	Betweenness Relationships	48.94	23.4	31.91	23.4	21.28
	Corner/Angle Positioning	44.44	30.56	41.67	16.67	36.11
	Directional Relations	58.14	46.51	58.14	34.88	37.21
	Proximity Gradients	39.53	20.93	25.58	16.28	18.6
Size and Scale	Distance-Size Correlation	48.0	32.0	36.0	20.0	22.0
	Perspective Distortion	48.0	24.0	22.0	20.0	12.0
	Relative Size Comparison	36.0	26.0	28.0	18.0	16.0
	Scale Consistency	43.14	21.57	23.53	19.61	9.8
	Shadow-Size Projection	37.25	27.45	23.53	23.53	15.69
Spatial Navigation	Accessibility Constraints	41.46	31.71	39.02	24.39	21.95
	Obstacle Avoidance	40.0	22.0	22.0	26.0	14.0
	Pathway Existence	23.91	15.22	21.74	10.87	19.57
	Spatial Sequence	52.0	32.0	34.0	24.0	20.0
	Viewpoint Visibility	28.0	22.0	32.0	20.0	10.0

Table 13: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Open-source Large Models. We bold and underline the best score within each model category.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-32b-instruct	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct	GLM-4.5V-106B-MoE
3D Geometry	Gravity Effects	14.0	22.0	12.0	30.0	24.0	28.0
	Shape Projection	10.0	20.0	10.0	20.0	16.0	20.0
	Spatial Containment	14.0	30.0	18.0	22.0	20.0	38.0
	Stability Prediction	23.81	30.95	16.67	21.43	26.19	38.1
	Volume Comparison	23.91	10.87	26.09	41.3	28.26	32.61
Depth and Occlusion	Complete Occlusion Inference	25.64	20.51	12.82	25.64	30.77	33.33
	Layering Order	8.57	15.71	10.0	14.29	15.71	10.0
	Partial Occlusion	12.0	16.0	8.0	14.0	20.0	16.0
	Reflective Surfaces	26.0	16.0	8.0	34.0	32.0	24.0
	Transparency Effects	18.0	14.0	10.0	20.0	22.0	26.0
Orientation	Cardinal Direction	32.0	34.0	20.0	38.0	28.0	30.0
	Facing Direction	11.9	26.19	14.29	21.43	11.9	16.67
	Object Rotation	26.0	16.0	10.0	18.0	28.0	20.0
	Stacking Orientation	17.14	25.71	17.14	25.71	5.71	37.14
	Tool Handedness	20.0	20.0	12.0	20.0	32.0	24.0
Relative Positioning	Alignment Patterns	23.26	46.51	20.93	46.51	30.23	30.23
	Betweenness Relationships	34.04	31.91	17.02	25.53	38.3	29.79
	Corner/Angle Positioning	27.78	36.11	11.11	27.78	27.78	16.67
	Directional Relations	30.23	39.53	25.58	34.88	25.58	32.56
	Proximity Gradients	9.3	18.6	9.3	18.6	18.6	20.93
Size and Scale	Distance-Size Correlation	20.0	26.0	12.0	26.0	30.0	26.0
	Perspective Distortion	10.0	18.0	10.0	18.0	12.0	20.0
	Relative Size Comparison	8.0	16.0	14.0	32.0	22.0	24.0
	Scale Consistency	11.76	21.57	3.92	19.61	21.57	23.53
	Shadow-Size Projection	17.65	21.57	9.8	27.45	23.53	27.45
Spatial Navigation	Accessibility Constraints	21.95	26.83	26.83	34.15	39.02	24.39
	Obstacle Avoidance	12.0	18.0	12.0	20.0	30.0	22.0
	Pathway Existence	19.57	21.74	6.52	17.39	15.22	17.39
	Spatial Sequence	22.0	28.0	12.0	22.0	24.0	34.0
	Viewpoint Visibility	18.0	16.0	12.0	12.0	24.0	24.0

9.3%—evidence that architectural scaling assists without guaranteeing coverage across the harder continuous-reasoning tasks. Much of this variation is plausibly attributable to heterogeneous training data: models exposed to richer depth and geometry corpora fare better on occlusion and containment, whereas others generalize poorly. The persistent weakness on Proximity Gradients indicates that these systems are not optimized for subtle spatial analogy and distance scaling. Inter-model variance is moreover higher in this group than among closed-source systems, a reflection of open-source development heterogeneity. Large open-source models thus display encouraging peaks alongside inconsistent coverage, reinforcing that scale alone is insufficient without targeted pre-training and fine-tuning.

Open-source Small Models (Table 14). Small open-source models predictably trail their larger counterparts, yet their results reveal noteworthy localized strengths. qwen-2.5-vl-3b-instruct achieves 18.0% in Perspective Distortion and 26.0% in Relative Size Comparison, while Gemma-3-4B-it leads

Table 14: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Open-source Small Models. We bold and underline the best score within each model category.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	4.0	6.0	14.0	24.0	16.0	22.0
	Shape Projection	2.0	8.0	10.0	4.0	10.0	0.0
	Spatial Containment	6.0	4.0	12.0	20.0	24.0	8.0
	Stability Prediction	2.38	11.9	14.29	19.05	21.43	19.05
	Volume Comparison	15.22	32.61	28.26	30.43	30.43	28.26
Depth and Occlusion	Complete Occlusion Inference	2.56	5.13	10.26	17.95	20.51	23.08
	Layering Order	5.71	7.14	10.0	14.29	11.43	14.29
	Partial Occlusion	14.0	8.0	12.0	16.0	4.0	10.0
	Reflective Surfaces	16.0	16.0	6.0	14.0	18.0	26.0
	Transparency Effects	10.0	20.0	4.0	28.0	14.0	8.0
Orientation	Cardinal Direction	12.0	16.0	16.0	26.0	20.0	24.0
	Facing Direction	9.52	4.76	28.57	14.29	7.14	19.05
	Object Rotation	4.0	10.0	12.0	8.0	14.0	16.0
	Stacking Orientation	17.14	11.43	8.57	17.14	20.0	28.57
	Tool Handedness	8.0	12.0	8.0	12.0	12.0	12.0
Relative Positioning	Alignment Patterns	13.95	27.91	32.56	11.63	41.86	34.88
	Betweenness Relationships	12.77	17.02	0.0	19.15	17.02	25.53
	Corner/Angle Positioning	22.22	30.56	2.78	22.22	19.44	22.22
	Directional Relations	11.63	25.58	9.3	34.88	23.26	37.21
	Proximity Gradients	9.3	18.6	9.3	11.63	16.28	18.6
Size and Scale	Distance-Size Correlation	6.0	10.0	14.0	16.0	20.0	12.0
	Perspective Distortion	8.0	12.0	18.0	16.0	12.0	16.0
	Relative Size Comparison	10.0	16.0	26.0	24.0	10.0	16.0
	Scale Consistency	7.84	9.8	11.76	11.76	7.84	11.76
	Shadow-Size Projection	13.73	11.76	21.57	13.73	25.49	23.53
Spatial Navigation	Accessibility Constraints	4.88	26.83	7.32	17.07	19.51	19.51
	Obstacle Avoidance	16.0	16.0	12.0	18.0	28.0	26.0
	Pathway Existence	0.0	8.7	2.17	8.7	17.39	8.7
	Spatial Sequence	12.0	22.0	8.0	26.0	20.0	28.0
	Viewpoint Visibility	16.0	18.0	16.0	24.0	14.0	16.0

in Spatial Containment (24.0%) and Stability Prediction (21.43%). These successes are offset by outright failures: InternVL3.5-2B scores 0.0% in Betweenness Relationships, and InternVL3.5-1B manages only 2.0% in Shape Projection. The extreme variability is most consistent with limited capacity and restricted training diversity, which together preclude robust representations of geometric and relational principles; sub-categories demanding precise reasoning about orientation or hierarchy trigger complete breakdowns, as small misinterpretations propagate into total failure. That these models nonetheless occasionally surpass larger ones on narrow tasks indicates that even small-scale training can instill isolated competencies—a testament to specialization that simultaneously confirms the structural ceiling of compact architectures. Small open-source models thus exhibit pockets of promise within pervasive brittleness, their overall accuracy bounded by size and data constraints.

Reasoning Models (Table 15). Reasoning-oriented models present a sharper internal divide, with Gemini-2.5-flash-thinking standing far above its peers. It records the best scores in nearly every sub-category, including Tool Handedness at 75.0% and Complete Occlusion Inference at 57.14%. Kimi-VL-A3B-Thinking-2506, by contrast, struggles markedly, scoring just 6.0% in Spatial Containment and lagging throughout. This gap is most readily explained by differences in how these models integrate multi-modal streams into a coherent reasoning framework: Gemini-2.5-flash-thinking behaves as though forming and querying an internal world model, which would allow it to infer object interactions and physical properties more effectively, whereas weaker models lean on surface-level pattern matching that proves inadequate for compositional or causal reasoning. Its consistent strength across categories indicates that explicit reasoning mechanisms offer tangible benefits for spatial understanding. Yet its failure to dominate Shape Projection and Betweenness Relationships shows that even advanced reasoning modules are not universally effective. Reasoning models therefore hold clear promise while the category itself remains heterogeneous and far from solved.

Spatial Reasoning Models (Table 16). Spatial reasoning models, though explicitly designed for the task, record the lowest results overall. Performance is highly fragmented: several models tie at 16.0% in Gravity Effects, 8.0% in Shape Projection, and 14.29% in Stability Prediction. SpaceQwen2.5-VL-3B-Instruct stands out modestly with 23.91% in Volume Comparison and 13.73% in Scale Consistency, yet these scores remain far below competitive thresholds. At the other extreme, Betweenness Relationships and Corner/Angle Positioning collapse entirely, at 0.0% and 2.78% respectively. Such results indicate that limited capacity and narrowly focused training severely constrain these models’ representational flexibility, and the failures are best read as evidence that specialization without sufficient capacity or data breadth cannot yield generalizable spatial reasoning: on geometric and positional tasks in particular, even slight representational errors precipitate total

Table 15: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	gemini-2.0-flash-thinking	gemini-2.5-flash-thinking	Kimi-VL-A3B-Thinking-2506	Step3
3D Geometry	Gravity Effects	32.0	40.0	14.0	38.0
	Shape Projection	24.0	0.0	10.0	30.0
	Spatial Containment	36.0	50.0	6.0	30.0
	Stability Prediction	26.19	42.86	14.29	33.33
	Volume Comparison	36.96	44.44	23.91	30.43
Depth and Occlusion	Complete Occlusion Inference	33.33	57.14	17.95	33.33
	Layering Order	20.0	20.0	14.29	14.29
	Partial Occlusion	28.0	50.0	8.0	14.0
	Reflective Surfaces	34.0	63.64	10.0	16.0
	Transparency Effects	26.0	57.14	22.0	20.0
Orientation	Cardinal Direction	40.0	20.0	18.0	26.0
	Facing Direction	35.71	37.5	14.29	28.57
	Object Rotation	18.0	40.0	4.0	22.0
	Stacking Orientation	22.86	0.0	5.71	34.29
	Tool Handedness	44.0	75.0	20.0	32.0
Relative Positioning	Alignment Patterns	46.51	66.67	25.58	41.86
	Betweenness Relationships	27.66	20.0	12.77	25.53
	Corner/Angle Positioning	36.11	50.0	11.11	11.11
	Directional Relations	39.53	44.44	27.91	37.21
	Proximity Gradients	23.26	16.67	11.63	18.6
Size and Scale	Distance-Size Correlation	32.0	57.14	14.0	22.0
	Perspective Distortion	18.0	11.11	10.0	30.0
	Relative Size Comparison	28.0	11.11	14.0	36.0
	Scale Consistency	33.33	20.0	11.76	31.37
	Shadow-Size Projection	35.29	18.18	13.73	29.41
Spatial Navigation	Accessibility Constraints	31.71	45.45	17.07	41.46
	Obstacle Avoidance	26.0	10.0	16.0	18.0
	Pathway Existence	23.91	25.0	13.04	19.57
	Spatial Sequence	34.0	12.5	30.0	36.0
	Viewpoint Visibility	32.0	12.5	16.0	26.0

Table 16: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Spatial Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	14.0	12.0	16.0
	Shape Projection	8.0	8.0	6.0
	Spatial Containment	8.0	10.0	10.0
	Stability Prediction	14.29	14.29	7.14
	Volume Comparison	19.57	23.91	23.91
Depth and Occlusion	Complete Occlusion Inference	7.69	10.26	7.69
	Layering Order	8.57	11.43	4.29
	Partial Occlusion	10.0	12.0	4.0
	Reflective Surfaces	4.0	4.0	2.0
	Transparency Effects	4.0	8.0	2.0
Orientation	Cardinal Direction	18.0	24.0	8.0
	Facing Direction	23.81	23.81	16.67
	Object Rotation	20.0	20.0	18.0
	Stacking Orientation	8.57	8.57	17.14
	Tool Handedness	0.0	4.0	8.0
Relative Positioning	Alignment Patterns	32.56	32.56	23.26
	Betweenness Relationships	4.26	2.13	0.0
	Corner/Angle Positioning	2.78	2.78	2.78
	Directional Relations	11.63	6.98	11.63
	Proximity Gradients	6.98	6.98	9.3
Size and Scale	Distance-Size Correlation	18.0	18.0	16.0
	Perspective Distortion	16.0	20.0	2.0
	Relative Size Comparison	22.0	20.0	8.0
	Scale Consistency	9.8	11.76	13.73
	Shadow-Size Projection	27.45	27.45	19.61
Spatial Navigation	Accessibility Constraints	7.32	7.32	9.76
	Obstacle Avoidance	10.0	8.0	14.0
	Pathway Existence	6.52	6.52	8.7
	Spatial Sequence	16.0	12.0	16.0
	Viewpoint Visibility	20.0	16.0	8.0

breakdown. Despite their explicit focus, then, spatial reasoning models prove least effective here, which underscores that both scale and diversity are prerequisites for robust performance.

Overall Discussion. Across all five groups—closed-source (Table 12), open-source large (Table 13), open-source small (Table 14), reasoning (Table 15), and spatial reasoning (Table 16)—the same pattern of fragmented competence recurs. Closed-source models perform best overall, GPT-5-mini reaching 58.14% in Directional Relations and 55.81% in Alignment Patterns, yet even they collapse on Proximity Gradients. Large open-source models post impressive peaks such as 60.0% in Depth and Occlusion but fall to 9.3% in Proximity Gradients, illustrating that scale without specialization remains insufficient. Small open-source models sharpen the contrast, pairing rare localized strengths (e.g., 26.0% in Relative Size Comparison) with lows as extreme as 0.0% in Betweenness Relationships. Reasoning models demonstrate that explicit reasoning mechanisms can materially improve results—Gemini-2.5-flash-thinking reaches 75.0% in Tool Handedness—yet peers such as Kimi-VL-A3B fall to single digits, exposing the fragility of less-integrated reasoning frameworks. Spatial reasoning models, notwithstanding their specialization, underperform the most, with scores often below 15% and total failures on Betweenness Relationships. These convergent observations point to three proximate causes: the absence of robust representations for continuous and analogical spatial cues; structural limitations tied to scale and training diversity; and brittle reasoning pipelines that impede cross-task generalization. Within the settings examined here, no single approach—proprietary scaling, open-source expansion, reasoning integration, or specialization—delivers comprehensive coverage, and the frequent within-model gaps exceeding 50 percentage points confirm that current progress is piecemeal. Future work should accordingly aim to unify robust geometric encodings, physics-informed reasoning, and cross-task transfer, advancing beyond isolated gains toward consistent spatial intelligence.

F.2 MODEL FAILURE ANALYSIS

The distribution in Table 17 exposes persistent weaknesses across categories, with a pronounced skew toward the lower bins (0–2 models correct). 3D Geometry is broadly challenging, Shape Projection (19 items with zero correct) and Stability Prediction (10 items with zero correct) faring especially poorly—an indication that reliable physical and geometric reasoning is absent. Depth and Occlusion is the hardest category overall, accumulating the highest zero-correct counts (72), concentrated in Layering Order (20 zero) and Transparency Effects (16 zero), both of which demand nuanced handling of occlusion and non-local cues. Orientation exhibits comparable difficulty: Cardinal Direction achieves some mid-range success, but Stacking Orientation (11 zero) and Tool Handedness (5 zero, none with perfect success) underscore the brittleness of embodied directional reasoning. Relative Positioning is widely inconsistent, with Betweenness Relationships and Proximity Gradients rarely solved by more than two or three models, whereas Corner/Angle Positioning shows low but comparatively balanced performance. Size and Scale is relatively stronger—several subcategories (e.g., Scale Consistency, with 6 full successes) show that some models internalize coarse perspective and scaling cues—though failures remain high in Shadow-Size Projection (17 zero). Spatial Navigation divides cleanly: Spatial Sequence and Obstacle Avoidance admit occasional full-model success, while Pathway Existence (16 zero) and Accessibility Constraints (10 zero) mark persistent difficulty in multi-step relational chaining. The grand total is decisive: across 356 items, zero-correct dominates (356), and only 74 items are solved by all five models—confirming that consistent spatial competence remains elusive.

These findings locate the root causes of failure less in dataset design than in the representational and reasoning limitations of current models. The concentration of errors in Depth and Occlusion reflects a reliance on superficial appearance features in place of structured scene representations: models latch onto surface texture rather than reasoning about hidden layers, which systematically corrupts Layering Order. Failures in Orientation tasks such as Tool Handedness expose the absence of embodied affordance modeling, since pretrained internet imagery rarely annotates left/right handedness or stacking physics. The brittleness in Relative Positioning and Spatial Navigation points to a deficiency in chaining multi-hop spatial relations—Pathway Existence, for instance, requires integrating local connectivity with global layout, which attention-based co-occurrence priors capture poorly. The partial success in Size and Scale, conversely, suggests that models exploit statistical regularities in perspective distortion and scale consistency, only to collapse on rarer cases such as shadows. Collectively, these patterns indicate that current models succeed when cues are salient and correlate with training priors but fail when reasoning demands counterfactual, compositional, or embodied inference. Addressing this will require architectural advances that explicitly model reference frames, occlusion layers, and physical dynamics, together with training data enriched by

Table 17: Distribution of question counts by the number of models (0–5) that correctly answered each item across categories and sub-categories in SPATIALAB-OPEN. “0” indicates no model answered correctly, while “5” indicates only five models succeeded to answer these among all models tested.

Category	Sub-Category	Number of Models Successfully Answered					
		0	1	2	3	4	5
3D Geometry	Gravity Effects	13	7	3	4	1	2
	Shape Projection	19	5	6	2	5	1
	Spatial Containment	10	4	3	6	2	3
	Stability Prediction	10	3	7	3	0	2
	Volume Comparison	11	3	4	3	4	2
3D Geometry Total		63	22	23	18	12	10
Depth & Occlusion	Complete Occlusion Inference	7	5	4	2	5	2
	Layering Order	20	14	10	6	3	3
	Partial Occlusion	14	8	5	5	4	2
	Reflective Surfaces	15	4	1	5	2	5
	Transparency Effects	16	7	7	3	1	2
Depth & Occlusion Total		72	38	27	21	15	14
Orientation	Cardinal Direction	8	10	0	1	5	2
	Facing Direction	13	5	5	1	3	2
	Object Rotation	12	10	6	4	1	2
	Stacking Orientation	11	5	1	2	1	1
	Tool Handedness	5	4	1	4	2	0
Orientation Total		49	34	13	12	12	7
Relative Positioning	Alignment Patterns	7	5	3	1	1	1
	Betweenness Relationships	11	5	7	2	1	3
	Corner/Angle Positioning	7	6	2	1	3	2
	Directional Relations	5	4	4	4	2	0
	Proximity Gradients	10	10	3	3	2	4
Relative Positioning Total		40	30	19	11	9	10
Size & Scale	Distance-Size Correlation	7	5	9	4	1	5
	Perspective Distortion	12	6	6	8	5	2
	Relative Size Comparison	14	7	4	2	2	4
	Scale Consistency	14	7	5	3	3	6
	Shadow-Size Projection	17	4	4	4	3	2
Size & Scale Total		64	29	28	21	14	19
Spatial Navigation	Accessibility Constraints	10	5	4	0	1	2
	Obstacle Avoidance	15	5	8	2	2	5
	Pathway Existence	16	6	7	3	1	1
	Spatial Sequence	14	3	3	3	3	5
	Viewpoint Visibility	13	3	9	5	3	1
Spatial Navigation Total		68	22	31	13	10	14
Grand Total		356	175	141	96	72	74

embodied simulation and counterfactual augmentation; absent these, spatial-reasoning progress is likely to remain fragmented—strong on surface-level cues, fragile on structured, real-world spatial logic.

G ANALYSIS OF MCQ VS. OPEN-ENDED PERFORMANCE GAPS

This section analyzes in detail the performance gap between the multiple-choice (MCQ) and open-ended evaluation formats across the spatial reasoning categories of SPATIALAB, as reported in Table 18.

Table 18: **Performance gap between MCQ and Open-ended (%) ($\downarrow$) across spatial categories on SPATIALAB.** A lower value indicates that a model performs comparably in both setups, whereas a higher value reflects greater disparity. We bold and underline the highest gap within each model group.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Proprietary Models							
GPT-4o-mini	22.40	23.76	16.51	31.74	27.85	20.50	23.53
GPT-5-mini	20.08	23.27	13.21	2.38	19.41	13.36	3.36
Gemini-2.0-Flash	30.89	26.73	26.89	28.18	22.37	25.07	15.13
Gemini-2.5-Flash	21.62	16.34	17.45	15.87	21.51	17.36	10.93
Claude 3.5 Haiku	23.16	21.78	20.29	15.47	24.89	20.29	16.39
Mistral Medium 3.1	30.50	25.74	32.54	26.59	24.89	26.93	21.43
Open-Source Models (Small)							
InternVL3.5-1B	22.78	13.37	23.58	22.62	20.67	22.00	27.73
InternVL3.5-2B	20.46	20.79	16.99	20.64	14.35	19.21	21.85
Qwen-VL2.5-3B-Instruct	27.03	30.69	29.24	28.97	29.96	28.50	25.63
InternVL3.5-4B	25.10	26.24	34.91	20.24	23.20	25.29	23.53
Gemma-3-4B-it	21.23	31.68	22.17	22.22	18.14	22.93	23.53
Qwen-VL2.5-7B-Instruct	22.01	22.27	18.40	26.19	15.61	22.14	27.73
Llama-3.2-11B-Vision-Instruct	13.51	-1.98	17.92	17.07	13.50	11.93	9.66
Gemma-3-27B-it	23.93	23.27	19.82	25.40	25.32	23.14	20.59
Qwen-VL2.5-32B-Instruct	30.50	31.68	28.30	35.32	28.27	29.85	24.79
InternVL3.5-72B	36.68	33.17	34.44	29.37	28.69	31.57	27.31
Qwen-VL2.5-72B-Instruct	27.80	26.73	23.59	19.05	28.27	24.22	20.17
Llama-3.2-90B-Vision-Instruct	28.95	29.21	30.66	25.00	21.52	26.36	23.53
Reasoning Models							
o4-mini	25.48	22.77	21.23	-3.18	17.30	15.35	10.50
Gemini-2-Flash-Thinking	13.90	10.39	11.32	21.03	13.50	13.00	6.73
Gemini-2.5-Flash-Thinking	8.22	16.61	19.46	33.42	31.37	20.16	8.66
Kimi-VL-A3B-Thinking-2506	27.02	28.71	33.50	26.98	22.78	27.92	29.41
Spatial Reasoning Models							
SpaceOm	31.66	32.18	25.95	24.21	27.00	28.43	29.83
SpaceThinker-Qwen2.5VL-3B	28.57	29.21	27.83	23.81	27.84	27.28	26.89
SpaceQwen2.5-VL-3B-Instruct	31.28	23.76	28.31	38.89	35.87	29.78	18.90

G.1 QUANTITATIVE SNAPSHOT

Across 25 models, the mean performance gap between the MCQ and open-ended formats is **23.0%**, with a standard deviation of 5.5%. Dataset-wide per-subtask means are: 3D geometry (24.57), depth & occlusion (23.45), orientation (23.70), relative positioning (23.11), size & scale (23.42), and spatial navigation (22.89), for an overall average of 23.01. The overall gap ranges from 11.93 to 31.57 (median $\approx$ 23.34). At the family level, spatial reasoning models exhibit the largest average gap (27.03) and reasoning-oriented models the smallest (19.11), while proprietary models average lower gaps (21.09) than many large open-source models (24.51); the human baseline itself registers a gap of 22.64. The largest gaps arise in InternVL3.5-72B (31.57%), Qwen-VL2.5-32B (29.85%), and SpaceQwen2.5-VL-3B-Instruct (29.78%), and the smallest in Llama-3.2-11B (11.93%), Gemini-2-Flash-Thinking (13.00%), and GPT-5-mini (13.36). Correlational analysis is unambiguous: spatial navigation dominates the overall gap (Pearson $r = 0.99$), with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) also strongly aligned.

G.2 PATTERNS AND OBSERVATIONS

Three broad patterns emerge. First, specialist spatial reasoning models exhibit the largest MCQ $\rightarrow$ open-ended gaps, a pattern consistent with optimization for classification-style supervision at the expense of robustness to free-form generation, whereas reasoning-focused models display smaller gaps, reflecting the stabilizing effect of chain-of-thought-style training on generative output. Second, spatial navigation is the strongest predictor of overall disparity: models that falter on open-ended navigation reasoning drive the aggregate gap, with orientation and 3D geometry contributing substantially and relative position and size/scale varying more idiosyncratically across models. Third, scale alone does not explain robustness—smaller models such as Llama-3.2-11B attain among the lowest gaps while very large models such as InternVL3.5-72B show the highest. Finally, several negative or near-zero gaps (e.g., depth & occlusion for Llama-3.2-11B at -1.98 , relative position for o4-mini at -3.18) indicate that open-ended evaluation is, in some cases, the more reliable format, most plausibly where MCQ distractors are misleading.

G.3 HYPOTHESES ON ROOT CAUSES

We hypothesize several interacting factors that underlie the observed MCQ→open-ended performance gaps in SPATIALAB, each supported by statistical observations:

1. **MCQ structural advantage.** MCQ format constrains the output space, allowing models to exploit surface cues or eliminate distractors. Across models, the mean MCQ score is systematically higher than the open-ended score, with the average gap at 23.0% and $\sigma = 5.5\%$. Negative gaps observed for Llama-3.2-11B in depth & occlusion (-1.98) and o4-mini in relative position (-3.18) indicate that open-ended generation can occasionally outperform MCQ when distractors are misleading, reinforcing the notion that MCQ can artificially inflate or distort measured performance.

Recommendation: To address this issue, we suggest complementing MCQ evaluations with open-ended tasks and subtask-level diagnostics, which can more accurately capture model competence and reduce overestimation caused by option-based cues.

2. **Specialization bias in spatial reasoning models.** Spatial reasoning models exhibit the largest average gaps (27.03%) despite being optimized for spatial tasks, suggesting that they may have been trained primarily on categorical or synthetic selection tasks rather than free-form generation. For instance, InternVL3.5-72B, a 72B parameter spatial reasoning model, shows the highest overall gap (31.57%), particularly in spatial navigation (36.68%) and orientation (34.44%), highlighting a reliance on classification-like supervision. Correlations between subtask gaps and overall gap (SpNav: $r = 0.99$, Orientation: $r = 0.83$) indicate that these format-dependent failures are concentrated in sequential and geometric reasoning tasks.

Recommendation: To mitigate this bias, we recommend fine-tuning or instruction-tuning on open-ended spatial generation tasks, which can improve model robustness to generative formats and reduce reliance on categorical supervision.

3. **Instruction-tuning and stepwise decoding.** Reasoning-oriented models (e.g., Gemini-2-Flash-Thinking, o4-mini) demonstrate smaller average gaps (around 19.11%) and lower variance across subtasks. For example, Gemini-2-Flash-Thinking has an overall gap of 13.0% and consistently modest subtask gaps (Difference in 3D Geometry: 13.9%, Difference Relative Positioning: 21.03%), suggesting that chain-of-thought or stepwise decoding helps models produce stable open-ended responses. This is further supported by reduced correlation between subtask gap variance and overall gap within this family, implying that instruction-tuned reasoning mitigates format sensitivity.

Recommendation: To address format sensitivity, we suggest adopting chain-of-thought or stepwise decoding strategies and instruction-tuning for open-ended spatial tasks, which can improve generative consistency and reduce MCQ dependence.

4. **Spatio-navigation stresses sequential grounding.** Spatial navigation gaps are the strongest predictor of overall MCQ→open-ended disparity (Pearson $r = 0.99$). Models with high performance gaps in spatial navigation, such as SpaceQwen2.5-VL-3B-Instruct (Spatial Navigation gap = 35.87%), also exhibit large overall gaps (29.78%), indicating that multi-step referential reasoning is disproportionately affected in open-ended formats. This suggests that MCQ options simplify the reasoning chain, masking underlying weaknesses in sequential grounding and relational computation.

Recommendation: To better capture sequential reasoning abilities, we recommend including dedicated open-ended spatio-navigation tasks and employing targeted multi-step reasoning supervision, which can improve grounding and reduce overestimation by MCQ.

5. **MCQ distractor artifacts.** Some negative or near-zero gaps indicate that poorly designed distractors can suppress measured MCQ performance. For example, o4-mini shows a negative relative position gap (-3.18%) while maintaining positive gaps in other subtasks, implying that distractor quality can artificially depress MCQ accuracy. Conversely, models like Qwen-VL2.5-32B-Instruct exhibit very high MCQ→open gaps (overall = 29.85%) across multiple subtasks, suggesting that MCQ distractors can also exaggerate perceived competence in selection-based tasks.

Recommendation: To address this, we suggest careful auditing and refinement of MCQ distractors and complementing them with open-ended evaluation, which can yield more accurate and unbiased measures of model spatial reasoning ability.

G.4 PRACTICAL IMPLICATIONS

These findings establish why both MCQ and open-ended results must be reported. Exclusive reliance on MCQ inflates estimates of practical spatial competence, with average disparities near 23%, and subtask-level breakdowns are indispensable because spatial navigation disproportionately drives the observed differences. Diagnostic evaluations that disentangle classification from generative reasoning thus isolate error modes more precisely than aggregate accuracy allows. Instruction-tuning and decoding strategies that encourage stepwise reasoning appear promising for reducing MCQ dependence, and careful design and auditing of MCQ distractors remain necessary for fair cross-format comparison.

H DETAILS ON IMPROVING VISUAL REASONING CAPABILITIES

To enhance spatial reasoning performance on our diverse visual question benchmark, we investigated a set of complementary inference- and training-time strategies, reported in turn below.

H.1 INHERENT REASONING MECHANISMS

We first isolate the role of built-in reasoning mechanisms by comparing standard models with their reasoning-enabled counterparts (Table 2, Table 3). Under MCQ evaluation, reasoning consistently improves performance, though its magnitude varies by category: Gemini-2.5-Flash-Thinking gains +4.64% overall over its non-reasoning variant (52.93% vs. 48.29%), with particularly strong improvement in Relative Positioning (+13.1%) and Orientation (+8.6%). The same trend holds for o4-mini-medium, which attains the highest overall MCQ accuracy (53.21%)—evidence that explicit reasoning scaffolds are especially effective in structured, discrete-answer settings.

Open-ended evaluation yields a more nuanced picture. Reasoning-equipped models again outperform their baselines, but unevenly. Gemini-2.5-Flash-Thinking improves marginally over Gemini-2.5-Flash (32.77% vs. 30.93%), whereas the largest leap belongs to o4-mini-medium (37.86%), which substantially surpasses all non-reasoning models. Reasoning mechanisms thus mitigate part of the instability characteristic of free-form generation, most visibly in complex categories such as 3D Geometry and Spatial Navigation. The benefit is not universal, however: reasoning models occasionally regress—Gemini-2.5-Flash-Thinking drops sharply in Size & Scale (21.74% vs. 26.59%)—echoing our earlier finding that generative settings amplify brittleness.

Taken together, these comparisons show that explicit reasoning modules substantially strengthen structured MCQ performance and can stabilize open-ended responses, yet their benefits remain category-dependent. Reasoning mechanisms reinforce logical consistency and relational chaining, but they do not resolve the underlying challenges of scale sensitivity, occlusion, and free-form linguistic grounding.

H.2 CHAIN-OF-THOUGHT (CoT) PROMPTING

To examine whether structured reasoning improves spatial performance, we evaluated models under Chain-of-Thought (CoT) prompting. For each of the 30 sub-categories in SPATIALAB we sampled 5 items (seed fixed to 42), constructing a balanced test set that covers all major spatial phenomena while keeping evaluation computationally tractable. Example CoT prompts, illustrating the step-by-step reasoning scaffolds supplied to the models, appear in Figure 7. This design isolates the contribution of CoT reasoning from the confounds of model scale and training.

Our analysis (Table 19) shows that Orientation is the only category in which CoT prompting reliably improves performance, while most others stagnate or decline. Reasoning therefore helps when a task reduces to the logical alignment of directions but fails when it demands richer perceptual priors—depth, scale, or relative positioning. The most parsimonious explanation is that these models lack internalized knowledge of 3D perception and geospatial relationships; they are not, in the human sense, “aware” of these concepts. Applied to such tasks, step-by-step reasoning does not correct misconceptions but entrenches them, propagating and amplifying errors across successive steps.

This result establishes that the benchmark is non-trivial: it cannot be solved simply by grafting multi-step reasoning onto existing models. Unlike textual reasoning tasks, where CoT reliably improves

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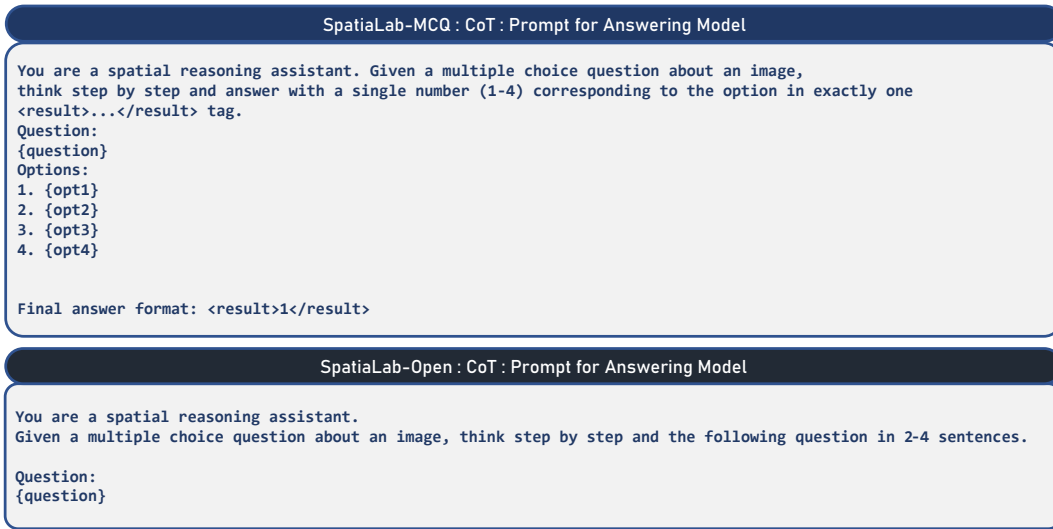


Figure 7: Chain-of-thought (CoT) Prompts.

Table 19: Effect of CoT prompting across categories for InternVL-3-78b and Gemini-2.5-Flash, under both MCQ and open-ended formats. Values are accuracies (%); Gain is the accuracy with CoT minus that without. Orientation is the only category with consistent positive gains, while most categories stagnate or decline, and Gemini’s MCQ accuracy drops by 18.00% overall.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain
3D Geometry	52.00	52.00	0.00	28.00	28.00	0.00	48.00	20.00	-28.00	60.00	28.00	-32.00
Depth & Occlusion	60.00	60.00	0.00	16.00	12.00	-4.00	48.00	28.00	-20.00	40.00	16.00	-24.00
Orientation	60.00	64.00	4.00	24.00	28.00	4.00	56.00	24.00	-32.00	0.00	32.00	32.00
Relative Positioning	80.00	72.00	-8.00	44.00	56.00	12.00	60.00	36.00	-24.00	50.00	32.00	-18.00
Size & Scale	56.00	44.00	-12.00	28.00	20.00	-8.00	48.00	60.00	12.00	25.00	8.00	-17.00
Spatial Navigation	64.00	64.00	0.00	48.00	40.00	-8.00	68.00	52.00	-16.00	40.00	28.00	-12.00
Overall	62.00	59.33	-2.67	31.33	30.67	-0.66	54.67	36.67	-18.00	38.24	24.00	-14.24

outcomes, SPATIALAB exposes limitations in perceptual and spatial grounding that additional reasoning depth does not address. In categories such as depth and size, an incorrect prior means that each additional reasoning step carries the model further from the correct solution, and the sharp drops—most pronounced for Gemini in MCQ settings—show how fragile reasoning becomes when it is not anchored by robust perceptual knowledge.

The task design is thus a deliberate strength: it moves beyond surface-level reasoning to probe whether a model genuinely understands geospatial and 3D relational concepts. Solving it requires more than logical chaining; it demands embeddings and training that capture core aspects of perception and spatial cognition. In this respect the benchmark challenges models at the level of representation and, in doing so, exposes a consequential gap in current architectures.

H.3 CHAIN-OF-THOUGHT (COT) PROMPTING WITH SELF-REFLECTION

We extend the CoT setup with a self-reflection stage in which the model is prompted to review and revise its initial reasoning before committing to an answer (see Figure 8). The evaluation protocol is otherwise identical to standard CoT, using 5 balanced samples per sub-category (seed 42), so that the comparison isolates the effect of self-reflection against the baseline CoT condition.

Augmenting CoT with self-reflection produced mixed, strongly model-dependent outcomes. For InternVL-3-78b, overall accuracy declined modestly (−2.66% MCQ, −8.67% open-ended), with sharp drops in depth, relative positioning, and navigation—an indication that reflection frequently reinforced errors rather than correcting them. Localized gains appeared, such as +12% in Size & Scale (MCQ), but they were inconsistent and offset by large degradations (−28% in relative

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Figure 8: Chain-of-thought (CoT) with Self-Reflection Prompts.

Table 20: CoT versus CoT with self-reflection (CoT-SR) across categories for InternVL-3-78b and Gemini-2.5-Flash, under both MCQ and open-ended formats. Values are accuracies (%); Gain is CoT-SR minus CoT. Self-reflection substantially improves Gemini’s MCQ accuracy (+20.66% overall) but does not transfer to open-ended generation (+0.67%), and degrades InternVL-3-78b in both formats.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain
3D Geometry	52.00	56.00	4.00	28.00	20.00	-8.00	20.00	52.00	32.00	28.00	24.00	-4.00
Depth & Occlusion	60.00	44.00	-16.00	12.00	16.00	4.00	28.00	52.00	24.00	16.00	16.00	0.00
Orientation	64.00	64.00	0.00	28.00	32.00	4.00	24.00	48.00	24.00	32.00	36.00	4.00
Relative Positioning	72.00	64.00	-8.00	56.00	28.00	-28.00	36.00	72.00	36.00	32.00	32.00	0.00
Size & Scale	44.00	56.00	12.00	20.00	12.00	-8.00	60.00	56.00	-4.00	8.00	12.00	4.00
Spatial Navigation	64.00	56.00	-8.00	40.00	24.00	-16.00	52.00	64.00	12.00	28.00	28.00	0.00
Overall	59.33	56.67	-2.66	30.67	22.00	-8.67	36.67	57.33	20.66	24.00	24.67	0.67

positioning open-ended). Gemini-2.5-Flash, by contrast, posted strong MCQ gains (+20.66%), with particularly large improvements in 3D Geometry (+32%) and Depth & Occlusion (+24%), while its open-ended improvement remained negligible (+0.67%). These results reveal a key asymmetry: reflection helps in multiple-choice contexts, where a model can prune unlikely options, but does not transfer to free-form generation, where uncertainty propagates into ungrounded rationales. The categories most improved by reflection are moreover those in which geometric consistency is central (orientation, depth, size), whereas tasks that require relational integration across multiple objects

(relative positioning, navigation) degrade. Reflection thus operates as a stabilizer in structured-choice settings without reliably enhancing naturalistic reasoning.

This inconsistency mirrors our findings with plain CoT: reasoning enhancements succeed only where the model possesses stable perceptual anchors on which to ground its steps. Self-reflection adds an explicit verification pass, but that pass is only as reliable as the model’s internal priors; when depth cues, occlusion ordering, or coordinate frames are poorly encoded, the second pass amplifies misconceptions rather than repairing them. This accounts for the asymmetry observed—Gemini, with stronger perceptual grounding, benefits under MCQ, whereas InternVL-3-78b degrades more frequently. The deeper cause is a pretraining mismatch: current VLMs import reasoning templates from language modeling yet lack the embedded 3D and geospatial representations that humans deploy. Absent such anchors, reflection cannot perform genuine error correction and instead operates as a linguistic filter that polishes reasoning syntax without improving perceptual grounding. Reflective loops therefore afford modest utility in discrete-answer formats but collapse under open-ended demands, where external factual verification is unavailable.

These findings indicate a substantive gap in current architectures: surface-level reasoning layers cannot compensate for absent spatial cognition, and meaningful progress will require deeper integration of geometric and perceptual structure. Because VLMs are trained predominantly on web data, moreover, the concepts that are well represented online are those on which models perform better, while under-represented spatial and 3D concepts fare worse—a distributional asymmetry that reinforces the disparities observed here.

H.4 SUPERVISED FINE-TUNING (SFT)

H.4.1 SETUP AND IMPLEMENTATION

We fine-tuned the Qwen 2.5 VL 3B instruction-following model (unsloth/Qwen2.5-VL-3B-Instruct) with the parameter-efficient LoRA method (Hu et al., 2021). To accommodate limited GPU resources, we loaded the model in 4-bit precision (`load_in_4bit=True`) and inserted LoRA adapters into the attention and feed-forward projection layers. The LoRA hyperparameters—rank, scaling factor, dropout, and target modules—are summarized in Table 21. Gradient checkpointing reduced memory footprint, and a fixed random seed ensured reproducibility.

For tokenization, we adopted the Qwen-2.5 chat template to keep formatting consistent between training and inference. Each training example was serialized into a single text sequence, and dynamic padding was applied via `DataCollatorForSeq2Seq`. Training used an effective batch size of 4 (per-device batch size 1 with gradient accumulation of 4); the trainer and optimization hyperparameters—learning rate, optimizer, weight decay, and scheduler—are listed in Table 22. The training loop comprised four single-epoch passes, yielding four saved checkpoints.

This configuration enabled efficient fine-tuning of a large vision–language model under constrained GPU resources while preserving high adaptation capacity.

Table 21: LoRA Hyperparameters for SFT

Hyperparameter	Value	Notes
LoRA rank (r)	16	Low-rank update capacity
LoRA alpha	16	Scaling factor
LoRA dropout	0	No dropout applied
Gradient checkpointing	Enabled	Library-specific flag
LoRA random seed	3407	Adapter initialization

H.4.2 SUPERVISED FINE-TUNING (SFT) PERFORMANCE

The learning trajectories in Figure 4a reveal a marked divergence between the MCQ and open-ended evaluation modes over training epochs. MCQ accuracy begins at a lower baseline of 25.63% but rises steadily, converging near 36.59% by the fourth epoch—an improvement indicating that the model benefits substantially from supervised fine-tuning on tasks that constrain answers to a limited

Table 22: Trainer and Optimization Hyperparameters for SFT

Hyperparameter	Value
Base model	unsloth/Qwen2.5-VL-3B-Instruct
Max sequence length	2048
Load in 4-bit	True
Per-device train batch size	1
Gradient accumulation steps	4
Effective batch size	4
Learning rate	2e-4
Optimizer	paged_adamw_8bit
Weight decay	0.01
LR scheduler	linear
Logging steps	1
Seed	42
Tokenizer template	Qwen-2.5

candidate space. Open-ended evaluation, by contrast, starts higher at 34.40% but declines sharply by the second epoch, falling to as low as 12.62%, and only partially recovers thereafter, reaching 35.48% at the end. This instability betrays a sensitivity in open-ended generation that the MCQ setting does not exhibit, where the structure of multiple-choice questions furnishes a more stable optimization signal. The fluctuations in open-ended performance are best read as evidence that fine-tuning overfits to MCQ-like reasoning patterns at the expense of free-form language generalization: the dotted baselines make plain that the MCQ mode advances beyond its initial standing while the open-ended mode remains approximately stagnant. In sum, the figure shows that SFT predominantly favors MCQ tasks in both stability and consistent accuracy gain.

The tabulated results in Table 4b complement the figure with a category-level before-and-after breakdown across the six spatial reasoning categories. Under MCQ evaluation, performance improves in every category, from a modest 1.59% in Relative Positioning to a substantial 17.11% in Size and Scale, for a Grand Total gain of 10.97%—confirming that the fine-tuning procedure effectively enhances structured-answer performance. The open-ended results are far more mixed, combining positive and negative changes. 3D Geometry and Spatial Navigation show small positive gains (3.50% and 2.82%, respectively), yet Size and Scale drops by −3.95% and Relative Positioning by −0.79%, evidence that fine-tuning can inadvertently impair certain reasoning modes in free-form contexts. The overall open-ended gain of 1.07% is negligible beside the MCQ improvement. These contrasts establish that the effectiveness of SFT is acutely sensitive to the evaluation paradigm, and the category-level decomposition pinpoints where training helps, where it fails, and how its benefits diverge between constrained and unconstrained outputs.

Closer inspection exposes striking disparities between categories and formats. Under MCQ, Size and Scale and Spatial Navigation yield the largest improvements, an indication that multiple-choice supervision is especially effective for scale discrimination and navigational reasoning. In open-ended form, however, Size and Scale actually declines, so the same learned representations do not transfer to generative contexts. Relative Positioning likewise barely improves under MCQ and worsens open-ended, marking a consistent weakness in relational reasoning that persists across formats, whereas Orientation shows strong MCQ gains (+13.33%) alongside small open-ended improvements, suggesting that some categories generalize more robustly. The divergence between formats most plausibly arises from the structural scaffolding that MCQs provide, which relieves the model of generating precise linguistic formulations; open-ended tasks, requiring reasoning and natural-language fluency in tandem, instead amplify latent weaknesses. Fine-tuning thus strongly enhances pattern recognition in discrete-choice settings while introducing volatility, and even degradation, once free expression is required—a duality demonstrating that interventions optimized for MCQs do not transfer seamlessly to unconstrained reasoning.

The implications of these findings are twofold. First, SFT as currently applied confers strong benefits on structured tasks but limited—or even negative—transfer to open-ended reasoning. This limitation is not merely an artifact of fine-tuning but is symptomatic of how contemporary VLMs

are pretrained: existing pipelines emphasize broad image–text alignment, object recognition, and captioning, while providing little explicit supervision over compositional spatial structure, occlusion layers, or geometric transformation. Models consequently excel at surface-level association without acquiring robust representations of depth, orientation, or scale. The instability observed in open-ended spatial reasoning indicates that the learned priors are fragile: absent multiple-choice scaffolding, they revert to shallow pattern matching in place of grounded inference. The categories in which declines are most pronounced—Size and Scale, Relative Positioning—point to inductive biases missing from pretraining, biases that humans acquire through embodied interaction with 3D environments. A further contributing factor is reliance on synthetic or instruction-following datasets that flatten spatial diversity into puzzle-like abstractions, which impedes generalization to real-world clutter, noise, and complexity. Together these results argue for rethinking both pretraining and fine-tuning for spatial reasoning around datasets and objectives that reflect naturalistic visual–spatial challenges. Absent such targeted reforms, current VLMs risk an algorithmic straitjacket—competent under constrained evaluation yet brittle and misaligned with the embodied, flexible reasoning humans bring to real-world spatial understanding.

H.4.3 SFT DYNAMICS ANALYSIS

To characterize the stability of Supervised Fine-Tuning (SFT) on spatial reasoning tasks, we conducted a multi-seed validation study ($N = 5$, seeds A–E) with the `Qwen-2.5-VL-3B-Instruct` model. The training trajectory, summarized in Table 23, reveals a pronounced divergence in performance dynamics between the Multiple-Choice Question (MCQ) and Open-Ended evaluation formats.

Divergent Learning Trajectories. As Table 23 and Figure 9 show, performance on SPATIALAB-MCQ improves monotonically, rising from a baseline of 25.63% to an average of 35.74% at Epoch 4. The model thus adapts to the discriminative structure of the task, learning to map visual features onto constrained output options with increasing reliability.

SPATIALAB-OPEN, by sharp contrast, traces a pronounced *U-shaped* trajectory across all five seeds. In place of steady improvement, the model suffers an immediate performance collapse in the early epochs (falling from 34.40% to an average of 19.26% at Epoch 1); although it recovers in later epochs (converging to $\approx 34.82\%$ at Epoch 4), the final gain over the zero-shot baseline is negligible (+0.42%).

Table 23: Multi-seed validation of SFT dynamics ($N = 5$). While MCQ performance improves steadily, Open-Ended performance exhibits a systematic “U-shaped” collapse and recovery, indicating representational instability in generative spatial tasks.

Epoch	Type	Seed A	Seed B	Seed C	Seed D	Seed E
0 (Base)	MCQ	25.63%	25.63%	25.63%	25.63%	25.63%
	Open	34.40%	34.40%	34.40%	34.40%	34.40%
1	MCQ	29.56%	35.83%	24.13%	27.63%	26.53%
	Open	22.50%	23.20%	12.50%	14.00%	24.10%
2	MCQ	34.33%	37.76%	27.33%	31.83%	34.43%
	Open	12.62%	25.30%	28.80%	23.80%	16.65%
3	MCQ	34.68%	37.33%	32.86%	32.23%	34.73%
	Open	28.69%	22.22%	30.34%	32.90%	28.30%
4	MCQ	36.59%	38.48%	34.43%	34.36%	34.83%
	Open	35.48%	34.59%	33.92%	34.70%	35.40%

Representational Brittleness and Alignment Tax. We attribute this phenomenon to the fragility of spatial representations in current VLMs under fine-tuning. The initial collapse in open-ended accuracy is consistent with *catastrophic forgetting* of the linguistic priors required for fluent, grounded generation: as the model optimizes for the specific instruction-following format of the training data, it overfits to the structural logic of the tasks—benefiting MCQ—at the expense of the flexible generative grounding that open-ended description demands.

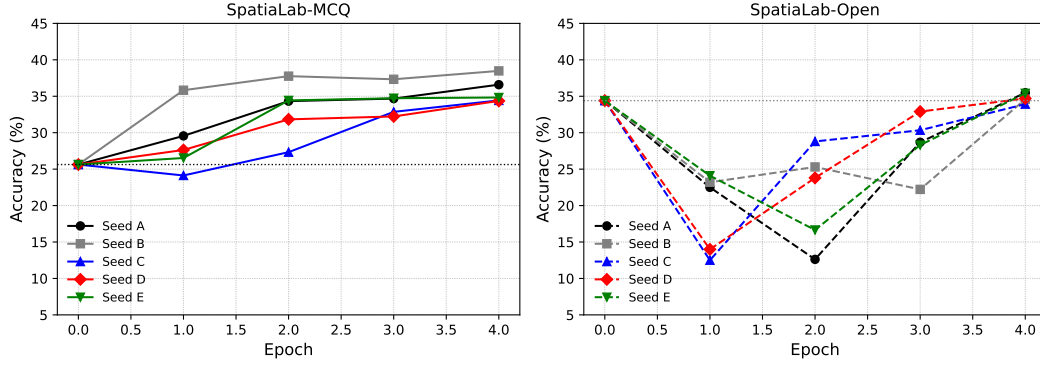


Figure 9: SFT Dynamics Analysis.

The subsequent recovery phase corresponds to the model re-learning to articulate spatial concepts within the new distribution, yet it fails to surpass the initial baseline by any meaningful margin. This dynamic sharpens the central finding: SFT is effective at teaching a model to *select* correct spatial answers, but it does not fundamentally enhance the underlying *generative spatial representation* that robust open-ended reasoning requires.

H.4.4 PERFORMANCE GAIN ON EXTERNAL BENCHMARKS

To determine whether the representations learned from SPATIALAB generalize beyond the benchmark’s specific format, we assessed the transferability of our fine-tuned model to three external spatial reasoning datasets—OMNISPATIAL, SPACE (Multimodal), and MIND THE GAP. We took the Qwen2.5-VL-3B-Instruct model fine-tuned on SPATIALAB under the SFT protocol detailed in Appendix H.4.1 and evaluated it on these external benchmarks in a zero-shot setting.

Table 24: Transfer learning performance of Qwen2.5-VL-3B-Instruct on external benchmarks after fine-tuning on SPATIALAB. The consistent gains indicate that the model learns generalizable spatial representations rather than merely overfitting to the source dataset’s format.

Benchmark	Base Accuracy	After SFT	Improvement
OmniSpatial	40.30%	47.35%	+7.05%
SPACE (Multimodal)	23.43%	28.67%	+5.24%
Mind The Gap	35.86%	47.56%	+11.70%

As Table 24 shows, fine-tuning on SPATIALAB yields substantial and consistent improvements across all three domains, from +5.24% to +11.70%. The largest gain, on MIND THE GAP (+11.70%), indicates that our benchmark’s emphasis on occlusion and spatial continuity directly addresses the reasoning gaps that dataset targets, while the improvements on OMNISPATIAL (+7.05%) and SPACE (+5.24%) confirm that the geometric and relational concepts encoded in our taxonomy align with broader definitions of spatial intelligence in the literature. These results speak to the quality of the SPATIALAB data. Although the work’s primary purpose is to serve as a rigorous diagnostic benchmark, the consistent transfer demonstrates that the dataset does not merely reward overfitting to a specific question template; rather, the diversity of its 30 sub-categories compels the model to acquire robust, transferable spatial representations that generalize to unseen distributions and task formats.

H.5 AI AGENTS FOR SPATIAL REASONING

We developed SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025), to perform structured spatial reasoning over images within the SPATIOXOLVER evaluation. The system decomposes complex visual reasoning into specialized sub-tasks, each delegated to a dedicated agent, so that objects, attributes, relations, symmetries, and transformations are processed in

focused isolation. This modular design improves accuracy and interpretability and supports multi-step reasoning, allowing the system to propagate low-level perceptual cues into higher-order relational and transformational inference. The architecture, its constituent agents, and their coordinated interaction are detailed below, with the full prompt set provided in Figures 10, 11, and 12.

H.5.1 AGENTS

Base Visual Analysis Agent. Base Visual Language Model (VLM) agent initiates the pipeline by generating a detailed textual description of the input image, capturing object types—including geometric shapes, natural elements, and artificial icons—alongside attributes such as size, orientation, color, and shading. Relative positions like leftmost, center, and top are encoded explicitly, and the prompt emphasizes exhaustive detail without summarization. Images are converted to data URIs and sent with the prompt to a Gemini-2.5-based model, whose output is parsed into a structured bullet-list using regex-based extraction. This structured description serves as the foundation for downstream agents, providing a comprehensive perceptual anchor that encodes both explicit and implicit spatial cues. The approach ensures that even subtle visual features, such as partial occlusions or minor shape variations, are retained for later relational and transformational reasoning.

Object Segmentation Agent. Object Segmentation agent converts the verbose textual description into a discrete list of uniquely identified objects, differentiating between visually similar instances. Each object receives a stable identifier (e.g., Obj1, Obj2) to maintain consistent references across subsequent reasoning steps. The agent uses carefully formatted prompts and calls the same Gemini-2.5 model to structure the description, while post-processing splits lines by object ID to enforce consistency. This structured representation reduces ambiguity in downstream attribute and relation extraction, enabling precise mapping of complex scenes. By explicitly isolating objects, the agent also facilitates detection of overlaps, repetitions, and hierarchical groupings that may not be evident in raw descriptions.

Attribute Extraction Agent. Attribute Extraction agent translates each object into a structured JSON capturing core properties such as shape, size, color, shading, orientation, and positional descriptors. Prompts enforce strict JSON output, and responses are parsed using `json.loads()`, with fallback defaults to ensure robustness against model inconsistencies. This structured encoding allows quantitative and qualitative comparisons of objects, supporting later stages of spatial reasoning. By preserving both absolute and relative measures, the agent maintains fidelity to the original perceptual content, while also normalizing descriptors for computational consistency. The JSON structure enables programmatic downstream operations, such as calculating symmetries or detecting transformations.

Spatial Relation Agent. The Spatial Relation agent encodes object interactions as triples (ObjectA,Relation,ObjectB), capturing directional relations (e.g., left_of, above, inside) and higher-order alignments like rows, grids, or repetition patterns. Consistent object IDs are enforced to prevent relational ambiguity, and parsing splits output lines starting with parentheses, followed by automated consistency checks. This step transforms static object descriptions into a relational graph that encodes spatial topology and hierarchy. The resulting structure allows reasoning over both pairwise interactions and more complex scene configurations, which is critical for tasks like symmetry detection, grouping, and multi-step navigation of spatial layouts.

Grouping and Symmetry Agent. Grouping and Symmetry agent abstracts higher-order patterns from object lists and spatial relations, identifying clusters, rows, grids, and symmetries—including vertical, horizontal, rotational, and translational. Outputs are requested in structured JSON with keys `groups` and `symmetries`, with empty structures as fallbacks to maintain downstream compatibility. This agent captures emergent scene regularities that are not apparent at the object or pairwise relation level, providing critical context for pattern recognition and reasoning. By formalizing structural redundancies and symmetries, the agent supports both perceptual inference and anticipatory reasoning for dynamic transformations.

Transformation Tracking Agent. For multi-frame sequences, the Transformation Tracking agent logs object-level changes including translations, scaling, rotations, color/shading shifts, and appearance or disappearance events. Structured logs are extracted line-by-line using regex patterns that

capture directional arrows, producing a temporal representation of the scene. This encoding allows the system to track not just static spatial configurations but also dynamic evolution, supporting reasoning about motion, causality, and object interactions over time. The agent ensures that transformations are linked to consistent object IDs, enabling coherent cross-frame reasoning and predictive modeling.

Representation Standardization Agent. Finally, the Representation Standardization agent consolidates all outputs into a unified JSON format containing objects, relations, groups, symmetries, and transformations. Prompts enforce consistent object IDs and structured formatting, while JSON parsing failures revert to empty dictionaries to ensure pipeline robustness. By centralizing all perceptual, relational, and temporal information, this agent creates a comprehensive scene representation that is both human-interpretable and machine-readable. The resulting unified structure serves as a versatile foundation for downstream tasks, including reasoning evaluation, spatial pattern analysis, and multi-frame prediction.

Open-Ended Spatial Reasoning Agent. The standardized representation feeds into the reasoning agent, which answers multiple-choice or pattern recognition questions. Inputs include objects, attributes, relations, groups, symmetries, and transformations. The model considers geometric patterns, relational alignment, and dynamic transformations to generate or select the correct answer.

H.5.2 TECHNICAL IMPLEMENTATION DETAILS

For our technical implementation, we use Gemini-2.5-Flash with a low temperature (0.1) to ensure deterministic outputs and perform multiple iterations for refinement. API calls are handled via OpenRouter, supporting both text and image inputs, while parsing of model responses relies on regex and JSON-based extraction to generate structured outputs. All interactions are logged in JSON format for reproducibility, and the final output is a unified JSON per image containing comprehensive information on objects, relations, symmetries, and transformations. This multi-agent decomposition allows each specialized model to focus on a constrained sub-task, enhancing overall accuracy, interpretability, and robustness in structured spatial reasoning.

H.5.3 EVALUATION AND ANALYSIS

The evaluation protocol remains identical to the standard CoT and CoT-SR setting, using 5 balanced samples per sub-category (seed 42), making a total of 150 samples.

Table 25: Agent Performance Analysis

Category	MCQ		Gain	Open Evaluation		Gain
	Normal	With Agents		Normal	Agent Results	
3D Geometry	48.00%	44.00%	-4.00%	60.00%	48.00%	-12.00%
Depth & Occlusion	48.00%	44.00%	-4.00%	40.00%	16.00%	-24.00%
Orientation	56.00%	64.00%	+8.00%	0.00%	36.00%	+36.00%
Relative Positioning	60.00%	60.00%	0.00%	50.00%	36.00%	-14.00%
Size & Scale	48.00%	52.00%	+4.00%	25.00%	24.00%	-1.00%
Spatial Navigation	68.00%	64.00%	-4.00%	40.00%	28.00%	-12.00%
Overall	54.67 %	57.33%	+2.66%	35.83%	31.33%	-4.50%

The evaluation of Gemini-2.5-Flash with agents highlights both strengths and limitations of agentic reasoning in perceptual-geometric tasks. Orientation is the only category where agents provide consistent and substantial gains, improving by +8.00% in MCQ and an impressive +36.00% in open evaluation. This pattern suggests that agent-based multi-step reasoning is effective when the task can be decomposed into sequential alignment of directional cues, where structured deliberation offers a clear advantage over single-step outputs.

In contrast, most other categories exhibit negative or neutral gains, revealing that agent reasoning does not universally enhance performance. For example, depth & occlusion suffers a severe -24.00%

drop in open evaluation, while spatial navigation decreases by -12.00%. These tasks require deeper perceptual grounding in three-dimensional space, and the absence of such priors means that reasoning steps tend to reinforce misconceptions rather than correct them. Similarly, relative positioning shows stagnation in MCQ and a -14.00% decline in open evaluation, indicating that agent loops fail to add robustness when relational understanding is inherently weak.

Some categories show marginal improvements, such as size & scale (+4.00% in MCQ), but the gains are not stable and do not transfer to open evaluation. Even in 3D geometry, agents fail to provide a boost, with -4.00% and -12.00% drops in MCQ and open evaluation, respectively. These results underscore that our benchmark is complex and not solvable simply by layering multi-step agentic reasoning on top of existing models. Instead, it requires a deeper integration of geospatial understanding and perceptual representations, which current architectures lack.

Overall, the mixed results emphasize that agentic workflows alone are insufficient for advancing performance in this domain. While orientation tasks benefit from structured reasoning, most categories expose the models’ lack of inherent awareness of perceptual concepts. This makes errors propagate through multi-step reasoning, amplifying performance drops. Consequently, our benchmark serves as a strong stress test, highlighting that future progress will require embedding genuine perceptual and geospatial priors rather than relying solely on agent-based reasoning.

I QUALITATIVE ERROR ANALYSIS AND ERROR PATTERNS

This section presents a focused qualitative analysis of model failures observed in our visual reasoning benchmark. We explain each error class, provide diagnostic root cause analysis, identify recurring patterns, and describe verification protocols. The discussion aims to be both interpretable and actionable, so it can inform future architectural and evaluation design.

I.1 OVERVIEW

Across the sample set, models achieve strong coarse perception, they can reliably recognize object categories, colors, and simple binary relations, but they break down systematically when multiple cues must be integrated. Failures are not random noise but cluster into a small set of interpretable classes: spatial mislocalization, perspective and scale mistakes, occlusion and ordering errors, attribute confusion, and ungrounded open-ended rationalization. These classes recur across architectures and prompting strategies, which suggests that the underlying causes are common inductive biases in current vision–language pipelines rather than isolated training artifacts. Importantly, the same error often appears in both large-scale proprietary systems and small open-source baselines, indicating structural weaknesses shared across the model family. In the following, we unpack each error class with representative examples and root-cause hypotheses.

I.2 ERROR TAXONOMY AND ANALYSIS

I.2.1 SPATIAL MISLOCALIZATION AND REFERENCE CONFUSION.

■ **Observation.** When queries require distinguishing one object from a set of visually similar candidates, models frequently misidentify the target. For example, if asked to point out “the leftmost red cube among three stacked cubes,” the model may incorrectly output the middle cube, influenced by lighting highlights or central positioning. This problem becomes particularly acute in cluttered environments such as kitchen counters or crowded table scenes, where multiple overlapping objects share the same color and shape. In such cases, the model’s errors are not random: it tends to privilege objects that appear most salient due to size, brightness, or centrality, even when these cues contradict the explicit spatial instructions in the query. These consistent patterns highlight that the failures are systematic, not accidental.

★ **Analysis.** The recurring mistakes suggest that models lack a robust object-centric representation, or what cognitive science would call an “index” that persists across reasoning steps. Instead of maintaining a stable pointer to the correct candidate throughout the reasoning chain, models appear to drift, often reassigning attention to whichever object seems most visually striking at intermediate layers. This leads to reasoning errors such as starting with the leftmost cube but ending the chain with the brightest or largest cube. The inability to anchor spatial reference consistently makes tasks

involving relative comparisons, like “the ball behind the second chair”, particularly fragile. In essence, the model operates more like a salience detector than a referential reasoner.

⊗ **Root cause.** At the architectural level, feature pooling operations, whether via attention or convolution, tend to mix contextual information from neighboring regions. While this is beneficial for global context, it erases the fine-grained, object-specific identity needed to distinguish between similar candidates. Once pooled, features of two adjacent red cubes may be nearly indistinguishable in the latent space. Without an explicit mechanism, such as slot-based representations or pointer tokens, the network collapses distinct entities into blended embeddings. This structural limitation prevents reliable referent tracking across reasoning steps and explains why models consistently confuse dense, overlapping objects.

I.2.2 PERSPECTIVE AND SCALE MISTAKES.

■ **Observation.** Tasks involving metric reasoning about relative size, fit, or scale are inconsistently handled. On simple comparisons like “is a car larger than a bicycle,” models typically succeed because such relationships align with strong statistical priors. However, when faced with near-threshold decisions such as “will the small sphere fit inside the slightly larger cylinder,” errors become frequent and unpredictable. A common pattern is reliance on canonical object sizes: for example, predicting that a door must be taller than a truck even in a rendering where the truck is scaled down. Another failure occurs in unusual visualizations, such as toy miniatures or architectural models, where familiar proportions are deliberately inverted. In these cases, models cling to their priors and ignore the actual geometric evidence visible in the image.

★ **Analysis.** These behaviors indicate that models rely heavily on distributional statistics of object categories rather than performing true geometric inference. Instead of reasoning about projection geometry, relative bounding box sizes, or depth cues, they substitute their world knowledge of typical scales. This shortcut works well in everyday contexts but fails catastrophically in edge cases that deviate from the training distribution. For instance, when an unusually small airplane is rendered next to a person, the model still predicts the airplane as larger. Such errors highlight the lack of grounding in pixel-level metric cues and the absence of consistent use of perspective or vanishing points.

⊗ **Root cause.** Current architectures do not include explicit geometric reasoning modules that preserve projection-aware relationships. Supervision for scale and fit tasks is limited, so models learn correlations between object category labels and size but not the underlying geometry. Furthermore, training objectives prioritize overall accuracy rather than fine-grained metric consistency, meaning that scale-sensitive errors are under-penalized. Without inductive biases for 3D reasoning or targeted supervision, the model defaults to memorized priors about canonical sizes rather than interpreting the actual perspective geometry in the image.

I.2.3 OCCLUSION AND ORDERING FAILURES.

■ **Observation.** Determining which object is in front, behind, or partially hidden remains a core weakness. Models often misreport the foreground object when two animals overlap, or when semi-transparent barriers like glass or fences are involved. For instance, in an image where a cat sits partially behind a curtain, models may incorrectly describe the curtain as being occluded by the cat. Thin or fragile structures, such as wires, tree branches, or railings, are particularly error-prone: they are frequently omitted altogether, leading to wrong answers in questions like “what object lies behind the fence.” These mistakes occur consistently, especially when occlusion cues are subtle or when the depth difference is small.

★ **Analysis.** Such failures indicate that models treat depth and occlusion as weak, secondary signals. While the pixel data contains clear occlusion cues (e.g., edge boundaries, T-junctions), the intermediate features learned by the model do not retain them with sufficient resolution. Aggregation layers blur boundaries and downweight thin structures, effectively removing the very cues necessary to resolve ordering. This is why models succeed in gross depth distinctions (foreground vs. background) but collapse in fine-grained cases where partial occlusion must be inferred. In practice, they guess rather than infer, with systematic biases toward treating the visually larger or brighter object as foreground.

⊗ **Root cause.** The architecture lacks explicit mechanisms for multi-scale feature retention and has no dedicated supervision for depth or occlusion reasoning. Thin or partially occluded signals are easily lost when features are pooled across large receptive fields. Furthermore, training data often underrepresents subtle occlusion cases, especially those involving transparency or fine boundaries.

As a result, the learned representations flatten layered scenes into 2D appearance maps, preventing robust construction of depth-aware relational reasoning.

I.2.4 ATTRIBUTE CONFUSION AND SEMANTIC SWAP.

■ **Observation.** Models sometimes conflate perceptual attributes (such as color, texture, or material) with functional or semantic roles. A classic example is misjudging whether a crosswalk is safe: models may base the answer solely on the presence of painted stripes while ignoring the state of the traffic light or approaching vehicles. Similarly, a surface with a metallic texture may be misclassified as “steel” even if it is painted plastic. In tool-use contexts, the presence of handles or familiar shapes may lead the model to infer functionality that the object does not possess. These confusions are frequent in tasks requiring fine-grained visual grounding of attributes.

★ **Analysis.** These errors reveal that language priors strongly bias the model’s predictions when visual input is ambiguous. The decoder favors plausible narratives based on co-occurrence in the training data: “crosswalks are safe when stripes are visible,” “shiny textures correspond to metal,” and so on. This shortcut allows models to produce semantically coherent answers even when they do not inspect the relevant visual evidence. The result is superficially correct reasoning in familiar contexts but systematic errors in atypical or adversarial scenarios. In practice, this manifests as overfitting to textual stereotypes rather than grounding in perceptual details.

⊗ **Root cause.** The underlying training objective, maximizing likelihood of fluent and plausible text, does not enforce visual grounding. As long as the generated answer is semantically reasonable, the loss is minimized, even if the reasoning is incorrect. Consequently, the model is incentivized to substitute common-sense priors for genuine perceptual inference. Without auxiliary losses or verification steps to tie textual claims to image regions, models will continue to swap perceptual and semantic attributes.

I.2.5 OPEN-ENDED RATIONALIZATION WITHOUT VERIFICATION.

■ **Observation.** For explanatory or why-style questions, models often generate coherent but unsupported narratives. For example, when asked “why is the object leaning,” a model might answer “because the surface is slippery” even though slipperiness cannot be visually inferred. In another case, a tilted shelf might be explained as “due to an earthquake,” a claim entirely outside the evidence in the image. These errors highlight a tendency to fill gaps with imaginative but ungrounded rationales, producing fluent explanations that mislead users about the true evidence.

★ **Analysis.** The issue stems from a decoupling between text generation and visual grounding. The model generates high-quality language that reads convincingly, but there is no mechanism ensuring that these rationales are tied to actual visual features. Confidence calibration compounds the problem: models often assign high certainty to fabricated explanations, giving the false impression of reliability. As a result, open-ended outputs may look authoritative while being factually baseless. This mode of failure is particularly concerning in safety-critical domains where plausible but false rationales could misguide decision-making.

⊗ **Root cause.** Training is dominated by language-modeling objectives that reward fluency and coherence but not factual grounding. Decoders are optimized to produce the most probable continuation of text given the prompt, without constraints linking the explanation back to specific image evidence. The lack of an explicit verification layer or grounding requirement means that models default to linguistic plausibility over perceptual truth. Without additional architectural or training constraints, open-ended rationalizations will continue to prioritize surface coherence at the expense of evidential accuracy.

I.3 ERROR PATTERNS AND CROSS-MODEL SIGNALS

When comparing failures across different models, several consistent patterns appear, revealing insights into the underlying mechanisms of errors. One clear trend is that disagreements between models are systematic rather than random: some models consistently rely on geometric cues such as relative size, depth ordering, and occlusion relationships, while others fall back on linguistic priors or common-sense assumptions derived from training text. This leads to predictable divergences, for example when two models give different answers about which object is “in front” despite seeing the same scene. Errors also spike when multiple cues must be integrated simultaneously. Tasks that require reasoning about occlusion, relative scale, and perspective together are particularly challenging; for

instance, understanding whether a larger object partially occluding a smaller one is in front or behind often triggers consistent mispredictions. Calibration is another noticeable issue: high-confidence predictions frequently correspond to incorrect answers, especially in open-ended explanations or complex spatial queries. These patterns indicate that errors arise from how models balance perceptual grounding and linguistic plausibility. Different architectures exhibit distinct tendencies, some over-rely on visual geometry, others overfit to language correlations, so failures are not just isolated mistakes but reflect deeper trade-offs in reasoning strategies. This also implies that any attempt to improve performance should account for both sources of bias simultaneously rather than treating errors as independent events.

I.4 MITIGATION STRATEGIES

Given these error patterns, several targeted strategies can be pursued to reduce failures without requiring complete architectural redesign. For instance, geometry-aware supervision can be incorporated by adding auxiliary tasks like depth prediction, occlusion ordering, or overlap estimation. This encourages models to respect spatial constraints and improves consistency across predictions. Multi-scale feature retention is another approach: by using edge-aware pooling or attention mechanisms, models can maintain information from thin or partially occluded structures that would otherwise vanish in standard downsampling. Coordinate anchoring can help resolve confusion between viewer-centric and object-centric frames by explicitly representing reference points for relative positioning. Counterfactual augmentation can also be applied, where distractor objects are systematically swapped or perturbed, forcing the model to rely on the true referent cues rather than heuristics. Lightweight verification modules can be added on top of existing outputs, cross-checking claims against visual or geometric constraints to filter implausible predictions. Each of these strategies can be implemented incrementally, providing clear paths to reduce systematic failures while keeping the core model largely unchanged.

In addition to these architectural and training augmentations, supervised fine-tuning (SFT) offers a practical pathway to instill more reliable spatial reasoning. By curating datasets that explicitly highlight failure cases, such as near-threshold scale comparisons, occlusion ambiguities, or reference resolution traps, SFT can directly expose models to the edge scenarios where they otherwise default to heuristics. Carefully designed instruction-style fine-tuning further encourages step-by-step reasoning, prompting the model to verbalize intermediate spatial relations rather than skipping to a guess. This not only strengthens grounding but also improves interpretability, as errors become traceable to specific reasoning steps. Moreover, SFT can be paired with counterfactual or adversarial augmentations, ensuring that the model does not simply memorize but instead learns invariances to distractors and context shifts. Importantly, these interventions require no wholesale changes to the architecture: they adapt existing models by reshaping their inductive biases through supervision. In this sense, SFT serves as a lightweight yet powerful complement to geometry-aware and verification-based strategies.

In this work, we additionally performed SFT using 40% of the training data and observed clear performance gains, particularly on multiple-choice questions. This suggests that targeted supervision helps models better internalize spatial cues and resolve distractor confusions when answer options are well-defined. However, improvements on open-ended responses were more limited. The model continues to produce fluent but weakly grounded explanations, highlighting that SFT alone cannot fully address rationalization without verification. These findings reinforce the need to combine SFT with geometry-aware supervision, counterfactual augmentations, and lightweight verifiers to achieve balanced progress across both structured and free-form tasks.

Our analysis shows that failures are not arbitrary but structured and interpretable, emerging from recurring patterns rather than random noise. This indicates that VLM errors follow systematic trajectories tied to the inductive biases of their architectures and training regimes. For example, geometric reasoning often collapses when occlusion cues are weak, and in such cases language priors take over, producing answers that sound plausible but lack visual grounding. Similarly, scale inference drifts when canonical object size priors dominate over pixel-level metric cues, leading to consistent misjudgments in near-threshold scenarios. These findings highlight that the gap is not simply one of capacity but of representation: current models lack explicit mechanisms to anchor visual features to grounded geometric relations.

By adopting the proposed verification protocols and targeted mitigation strategies, researchers can address these weaknesses directly. Geometry-aware supervision and counterfactual augmentation

2916 would help models resist shortcuts, while multi-scale feature retention ensures thin structures and
2917 partial occlusions are not discarded. SFT, when carefully applied, can further strengthen structured
2918 reasoning by aligning model predictions with task constraints, though it must be combined with
2919 grounding objectives to avoid overfitting to discrete-choice formats. Importantly, verification modules
2920 provide a lightweight way to enforce consistency between claimed relations and image evidence,
2921 counteracting the tendency to rationalize without checking. Taken together, these steps chart a
2922 practical pathway toward systems that reason more like humans: robust to clutter, interpretable in
2923 their failures, and capable of explaining not only what they predict but why.

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Base VLM Agent Prompt

You are a Visual Scene Describer. Produce a raw, exhaustive text description of the given image.

Input image:
{image_input}

Your focus:

- Object Identification:
 - Shapes (circle, square, triangle, polygon, etc.)
 - Natural elements (tree, sun, water, etc.)
 - Artificial objects (arrows, patterns, icons)
- Attribute Noticing:
 - Size (small, medium, large, relative comparison)
 - Orientation (upright, rotated, tilted)
 - Shading or texture (solid, striped, shaded, transparent)
 - Color (if available)
- Positioning:
 - Relative location (leftmost, rightmost, top, bottom, center)
 - Proximity to other objects
- Level of Detail:
 - Do not summarize
 - Be verbose and explicit about every visible detail

Object Segmentation Agent Prompt

You are an Object Segmentation Specialist. Convert a verbose image description into a structured list of distinct objects.

Input description:
{visual_description}

Your focus:

- Object Isolation:
 - Treat each unique instance as a separate object
 - If multiple objects are similar, assign separate IDs
- Labeling:
 - Assign IDs: Obj1, Obj2, Obj3...
 - Keep descriptions short but complete
- Example Output:
Obj1: Small shaded circle (leftmost)
Obj2: Large unshaded square (center)
Obj3: Triangle rotated 45° (rightmost)

Attribute Extraction Agent Prompt

You are an Attribute Extraction Specialist. For each object, extract its properties into structured JSON.

Your focus:

- Attributes to Capture:
 - Shape
 - Size (relative if necessary)
 - Color or shading
 - Orientation or rotation
 - Position (leftmost, rightmost, center, top, bottom, etc.)
- Output Format:
JSON array, e.g.:
[
 {{ "ID": "Obj1", "Shape": "Circle", "Size": "Small", "Shading": "Yes", "Rotation": "0°", "Position": "Leftmost" }},
 {{ "ID": "Obj2", "Shape": "Square", "Size": "Large", "Shading": "No", "Rotation": "45°", "Position": "Center" }}
]

Figure 10: Prompts for SPATIOXOLVER (Part 1).

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Spatial Relation Agent Prompt

You are a Spatial Relation Specialist. Extract spatial relationships among objects.

Your focus:

- Core Relations:
 - left_of, right_of, above, below
 - inside, surrounding
 - touching, overlapping
 - aligned, centered
- Precision:
 - Always specify directionality (ObjA left_of ObjB is not the same as ObjB left_of ObjA)
 - Use consistent object IDs
- Structural Patterns:
 - Detect repeated relations (e.g., "each circle is above a square")
 - Note spacing and alignment
- Output Format:

Triples in form: (ObjectA, Relation, ObjectB)

Example:

(Obj1, left_of, Obj2)

(Obj3, inside, Obj4)

Grouping Symmetry Agent Prompt

You are a Grouping and Symmetry Specialist. Identify higher-order structures from objects and relations.

Relations:
{relation_list}

Your focus:

- Grouping:
 - Clusters of objects (rows, columns, triangles, grids)
 - Repeated patterns
- Symmetry:
 - Vertical, horizontal, rotational
 - Mirror or reflective symmetry
 - Translational repetition
- Output Format:

JSON with two fields: "groups" and "symmetries"

Example:

```
{  
  "groups": ["Obj1, Obj2, Obj3 form a horizontal row"],  
  "symmetries": ["Vertical symmetry across Obj2"]  
}
```

Transformation Tracking Prompt

You are a Transformation Tracker. Analyze changes across multiple frames of an image sequence.

Your focus:

- Types of Transformations:
 - Movement (left, right, up, down)
 - Scaling (larger, smaller)
 - Rotation (degrees)
 - Shading or color change
 - Shape change (square → circle)
 - Appearance/disappearance
- Output Format:

Per-object transformation logs:

Obj1: Frame1 → Frame2 (rotated 90° clockwise)

Obj2: Frame2 → Frame3 (moved right)

Figure 11: Prompts for SPATIOXOLVER (Part 2).

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Representation Standardization Agent Prompt

You are a Symbolic Abstraction Specialist. Consolidate all perception results into a standardized structured format.

Inputs:

- Objects: {object_list}
- Attributes: {attributes_json}
- Relations: {relation_list}
- Groups & Symmetries: {grouping_symmetry}
- Transformations: {transformations}

Your focus:

1. Create a unified JSON representation containing:
 - Objects with attributes
 - Relations as triples
 - Groups and symmetries
 - Transformations across frames
2. Ensure consistency in IDs and formatting
3. Example Output:

```
{
  "objects": [...],
  "relations": [...],
  "groups": [...],
  "symmetries": [...],
  "transformations": [...]}

```

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and select the correct option for the multiple-choice question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Options: {options_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.
- Select only the correct option Only (1, 2, 3, 4).
- Do NOT provide explanations, reasoning, or extra text. Only the option letter.

Output format: "<option_number>"

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and give correct answer for the question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.

Figure 12: Prompts for SPATIOXOLVER (Part 3).

J REPRESENTATIVE BENCHMARK SAMPLES ACROSS ALL SUB-CATEGORIES

We present representative benchmark samples for all 30 sub-categories in SPATIALAB, illustrating the diverse spatial reasoning challenges covered. These samples highlight the variety of visual phenomena, including 3D geometry, occlusion, orientation, relational positioning, size and scale, and navigation tasks. Together, they provide a concrete view of the types of reasoning each model must perform to succeed across the benchmark.

3D Geometry: Gravity Effects



Question: From which direction would the wind cause the liquid in the glass which is in the front to spill onto the ground first?

Options

1. From the left side
2. From the right side
3. From the front
4. From the back

Gold MCQ Answer: 1

Gold Open ended Answer: From the left side

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** A gust blowing from the back of the patio (the top of the image, toward the viewer) would push the liquid forward over the near edge and make it spill first. Winds from the sides would tend to slosh it sideways rather than directly over the front rim.
- **Gemini-2.5-Flash:** The glass in the front is the one on the right side of the coffee table. Given its position, a wind blowing from the left (from the direction of the wooden cabinet and tall plant) would likely cause the liquid to spill first. This is because the glass is positioned with its opening facing somewhat towards the left, making it more susceptible to a gust from that direction.
- **InternVL3.5-72B:** The wind would need to come from the upper-right direction to cause the liquid in the glass on the front right of the table to spill first. This is due to the positioning of the glass towards the front edge of the table, closest to the right side of the image.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about the direction the wind would have to come from to spill the liquid in the front glass. Looking at the position of the glass: it's on the left front edge of the coffee table. The table seems to have a flat surface. If the wind blows from the left side (from the viewer's left, which would be the front of the image if the viewer.

Figure 13: Some examples from the benchmark (3D Geometry: Gravity Effects) and evaluation results.

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3D Geometry: Shape Projection



Question: If the hanging light casts a shadow on the wall, which outline might appear?

Options

1. A rectangle narrowing at the bottom From the right side
2. A vertical cylinder
3. A circle with fuzzy edges
4. A triangle pointing downward

Gold MCQ Answer: 2

Gold Open ended Answer: A vertical cylinder

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: You would see the pendant's dome-shaped profile: a rounded semicircle or oval shadow with a thin vertical line or narrow neck above where the cord hangs. The shadow would fall on the wall to the right of the window, roughly over the nightstand.
- Gemini-2.5-Flash: The hanging light is a simple, cylindrical pendant. If it casts a shadow on the wall, the outline would likely be a circular or elliptical shape, depending on the angle of the light source relative to the wall. Given its position near the window, the shadow might also be influenced by ambient light or other objects in the room.
- InternVL3.5-72B: The hanging light fixture is likely to cast a shadow resembling its own shape onto the wall. Given its cylindrical form, the shadow would appear as a soft-edged, elongated circle or oval, depending on the angle of the light source relative to the wall.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user is asking which outline a hanging light would cast on the wall if it casts a shadow. The wall near the window is light-colored, so the shadow would be visible. Since the shade is cylindrical, the shadow might be a cylinder. But maybe the outline is of the lampshade's perimeter. Alternatively, if the light is hanging from a chain, the shadow could be more elongated. But based on typical light fixtures, a cylindrical shade would cast a circular or oval shadow. The answer should be the outline of the hanging light's shade.

Figure 14: Some examples from the benchmark (3D Geometry: Shape Projection) and evaluation results.

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3D Geometry: Spatial Containment



Question: Look at the tall pot in the background with the wide mouth (top left). could the small jug from the center of the photo fit completely inside it?

Options

1. Yes, easily with room to spare
2. No, the small jug hand will stuck
3. No, the small jug is too tall.
4. Yes, but it would be a perfect fit with no room left

Gold MCQ Answer: 2

Gold Open ended Answer: No, the small jug hand will stuck

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** No — the tall pot's mouth appears too narrow compared with the small jug's widest part and its handle. The jug wouldn't pass through the opening to sit completely inside.
- **Gemini-2.5-Flash:** No, the small jug from the center of the photo could not fit completely inside the tall pot in the background. While the tall pot has a wide mouth, the jug's overall height and width appear to be too large to be fully contained within the pot's interior. The jug's handle would also likely prevent it from fitting snugly.
- **InternVL3.5-72B:** The tall pot in the background with the wide mouth appears large enough to potentially accommodate the small jug from the center of the photo. Considering the proportions and the wide opening of the tall pot, it is likely that the small jug could fit completely inside it.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about whether the small jug in the center of the photo could fit completely inside the tall pot in the background with the wide mouth (top left). However, without explicit scale references, this conclusion relies on typical proportional relationships between similar pottery forms. Based on typical proportions, the small jug's height is likely shorter than the tall pot's, and its width seems narrow enough to fit.

Figure 15: Some examples from the benchmark (3D Geometry: Spatial Containment) and evaluation results.

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3D Geometry: Volume Comparison



Question: Compare the internal cooking volume of the microwave with the internal cooking volume of the larger oven below the stove.

Options

1. They both have the same volume A vertical cylinder
2. The microwave has slightly more volume
3. The oven below the stove has much more volume
4. The oven has more width, but the microwave has more height, so they are equal

Gold MCQ Answer: 3

Gold Open ended Answer: The oven below the stove has much more volume

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 3
- **InternVL3.5-72B:** 3
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** The microwave’s cooking cavity is much smaller than the larger oven below the cooktop — it’s only a small fraction of the oven’s internal volume. The countertop microwave is sized for single plates or small dishes, whereas the built-in oven can hold full baking trays and multiple racks for larger meals.
- **Gemini-2.5-Flash:** The internal cooking volume of the microwave appears to be significantly smaller than that of the larger oven below the stove. The microwave is a compact countertop model, suitable for reheating or cooking smaller items..
- **InternVL3.5-72B:** The internal cooking volume of the microwave on the right is likely smaller than that of the larger oven located below the stove. This is because microwaves typically have more compact dimensions to fit on countertops, while ovens have a greater internal capacity to accommodate larger dishes and multiple items. The size and design visible in the image suggest the oven has a deeper and wider cooking area compared to the microwave.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants me to compare the internal cooking volumes of the microwave and the larger oven below the stove in the provided image. So, the internal cooking volume of the oven below the stove is larger than that of the microwave. Based on their relative sizes in the image, the internal cooking volume of the oven below the stove is substantially larger than that of the microwave

Figure 16: Some examples from the benchmark (3D Geometry: Volume Comparison) and evaluation results.

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3D Geometry: Shape Projection



Question: If you look at the seating arrangement from above, what shape does it form?

Options

1. A perfect square
2. A round shape
3. A rectangle
4. An irregular shape

Gold MCQ Answer: 4

Gold Open ended Answer: An irregular shape

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: Viewed from above, the seating is arranged in a U-shape (a horseshoe) around the central fire pit. The L-shaped sectional plus the two facing chairs create that U configuration.
- Gemini-2.5-Flash: The seating arrangement, when viewed from above, primarily forms an L-shape with the large sectional sofa. This L-shape is complemented by two individual chairs placed opposite the open ends of the sectional. Together, the furniture creates a U-shaped configuration around the central fire pit, designed for communal gathering.
- InternVL3.5-72B: From above, the seating arrangement forms a U-shape around the central fire pit. The sectional sofa curves around the pit, leaving an open side opposite the tall fence, creating a semi-enclosed space.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user wants to know the shape formed by the seating arrangement when viewed from above. When viewed from above, the configuration clearly forms a right-angled L-shape, combining two parallel sections at a 90-degree angle. The symmetry of the layout and the placement of furniture around the fire pit reinforce this geometric form. The seating arrangement forms a right-angled L-shape when viewed from above.

Figure 17: Some examples from the benchmark (3D Geometry: Shape Projection) and evaluation results.

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Spatial Navigation: Accessibility Constraints



Question: The notebook at the leftmost corner of the table is inaccessible because of which object?

Options

1. Chair
2. Vase
3. Table
4. Mug with pens and pencil in it

Gold MCQ Answer: 4

Gold Open ended Answer: Mug with pens and pencil in it

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** The leftmost corner notebook is blocked by a wooden chair. The chair sits directly in front of the desk, preventing easy access to the notebook.
- **Gemini-2.5-Flash:** The notebook at the leftmost corner of the table is inaccessible because of the chair. The chair is positioned directly in front of the notebook, blocking direct access to it. To reach the notebook, one would first need to move the chair out of the way.
- **InternVL3.5-72B:** The notebook at the leftmost corner of the table is inaccessible because of the chair, which is positioned in front of it. The chair's backrest and seat block direct access to the notebook.
- **Kimi-VL-A3B-Thinking-2506:** Alright, let's tackle this spatial reasoning question. So, the question is asking why the notebook at the leftmost corner of the table is inaccessible. The chair is positioned near the leftmost corner of the table, and given its placement, it's reasonable to conclude that the chair is blocking access to the notebook. The chair's presence in that space would make it difficult to reach the notebook from the left side without moving the chair first. The chair is the object that's making the notebook at the leftmost corner inaccessible.

Figure 18: Some examples from the benchmark (Spatial Navigation: Accessibility Constraints) and evaluation results.

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Spatial Navigation: Obstacle Avoidance



Question: There is a clear direct path to the end of the hallway without colliding with any object.

Options

1. TRUE
2. FALSE
3. Maybe
4. Unable to determine

Gold MCQ Answer: 1

Gold Open ended Answer: TRUE

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

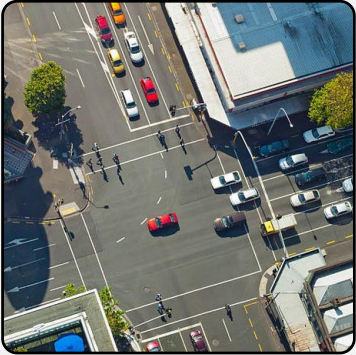
Open-ended Answers:

- **GPT-5-mini:** Yes. The central aisle is unobstructed from the foreground to the far end, with shelving stacked along both sides and no items blocking the floor. You can walk straight down the hallway without colliding with any objects.
- **Gemini-2.5-Flash:** Yes, there is a clear direct path to the end of the hallway without colliding with any object. The image shows a wide, unobstructed aisle between two rows of tall shelving units filled with boxes. The floor is clear, indicating a safe passage for movement.
- **InternVL3.5-72B:** The image shows a large warehouse with high shelving units full of boxes on both sides of a central aisle. The aisle appears to be clear and unobstructed, leading directly to the end of the hallway. There are no objects or obstacles in the pathway, ensuring a direct and clear route to the end of the aisle.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The user says there's a clear direct path to the end of the hallway without colliding with any object. A clear direct path exists because the shelving units are positioned symmetrically, leaving a central aisle free of objects. There are no visible hazards like hanging items, loose debris, or protruding elements in the visible portion of the hallway, allowing safe, uninterrupted movement to the end. The path is clear because the shelving units are symmetrically arranged, leaving the central walkway obstacle-free, with no visible hazards in the visible hallway.

Figure 19: Some examples from the benchmark (Spatial Navigation: Obstacle Avoidance) and evaluation results.

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Spatial Navigation: Pathway Existence



Question: Which walkways are the safest to cross according to the picture?

Options

1. Top and bottom
2. Top and left
3. Left and right
4. Bottom and right

Gold MCQ Answer: 1

Gold Open ended Answer: Top and bottom

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

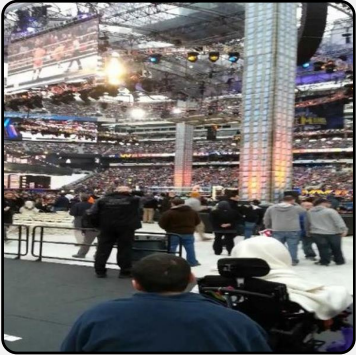
Open-ended Answers:

- **GPT-5-mini:** The marked white crosswalks at each corner of the intersection (the painted lines across the road) are the safest to use. They are signalized and where pedestrians are waiting, unlike cutting across mid-block or diagonally through the intersection.
- **Gemini-2.5-Flash:** The walkways marked with white lines, commonly known as crosswalks, are the safest to cross. These designated areas indicate where pedestrians have the right-of-way or are expected to cross the street. In the image, several people are visible using these marked crosswalks, suggesting they are the intended and therefore safest routes for pedestrians.
- **InternVL3.5-72B:** The crosswalks aligned with the pedestrian signals at the corners are the safest places to cross, as they are equipped with traffic signals ensuring vehicles are stopped. The designated pedestrian areas with painted lines across the streets are clearly marked for safety. Additionally, sidewalks provide a safe space for walking along the roads.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants to know which walkways are the safest to cross based on the image provided. The safest walkways to cross, based on the image, are the marked crosswalks equipped with pedestrian signals. Specifically, the crosswalk at the bottom-left corner of the intersection is safest because it features a visible pedestrian signal indicating walk, reducing ambiguity with traffic flow.

Figure 20: Some examples from the benchmark (Spatial Navigation: Pathway Existence) and evaluation results.

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Spatial Navigation: Viewpoint Visibility



Question: The human standing the closest (blue jacket) has clear view of the stadium, how?

Options

1. Seeing directly
2. Watching on the big screen on the left side
3. There is no stadium
4. Unable to determine

Gold MCQ Answer: 2

Gold Open ended Answer: Watching on the big screen on the left side

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- **Gemini-2.5-Flash:** The person in the blue jacket is seated in a wheelchair, which places them at a lower vantage point than the standing crowd. However, they have a clear view of the stadium because they are positioned at the very front of a section, likely an accessible seating area. This front-row placement ensures an unobstructed line of sight over the heads of any standing individuals further back.
- **InternVL3.5-72B:** The person in the blue jacket is standing near the front of the arena, which allows them a clear, unobstructed view of the stadium. Their position is close to the barrier separating the audience from the performance area, providing a direct line of sight towards the stage and the large screens overhead. This proximity minimizes visual obstructions from other spectators or structures.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking how the person in the blue jacket (who is standing the closest) has a clear view of the stadium. The person's elevated seating, minimized obstructions from nearby spectators, and potential alignment with stadium features (e.g., screens or lighting) enable their clear view despite being relatively close to the ground.

Figure 21: Some examples from the benchmark (Spatial Navigation: Viewpoint Visibility) and evaluation results.

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Spatial Navigation: Spatial Sequence



Question: Which path is the easiest for the human sitting in the bench(reading paper) to follow to get on the train?

Options

1. Go straight to the first red colored door
2. To make a turn and go to the second floor
3. Go through the door at the end of the train
4. No viable option

Gold MCQ Answer: 1

Gold Open ended Answer: Go straight to the first red colored door

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- **Gemini-2.5-Flash:** The simplest way for the person to board the train is to rise from the bench, turn right, and walk a short distance along the platform. They would then cross the yellow safety line and step through one of the train doors. This route is straightforward and involves minimal effort.
- **InternVL3.5-72B:** The person sitting on the bench can follow a straightforward path to board the train by moving forward along the platform, parallel to the train tracks, while keeping to the right of the yellow safety line. This path will lead them directly to the nearest open train door. They should avoid crossing the yellow line until they are ready to enter the train to ensure safety.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about determining the easiest path for the person sitting on the bench to get on the train. First, I need to look at the image provided. The person is sitting on the bench which is alongside the platform walls. The train is on the opposite side. To get to the train, the person would..

Figure 22: Some examples from the benchmark (Spatial Navigation: Spatial Sequence) and evaluation results.

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Size and Scale: Scale Consistency



Question: With the electric pole's capacity can all this wires be redesigned to carry properly in aligned structured design?

Options

1. The size of wires are too big
2. The wires need to be move into another pole
3. Wires can be set there following a layout
4. No, the bulk size of the wires are too much here for one pole

Gold MCQ Answer: 4

Gold Open ended Answer: No, the bulk size of the wires are too much here for one pole

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** Not safely in its current state — the pole is clearly overloaded and the wires are tangled, so simply re-bundling them on the same pole would exceed mechanical, clearance and safety limits. A proper redesign is possible but requires removing redundant lines, separating circuits (high-voltage vs. telecom) onto organized crossarms, trays or underground ducts, and likely installing stronger or additional poles and anchors.
- **Gemini-2.5-Flash:** While it's technically possible to redesign the wiring for better alignment, the sheer volume and entanglement of wires in the image suggest that the current electric pole's capacity is severely overtaxed. Simply reorganizing the existing wires on the same pole would not address the underlying capacity issues or the safety hazards presented by such a dense and disorganized network.
- **InternVL3.5-72B:** Redesigning the extensive web of wires around the electric pole into a properly aligned and structured design is feasible but would likely require significant effort and resources. Proper planning and coordination with multiple service providers would be essential. Implementing such changes would enhance safety and aesthetics, but it involves logistical challenges and coordination to minimize service disruptions.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking if all the wires on the electric pole can be redesigned to be properly aligned in a structured way. A redesigned system is feasible but dependent on evaluating regulatory, technical, and safety constraints. Prioritizing organized routing for essential cables while removing redundant ones could create.

Figure 23: Some examples from the benchmark (Size and Scale: Scale Consistency) and evaluation results.

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Size and Scale: Shadow-Size Projection



Question: If one light is rotated so that they are perpendicular to each other, how will the shadow look over the wall?

Options

1. It will cover almost all of the wall
2. The shadow will only take the size of the light frame as it's actual size
3. There won't be any shadow
4. Shadow will smaller than the actual light body

Gold MCQ Answer: 1

Gold Open ended Answer: It will cover almost all of the wall

MCQ Answers:

- **GPT-5-mini:** 4
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** If one lamp is rotated 90° so the two beams are perpendicular, you'll get two elongated oval patches of light on the wall at right angles to each other. Where the ovals overlap it will be brightest; outside them the wall will be darker, producing an L-shaped darker region and two perpendicular shadow lobes cast by the fixture.
- **Gemini-2.5-Flash:** If one light is rotated to be perpendicular to the other, the shadow cast on the wall would change significantly. The current shadows are likely elongated and somewhat parallel due to the lights' current orientation. With one light rotated, the shadows would intersect at a more acute angle, creating a more complex and possibly darker overlapping shadow where the light beams cross.
- **InternVL3.5-72B:** If one light is rotated so that the two lights are perpendicular to each other, the shadows on the wall will intersect at a right angle, creating intersecting patterns. The light beams will cast shadows in perpendicular directions, resulting in a cross-like effect where the shadows overlap. The intensity and shape of the shadows will depend on the orientation and angle of the lights relative to the wall.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's try to figure out this question. The user is asking about rotating a light so that they're perpendicular to each other and how the shadow on the wall would look. If they're further away, the shadow would be smaller.

Figure 24: Some examples from the benchmark (Size and Scale: Shadow-Size Projection) and evaluation results.

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Size and Scale: Perspective Distortion



Question: If similar shaped pipes are stacked together separately will it take up less space?

Options

1. Yes
2. No
3. Take more
4. Similar

Gold MCQ Answer: 1

Gold Open ended Answer: Yes

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** Yes - when like-shaped pipes are grouped you can pack them in their most efficient pattern ,which reduces wasted voids and the total footprint. The exact savings depend on orientation and dimensions; in some cases clever mixing or nesting of different shapes can beat separate stacks, but as a general rule same-shape grouping is more space-efficient.
- **Gemini-2.5-Flash:** Yes, if similarly shaped pipes are stacked together separately, it would likely take up less space. This is because organizing them by shape allows for more efficient packing and reduces wasted space between different forms. For example, square pipes can be stacked flush against each other, and circular pipes can be nested or arranged in a hexagonal pattern for optimal density.
- **InternVL3.5-72B:** Yes, stacking similar shaped pipes together separately will generally take up less space. Organizing them by shape allows for more efficient packing, reducing gaps and empty spaces between different sizes and types. This method optimizes the use of available storage area and improves accessibility.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is asking if stacking similar shaped pipes separately would take up less space. Stacking similar-shaped pipes separately can potentially optimize space usage. When pipes of the same diameter and shape (e.g., all round or rectangular) are grouped together, they can be arranged more tightly, minimizing gaps between

Figure 25: Some examples from the benchmark (Size and Scale: Perspective Distortion) and evaluation results.

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Size and Scale: Relative Size Comparison



Question: The ash-colored, thicker coiled spring will be surpassed by which coiled springs if all are kept in the standing position?

Options

1. Right bottom corner
2. Right top corner
3. Left top corner
4. Bottom mid spring

Gold MCQ Answer: 1

Gold Open ended Answer: Right bottom corner

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: The ash-colored thicker spring in the center will be shorter than the thin, taller silver spring just behind it (top-center) and the long, dark-gray spring at the right edge when all are stood upright. Those two coils extend above the ash-colored spring.
- Gemini-2.5-Flash: The ash-colored, thicker coiled spring, located near the center-right of the image, would be surpassed by the longer, thinner coiled spring positioned in the bottom right corner. Additionally, the two smaller, conical springs in the upper left quadrant would also be taller if fully extended and standing. These springs vary in height and coil density, indicating different potential applications.
- InternVL3.5-72B: The ash-colored, thicker coiled spring in the image will be surpassed in height by the two taller springs on its right. One of these taller springs is also ash-colored, while the other has a slightly darker hue. Both of these springs have more coils and are visibly longer when standing upright.
- Kimi-VL-A3B-Thinking-2506: Okay, let's try to figure out this question. The image shows a bunch of different coiled springs. The question is asking which coiled springs would be surpassed by the ash-colored, thicker one if all are kept in the standing position The thinner, tightly coiled springs (smaller in diameter or fewer coils) would be surpassed in height by the ash-colored, thicker spring when all are kept upright.

Figure 26: Some examples from the benchmark (Size and Scale: Relative Size Comparison) and evaluation results.

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Size and Scale: Relative Size Comparison



Question: If the red bow is taken out of it's place and kept far away in the behind how will it look and what will be it's capacity comparing with it's other similar clips at front?

Options

1. Large, same
2. Same, less
3. Small, same
4. Small, large

Gold MCQ Answer: 3

Gold Open ended Answer: Small, same

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 2
- **InternVL3.5-72B:** 2
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** If you move the red bow to the back it will appear smaller and more obscured by the bows in front because of perspective and overlap, and its red color will look less vivid from that angle. Physically nothing about the bow changes — its size, shape and clasp strength remain the same as the other similar clips in front, so its holding capacity is unchanged.
- **Gemini-2.5-Flash:** If the red bow were taken out and placed far behind, it would appear smaller and less distinct due to perspective. Its capacity, referring to its visual prominence or ability to hold hair, would remain the same as the other similar clips. However, its perceived capacity might diminish as it would no longer be in the foreground, drawing less attention.
- **InternVL3.5-72B:** If the red bow is moved to the back, it will be partially obscured by the other bows in the foreground, making it less prominent. Its size and capacity to hold hair will remain the same as the others, but visually, it won't stand out as much as it does currently when placed in the front.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about removing the red bow and placing it behind, then comparing its appearance and capacity to the other similar clips in front. Removing the red bow would reduce its central visual impact, making the remaining front-facing bows appear denser but less balanced. Its capacity (physical or visual space) in the front area would decrease, potentially limiting the number of similar bows that can be displayed

Figure 27: Some examples from the benchmark (Size and Scale: Relative Size Comparison) and evaluation results.

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Depth & Occlusion : Complete Occlusion Inference



Question: What is creating the striped shadow on the wall?

Options

1. The sunflower
2. Blinds
3. Window
4. Not applicable

Gold MCQ Answer: 2

Gold Open ended Answer: Blinds

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- **GPT-5-mini:** Sunlight streaming through window blinds (slatted shades) is casting the diagonal striped shadow on the wall. The slats break the light into parallel bands, producing the striped pattern.
- **Gemini-2.5-Flash:** The striped shadow on the wall is created by sunlight passing through a window with horizontal blinds. The blinds block portions of the light, casting distinct parallel lines onto the wall. This effect is common when direct sunlight enters a room through slatted window coverings.
- **InternVL3.5-72B:** The striped shadow on the wall is created by blinds or window slats. The sunlight is passing through the gaps between the horizontal blinds, casting parallel lines of light and shadow. This pattern is often seen when light is filtered through window coverings with narrow openings. The alignment and spacing of the lines suggest that the blinds are partially open, allowing the sun to project the shadow pattern onto the wall.
- **Kimi-VL-A3B-Thinking-2506:** The striped shadow on the wall is created by sunlight passing through a set of window blinds or shutters with narrow, evenly spaced slats. The parallel lines of the shadow result from the angular interaction between the light source (likely the sun) and the rigid, horizontal slats of the blinds, casting consistent geometric patterns onto the wall. The uniformity and sharpness of the shadows suggest the light is direct and possibly filtered through the gaps between the slats.

Figure 28: Some examples from the benchmark (Depth and Occlusion: Complete Occlusion Inference) and evaluation results.

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Depth & Occlusion : Layering Order



Question: What is the frontmost object in the picture?

Options

1. Trolley
2. Box
3. Man
4. Woman

Gold MCQ Answer: 1

Gold Open ended Answer: Trolley

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 29: Some examples from the benchmark (Depth and Occlusion: Layering Order) and evaluation results.

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Depth & Occlusion : Partial Occlusion



Question: What is the color of the ball behind the white colored square shaped device?

Options

1. Cyan
2. Yellow
3. Red
4. Not applicable

Gold MCQ Answer: 1

Gold Open ended Answer: Cyan

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The ball directly behind the white, square-shaped device is blue (a cyan/teal colored ball). It sits slightly to the left and behind the device.
- Gemini-2.5-Flash: The ball directly behind the white-colored square-shaped device (GameBoy) is teal. There is also a yellow-green ball further to the right of the GameBoy.
- InternVL3.5-72B: The ball behind the white, square shaped device is blue.
- Kimi-VL-A3B-Thinking-2506: The color of the ball behind the white square-shaped device is blue.

Figure 30: Some examples from the benchmark (Depth and Occlusion: Partial Occlusion) and evaluation results.

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Depth & Occlusion : Reflective Surfaces



Question: Which object is the shiniest among all the objects in the image?

Options

1. Kettle
2. Tea glass
3. Table clock
4. Lamp chimney

Gold MCQ Answer: 4

Gold Open ended Answer: Lamp chimney

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 31: Some examples from the benchmark (Depth and Occlusion: Reflective Surfaces) and evaluation results.

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Depth & Occlusion : Transparency Effects



Question: How many objects has filaments inside them in the image?

Options

1. 7
2. 6
3. 5
4. 4

Gold MCQ Answer: 4

Gold Open ended Answer: 4

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: Three. The glowing bulb being held clearly shows a filament, and two more filament bulbs are visible inside the round box on the left.
- Gemini-2.5-Flash: There are three objects with filaments inside them. One is the illuminated light bulb being held, and the other two are unlit light bulbs inside the "Sunny Day Kit" box. All three are incandescent light bulbs, which contain a filament that glows when heated.
- InternVL3.5-72B: In the image, there are four objects with visible filaments. These include the lit light bulb being held, and three unlit bulbs inside the container labeled "Sunny Day Kit."
- Kimi-VL-A3B-Thinking-2506: There are 3 objects with filaments in the image - one held in hand and two in the "Sunny Day Kit" container.

Figure 32: Some examples from the benchmark (Depth and Occlusion: Transparency Effects) and evaluation results.

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Orientation : Cardinal Direction



Question: If you go straight following route 6, which direction are you traveling?

Options

1. North
2. South
3. East
4. West

Gold MCQ Answer: 3

Gold Open ended Answer: East

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: You would be traveling east. The Route 6 shield is labeled "EAST" and the straight-ahead arrow beneath it shows that going straight follows eastbound Route 6.
- Gemini-2.5-Flash: If you go straight following route 6, you will be traveling East. The sign for Route 6 is directly below the "EAST" sign, and there is an arrow pointing straight up beneath the Route 6 shield, indicating that going straight leads to Route 6 East
- InternVL3.5-72B: If you go straight following Route 6, you are traveling east. This is indicated by the sign with the number "6" and the arrow pointing forward.
- Kimi-VL-A3B-Thinking-2506: Standard highway signs usually have the path of the route. So if it's route 11 East, the arrow would indicate the direction. But in the image, the left arrow is under the NORTH 11 EAST sign. Maybe that's indicating that route 11 East goes left? But the sign says East. Alternatively, maybe the arrows are indicating the direction of the route numbers. Like, route 11 North is going straight, route 11 East is turning left, etc. But then the route 6 sign has an upward arrow. So if you take route 6 straight, which direction is that? The arrow on the route 6 shield is pointing straight up. But the Question: is about direction—like North, South, East, West. Wait, the signs here are for routes, not cardinal directions. So maybe the answer is that going straight on route 6 would be in the direction indicated by the upward arrow. But the upward arrow might be part of the route 6 sign.

Figure 33: Some examples from the benchmark (Orientation: Cardinal Direction) and evaluation results.

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Orientation : Facing Direction



Question: Suppose, Rabindranath Tagore's portrait is hung on the left wall and it faces directly into the room, what is the direction of the portrait with reference to the wall?

Options

- Up
- Down
- Away
- Toward

Gold MCQ Answer: 3

Gold Open ended Answer: Away

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: If the portrait is on the left wall and faces directly into the room, it is oriented perpendicular to that wall and pointing inward. In other words, it faces away from the wall toward the center of the room.
- Gemini-2.5-Flash: The portrait of Rabindranath Tagore is located on the wall to the left, as seen in the image. If it faces directly into the room, its direction with reference to the wall would be perpendicular to the wall. This means the portrait is facing straight out from the wall into the space of the room.
- InternVL3.5-72B: If Rabindranath Tagore's portrait is hung on the left wall and faces directly into the room, it would be positioned perpendicular to the wall. This means the portrait is oriented so that viewers inside the room can face it directly when looking toward the wall on which it is hung.
- Kimi-VL-A3B-Thinking-2506: The portrait is hung on the left wall and faces directly into the room, meaning it is oriented inward toward the interior space.

Figure 34: Some examples from the benchmark (Orientation: Facing Direction) and evaluation results.

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Orientation : Object Rotation



Question: What is the approximate angle of the cases of the pocket watch?

Options

- 90
- 45
- 180
- 120

Gold MCQ Answer: 3

Gold Open ended Answer: 180

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: About 180°. The two covers are opened in opposite directions and lie nearly flat, forming an almost straight line.
- Gemini-2.5-Flash: The cases of the pocket watch are open, revealing the intricate mechanism inside. The front cover is open to approximately a 120-degree angle relative to the watch face. The back case is also open, but its angle is less clear, appearing to be around 90 degrees or slightly more.
- InternVL3.5-72B: The pocket watch in the image appears to be partially open, with the angle between its two cases estimated to be around 60 degrees. The case opens hinge-wise, and the visible gap suggests that it is made to display the internal mechanical components of the watch. This angle provides a good view of the watch face and some of its intricate gears..
- Kimi-VL-A3B-Thinking-2506: When the watch is open, the cases form an angle. If the covers are both open to 180 degrees relative to each other, that would be a straight line. But in this image, the back cover isn't fully open. It looks like it's at around 90 degrees from the front cover. So the angle between the cases would be approximately 90 degrees.

Figure 35: Some examples from the benchmark (Orientation: Object Rotation) and evaluation results.

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Orientation : Stacking Orientation



Question: What is the relationship between the road and the other objects in the image?

Options

1. Behind the building and plants
2. In front of the plants and leads towards the building
3. The road is beside the building, with no clear direction
4. The road is stacked above the building and plants

Gold MCQ Answer: 2

Gold Open ended Answer: In front of the plants and leads towards the building

MCQ Answers:

- **GPT-5-mini:** 2
- **Gemini-2.5-Flash:** 2
- **InternVL3.5-72B:** 2
- **Kimi-VL-A3B-Thinking-2506:** 2

Open-ended Answers:

- **GPT-5-mini:** The road runs in front of the large white building, curving around a central planted island and seating area. It separates and encircles the entrance and decorative planters, forming a driveway/roundabout in the courtyard.
- **Gemini-2.5-Flash:** The road in the image serves as an access path, leading towards the grand white building. It curves around a central island feature, which contains a decorative fountain and some potted plants. This suggests the road is a driveway or entrance loop designed to facilitate vehicle access to the building.
- **InternVL3.5-72B:** The road in the image forms a semi-circular driveway in front of a large white building with columns. It encircles a landscaped garden area with plants, a stone sculpture, and a fountain at the center, leading up to the building's entrance. The road is bordered by a curb that separates it from the garden and pedestrian space. Lamp posts are situated alongside the road, enhancing the building's stately appearance.
- **Kimi-VL-A3B-Thinking-2506:** The road curves around the front of the grand white building, acting as a transitional pathway that guides movement and attention toward the entrance while dividing space between the main structure and other elements like the pergola. It establishes scale and perspective while integrating with the surrounding landscape.

Figure 36: Some examples from the benchmark (Orientation: Stacking Orientation) and evaluation results.

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Orientation : Tool Handedness



Question: Considering the likely process of creating the mural, which tool's handedness would be most evident in the smooth, curved lines of the bengali script?

Options

1. Paint roller
2. Paint brush
3. Tape measure
4. Stencil

Gold MCQ Answer: 2

Gold Open ended Answer: Paint brush

MCQ Answers:

- **GPT-5-mini:** 2
- **Gemini-2.5-Flash:** 2
- **InternVL3.5-72B:** 2
- **Kimi-VL-A3B-Thinking-2506:** 2

Open-ended Answers:

- **GPT-5-mini:** Those flowing, tapered curves are most characteristic of a paintbrush (or broad paint marker) rather than a spray can. The handedness of the brush user — most likely right-handed — would be most evident, since right-handed strokes tend to pull and taper in predictable left-to-right sweeps visible in the Bengali script.
- **Gemini-2.5-Flash:** The smooth, curved lines of the Bengali script suggest the use of a brush, likely a fine-tipped one, to achieve such fluidity. The artist's handedness would be most evident in the consistent direction and pressure applied to the brushstrokes.
- **InternVL3.5-72B:** In creating the mural, the smooth, curved lines of the Bengali script would most likely show the artist's handedness, particularly if the lines consistently curve more naturally in one direction. If the curves are more fluid and precise on one side, it could suggest that the artist is either right-handed or left-handed, depending on the direction of the dominant curves.
- **Kimi-VL-A3B-Thinking-2506:** If the artist is left-handed, they might use the brush in their left hand, leading to strokes that curve more naturally. Alternatively, the tool itself (like a specific brush tip) might be designed for left or right handed use.

Figure 37: Some examples from the benchmark (Orientation: Tool Handedness) and evaluation results.

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Relative Positioning : Alignment Patterns



Question: Which of the objects is the closest to the centerline of the image?

Options

1. Rice
2. Nut
3. Avocado
4. Berries

Gold MCQ Answer: 4

Gold Open ended Answer: Berries

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 3

Open-ended Answers:

- GPT-5-mini: The red bowl of berries (raspberries and blueberries) is closest to the image's centerline. It sits almost directly in the middle of the composition, slightly left of the exact center..
- Gemini-2.5-Flash: The red bowl containing raspberries and blueberries is the closest object to the centerline of the image. Its position is almost perfectly centered horizontally, and it is located in the upper-middle section of the image vertically. While other objects like the avocado halves are also quite central, the red bowl's overall placement makes it the most aligned with the image's central axis.
- InternVL3.5-72B: In the image, the avocado halves are the closest to the centerline. One avocado is placed near the top center, and the other is slightly below it to the right, straddling the vertical centerline of the image.
- Kimi-VL-A3B-Thinking-2506: The bowl of mixed berries is closest to the centerline.

Figure 38: Some examples from the benchmark (Relative Positioning: Alignment Patterns) and evaluation results.

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Relative Positioning : Betweenness Relationships



Question: What is the object between the building and the street lamp?

Options

1. A monument of various statues
2. Statue
3. Tree
4. Humans

Gold MCQ Answer: 1

Gold Open ended Answer: A monument of various statues

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** Between the building and the street lamp stands a stone statue of a person. It appears to be one of the sculpted figures that are part of the fountain in the foreground.
- **Gemini-2.5-Flash:** The object located between the building and the street lamp is a large, ornate monument. This monument features multiple levels adorned with various statues and intricate carvings. At its very top, there is a prominent statue of a person.
- **InternVL3.5-72B:** The object between the building and the street lamp is a detailed stone monument or fountain. It features several statues, including a prominent figure standing on top and additional figures around its structure. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.
- **Kimi-VL-A3B-Thinking-2506:** The object between the building and the street lamp is a detailed stone monument or fountain. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.

Figure 39: Some examples from the benchmark (Relative Positioning: Betweenness Relationships) and evaluation results.

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Relative Positioning : Corner/Angle Positioning



Question: What is in the top left corner of the plant?

Options

1. A lobster
2. A crab
3. An octopus
4. Nothing

Gold MCQ Answer: 2

Gold Open ended Answer: A crab

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: B

Open-ended Answers:

- GPT-5-mini: A crab is in the top left corner of the plant, perched on the rim of the fishbowl. It's hanging over the water with its claws visible.
- Gemini-2.5-Flash: The plant in the image is a type of seaweed or coral, and it is contained within a glass bowl. In the top left corner of this plant, there is a branch of the plant extending upwards and slightly to the left. This branch is a dark, greenish-brown color.
- InternVL3.5-72B: In the top left corner of the plant, there is a small crab that is perched on edge of the bowl, outside the water. It is gripping the rim with its legs, and its body is partially inside the bowl. The background shows the open water, providing contrast to the scene.
- Kimi-VL-A3B-Thinking-2506: The answer is the crab

Figure 40: Some examples from the benchmark (Relative Positioning: Corner/Angle Positioning) and evaluation results.

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Relative Positioning : Directional Relations



Question: What is located directly behind the cheetah?

Options

1. Person
2. Tree
3. Safari vehicle
4. Cheetah

Gold MCQ Answer: 3

Gold Open ended Answer: Saffari vehicle

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Directly behind the cheetah is a white open-sided tour vehicle or tram. Several people are seated inside the tram, watching and taking photos.
- Gemini-2.5-Flash: Directly behind the cheetah, there is a white open-sided vehicle, likely a safari or tour cart. Inside this vehicle, several people are seated, observing the cheetah. Some of these individuals are holding up cameras or phones, seemingly taking pictures or videos.
- InternVL3.5-72B: Directly behind the cheetah, there is a group of people sitting and standing around, appearing to watch or interact with the cheetah. They are positioned near a small covered vehicle, which looks like a safari cart or shuttle. The area is a grassy patch, and there are trees and a hill in the further background.
- Kimi-VL-A3B-Thinking-2506: People seated on benches under a canopy are located directly behind the cheetah.

Figure 41: Some examples from the benchmark (Relative Positioning: Directional Relations) and evaluation results.

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Relative Positioning : Proximity Gradients

Question: Which book is closer to the book "american marriage" - "atonement" or "cloud atlas"?

Options

1. Cloud Atlas
2. Atonement
3. Crossed
4. nan

Gold MCQ Answer: 2

Gold Open ended Answer: Atonement

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- **GPT-5-mini:** "Atonement" is closer to "american marriage" than "Cloud Atlas." "Atonement" sits just to the right with only one small book between them, while "Cloud Atlas" is further to the right on the shelf.
- **Gemini-2.5-Flash:** "Atonement" is closer to "An American Marriage" than "Cloud Atlas." "Atonement" is directly adjacent to "An American Marriage" on its right side. "Cloud Atlas" is further to the right, with "Americanah" separating it from "Atonement."
- **InternVL3.5-72B:** The book "atonement" is closer to "american marriage" than "cloud atlas." In the image, "atonement" is directly next to "american marriage," while "cloud atlas" is located further to the left..
- **Kimi-VL-A3B-Thinking-2506:** Cloud Atlas is closer to "American Marriage" than "Atonement."

Figure 42: Some examples from the benchmark (Relative Positioning: Proximity Gradients) and evaluation results.

K STATISTICAL ROBUSTNESS AND DATASET STABILITY

To establish that the benchmark results are not artifacts of random initialization or dataset noise, we quantify both model robustness across runs and dataset stability across repeated trials. Such analyses are especially consequential for small- to medium-scale benchmarks, where minor fluctuations can otherwise propagate into misleading fine-grained conclusions (Drummond, 2009; Drummond & Japkowicz, 2010).

Let N denote the number of items *questions* in the benchmark and $R = 3$ the number of independent runs of the same model. Each run uses a distinct random seed while preserving all other hyperparameters.

K.1 MODEL ROBUSTNESS

For checking model evaluation robustness, we perform several statistical studies, as described below.

K.1.1 MULTIPLE RUN AVERAGES AND DEVIATIONS

Let $a_{i,r} \in 0, 1$ denote the correctness of item $i \in 1, \dots, N$ in run $r \in 1, 2, 3$. The per-item average accuracy is:

$$\bar{a} * i = \frac{1}{R} \sum_{r=1}^R a_{i,r} \quad (1)$$

and its corresponding standard deviation is:

$$\sigma_i = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (a_{i,r} - \bar{a}_i)^2} \quad (2)$$

The overall mean accuracy per run is:

$$A_r = \frac{1}{N} \sum_{i=1}^N a_{i,r} \quad (3)$$

and its standard deviation across runs is:

$$\sigma_A = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_r - \bar{A})^2} \quad (4)$$

where $\bar{A} = \frac{1}{R} \sum_{r=1}^R A_r$ is the overall mean accuracy across all runs.

K.1.2 INTRA-CLASS CORRELATION (ICC)

To quantify agreement between runs, we compute the Intra-Class Correlation coefficient (Shrout & Fleiss, 1979), using a two-way mixed-effects model:

$$\text{ICC}(3, k) = \frac{\sigma_{\text{between}}^2}{\sigma_{\text{between}}^2 + \frac{\sigma_{\text{within}}^2}{k}} \quad (5)$$

where $\sigma_{\text{between}}^2$ is the variance between items and σ_{within}^2 is the residual variance across runs. ICC values above 0.75 indicate good reliability, while values above 0.9 imply excellent consistency (Koo & Li, 2016).

K.2 DATASET STABILITY AND INTERNAL CONSISTENCY

We also analyze the stability of the dataset itself, whether question difficulty remains consistent across runs.

K.2.1 RESAMPLING STUDY

To further assess **robustness**, we perform a resampling analysis. From the total of $C = 30$ subcategories, we randomly sample subsets of size $S \in 20, 25$ for each sub-category from each of the $R = 3$ runs, across both models and evaluation types (MCQ and open-ended).

For each run r and subset size S , the accuracy is computed as:

$$A_{r,S} = \frac{1}{S} \sum_{i=1}^S a_{i,r,S}, \quad (6)$$

where $a_{i,r,S}$ denotes the accuracy for subcategory i in run r under subset size S .

The mean accuracy across runs for each subset size is:

$$\bar{A} * S = \frac{1}{R} \sum_{r=1}^R A_{r,S}, \quad (7)$$

and the corresponding standard deviation:

$$\sigma_S = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_{r,S} - \bar{A}_S)^2}. \quad (8)$$

As a complementary check, the Wilcoxon signed-rank test assesses median differences between paired samples. Let $d_r = A_{r,20} - A_{r,25}$ be the difference in accuracies for run r , excluding ties ($d_r \neq 0$). The test statistic is computed as:

$$W = \min(W^+, W^-), \quad (9)$$

where W^+ and W^- are the sums of signed ranks of positive and negative differences, respectively. The standardized z -score is given by:

$$z = \frac{W - \frac{n(n+1)}{4}}{\sqrt{\frac{n(n+1)(2n+1)}{24}}}, \quad (10)$$

where n is the number of nonzero pairs.

If the resulting p -value satisfies $p > 0.05$, we again fail to reject H_0 , confirming that the accuracies from 20- and 25-sample subsets are statistically indistinguishable.

K.2.2 ITEM-LEVEL CONSISTENCY

For each item i , its difficulty can be defined as $d_i = 1 - \bar{a} * i$. If item difficulty is consistent, the correlation between runs should be high:

$$\rho * r_1, r_2 = \frac{\text{Cov}(a_{\cdot, r_1}, a_{\cdot, r_2})}{\sigma_{r_1} \sigma_{r_2}} \quad (11)$$

A high mean pairwise correlation ($\bar{\rho} > 0.8$) suggests stable item difficulty and reliable dataset behavior.

K.2.3 CRONBACH'S ALPHA

To measure overall internal consistency, we compute Cronbach's alpha (Cronbach, 1951):

$$\alpha = \frac{R}{R-1} \left(1 - \frac{\sum_{r=1}^R s_r^2}{s_T^2} \right) \quad (12)$$

where s_r^2 is the variance of run r and s_T^2 is the variance of the total scores aggregated across runs. Values of $\alpha \geq 0.9$ indicate excellent internal consistency (Tavakol & Dennick, 2011).

K.3 EMPIRICAL RESULTS

We perform the analysis on two models: one large proprietary model, **Gemini-2.5-Flash**, and one small open-source model, **Qwen-2.5-vl-7b-Instruct**, thereby covering both large- and small-scale models as well as proprietary and open-source paradigms. We also conduct all tests in both multiple-choice (MCQ) and open-ended evaluation formats (treated as binary: correct or incorrect). For statistical analysis, we used the following Python packages and functions: `scipy.stats`² (Virtanen et al., 2020) for `wilcoxon` test; `pingouin`³ (Vallat, 2018) for `intraclass correlation` and `cronbach alpha`; and `numpy`⁴’s (Harris et al., 2020) `numpy.corrcoef` for correlations.

K.3.1 MODEL EVALUATION ROBUSTNESS

Multiple Run Averages and Deviations

We analyzed the stability of model accuracy across $R = 3$ independent runs. The standard deviation of accuracy (σ_A) across runs is negligible ($< 0.6\%$), indicating that the reported scores are precise estimates unaffected by random seeding noise.

Table 26: Model Accuracy and Variance across 3 Independent Runs.

Model	Format	Mean Accuracy ($\bar{A}$)	Std Dev (σ_A)
Qwen-2.5-vl-7b	MCQ	0.4079	0.0021
Gemini-2.5-Flash	MCQ	0.5193	0.0054
Qwen-2.5-vl-7b	Open	0.1936	0.0047
Gemini-2.5-Flash	Open	0.3124	0.0042

The uniformly low standard deviations across models and evaluation formats indicate that the performance estimates are stable and reproducible, with minimal sensitivity to stochastic run-to-run variation. This stability supports reading the observed accuracy differences as reflections of underlying model capability rather than as noise introduced by random initialization or decoding. Although open-ended evaluation exhibits marginally higher variance than MCQ, its absolute magnitude remains negligible, so free-form generation is assessed no less stably than the discriminative format. Because dataset size and scoring methodology together suffice to yield convergent estimates, single-run evaluation is adequate for large-scale comparison, and multi-run analysis serves primarily to corroborate that robustness. The net effect is to strengthen the empirical credibility of the reported results and to license cross-model and cross-format comparison with correspondingly greater confidence.

Intra-Class Correlation (ICC)

We computed ICC(3,k) to quantify the reliability of the scoring mechanism.

Table 27: Intra-Class Correlation (ICC) Reliability Scores.

Model	Format	ICC(3,k)	Interpretation
Qwen-2.5-vl-7b	MCQ	0.988	Excellent
Gemini-2.5-Flash	MCQ	0.990	Excellent
Qwen-2.5-vl-7b	Open	0.983	Excellent
Gemini-2.5-Flash	Open	0.981	Excellent

The uniformly high ICC(3,k) scores show that the evaluation exhibits near-perfect reliability across independent runs, irrespective of model size or response format. Item-level difficulty ordering is thus preserved, and run-to-run fluctuations do not perturb the relative ranking of samples. This degree of agreement indicates that the scoring pipeline is internally consistent and largely insensitive to stochastic variation in model outputs, which in turn discounts random guessing as a dominant factor even in open-ended generation, where variability is ordinarily higher. For evaluation purposes, the benchmark therefore furnishes a dependable signal of model capability, and longitudinal comparisons and fine-grained ablations can be conducted with corresponding confidence in measurement stability.

²<https://scipy.org/>

³<https://pingouin-stats.org/build/html/index.html>

⁴<https://numpy.org/>

Taken together, the multiple-run variance analysis and the ICC reliability estimates converge on the same conclusion: the evaluation is robust to stochastic effects along both axes. Absolute performance (low standard deviation) and relative performance (near-perfect ICC) remain stable across independent runs, so the reported results cannot be attributed to random seeding or decoding noise. This robustness is what allows the observed performance gaps across models and formats to be read as genuine capability differences, and it establishes the evaluation framework as a reliable, reproducible basis for comparative benchmarking.

K.3.2 DATASET STABILITY AND INTERNAL CONSISTENCY

Resampling Study To ensure statistical reliability, this resampling procedure is repeated over **1,000 independent trials**, each time drawing new random subsets of size $S = 20$ and $S = 25$ from the available subcategories. Repeating the test at this scale stabilizes the distribution of accuracy differences and p -values, ensuring that conclusions are not driven by a particular random draw. The consistent failure to reject H_0 across all trials therefore provides strong evidence that the observed robustness is systematic rather than incidental.

Table 28: Resampling Study (1,000 Trials): Wilcoxon Signed-Rank Test ($S = 20$ vs. $S = 25$).

Model	Format	Mean p -value	% Trials $p > 0.05$	Conclusion
Qwen-2.5-v1-7b	MCQ	0.290	100%	No Sig. Diff.
Gemini-2.5-Flash	MCQ	0.285	100%	No Sig. Diff.
Qwen-2.5-v1-7b	Open	0.294	100%	No Sig. Diff.
Gemini-2.5-Flash	Open	0.294	100%	No Sig. Diff.

The resampling results provide strong empirical evidence that accuracy estimates have already converged at comparatively small subcategory sizes. The uniformly high mean p -values and the absence of any statistically significant difference across all 1,000 trials show that the performance metrics are insensitive to moderate reductions in sample size, and this insensitivity holds across models and evaluation formats alike, so neither model scale nor output structure introduces additional sampling uncertainty. The variance in accuracy is therefore governed by model behavior rather than by data sparsity at the subcategory level. It follows that the current subcategory size (≈ 50) is statistically comfortable—affording a substantial margin beyond the minimum required for reliable estimation—which both corroborates the benchmark design and indicates that future extensions could reallocate samples across categories without compromising statistical robustness.

Item-Level Consistency and Cronbach’s Alpha We measured the mean pairwise Pearson correlation ($\bar{\rho}$) between runs and Cronbach’s Alpha to assess internal consistency.

Table 29: Item-Level Consistency Metrics.

Model	Format	Mean Pairwise $\bar{\rho}$	Cronbach’s α
Qwen-2.5-v1-7b	MCQ	0.966	0.988
Gemini-2.5-Flash	MCQ	0.972	0.990
Qwen-2.5-v1-7b	Open	0.950	0.983
Gemini-2.5-Flash	Open	0.946	0.981

The very high mean pairwise correlations (> 0.94) evidence strong item-level stability: the relative difficulty of individual questions is preserved across independent runs, so model responses are systematically structured rather than randomly variable even under stochastic decoding. The commensurately high Cronbach’s α values (> 0.9) confirm excellent internal coherence, with items contributing reliably to the aggregate accuracy measure. The slight reductions observed under open-ended evaluation are expected given its greater expressive latitude, yet they remain comfortably within the range of excellent reliability. The benchmark thus behaves as a unified, internally consistent measurement instrument, which is what warrants attributing model performance differences to genuine capability variation rather than to instability at the item level.

4968 K.3.3 OVERALL DISCUSSION

4969
4970 Taken together, our expanded statistical testing establishes that both the evaluation protocol and the
4971 dataset are robust along every axis examined.

4972 The three lines of evidence reinforce one another. **ICC and Cronbach's α scores consistently**
4973 **exceed 0.98**, indicating near-perfect reliability and internal consistency; the **standard deviation**
4974 **of accuracy across runs remains negligible** ($< 0.6\%$), confirming that reported performance is
4975 stable rather than driven by stochastic effects such as random seeding or decoding variability; and the
4976 **resampling study** demonstrates that even substantially smaller subsets ($N = 20$ and $N = 25$) yield
4977 accuracy estimates statistically indistinguishable from the full set, placing the chosen subcategory
4978 size (~ 25) well beyond the threshold required for convergence. Because both absolute scores and
4979 relative rankings remain stable across runs and sampling regimes, the fine-grained comparisons across
4980 models, formats, and subcategories are meaningful, and the granular analyses reported in the paper
4981 reflect genuine differences in model capability rather than artifacts of sampling variance or evaluation
4982 noise.

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